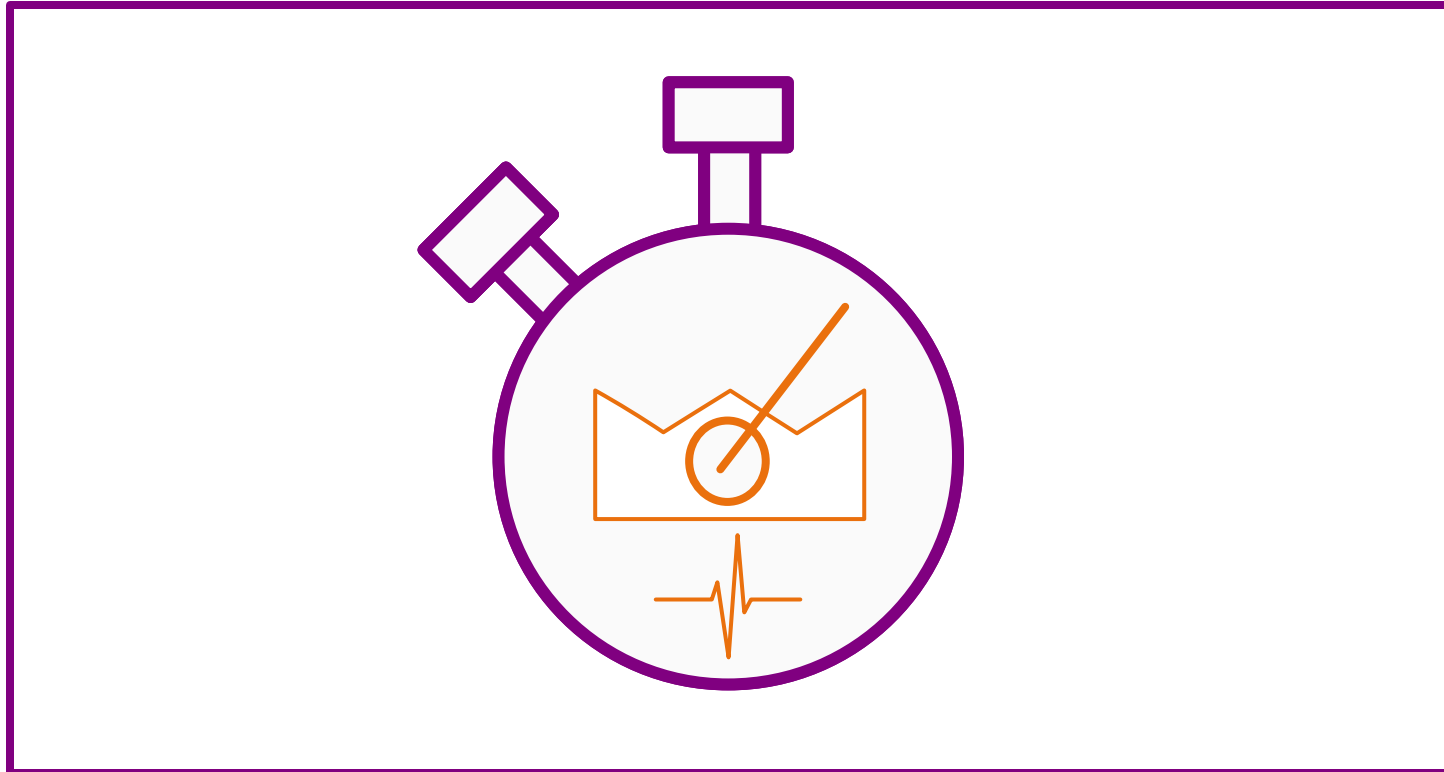


Time Maps

Follow a simple restaurant (Salsa Squares) example to learn the key concepts of the eVSM Quick stencils.
The content is organized to follow the continuous improvement process.



How to Use this File

This file contains the reading materials and the exercise pages from the course (title on previous page). While the course can only be taken on a computer, this booklet can be useful for note taking and later for refresher training.

This booklet is designed for on screen and print use. For on screen use, we recommend Acrobat Reader with the page display set to "Single Page View". If you are using this booklet on-screen while going through the exercises in eVSM, a second monitor is very helpful.

For hardcopy use, print the file on 8.5x11 or A4, and bind along the long edge.

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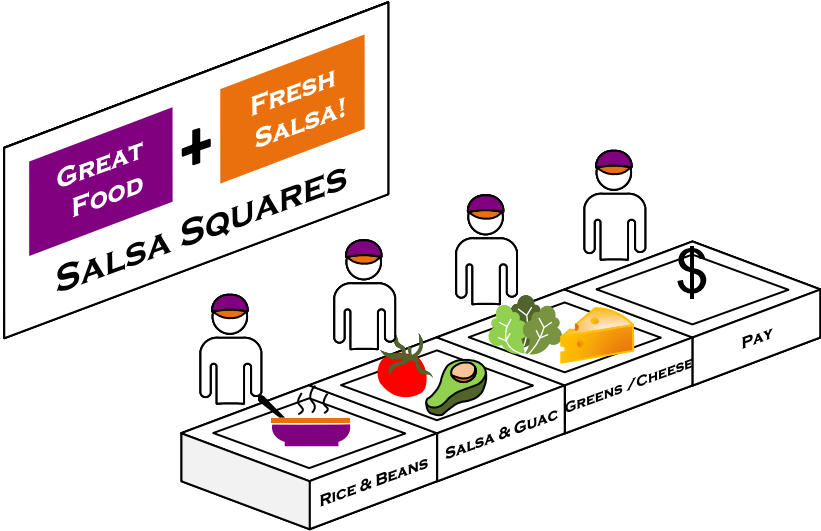
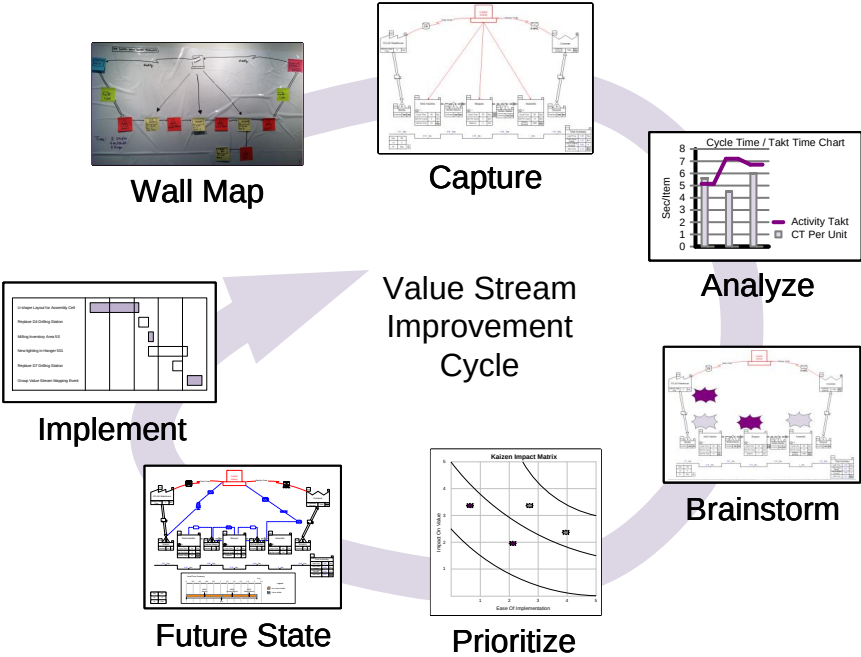
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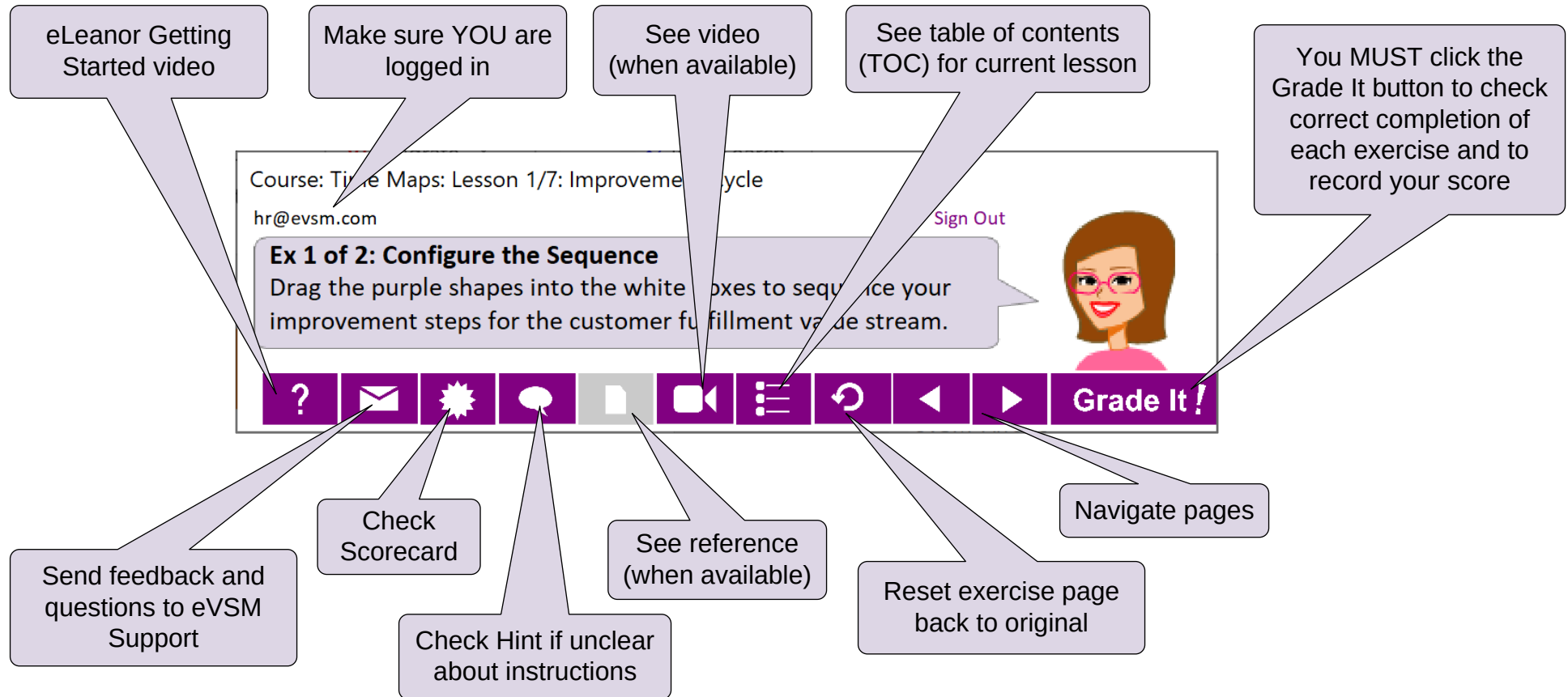
Improvement Cycle

Salsa Squares is a new restaurant concept that prides itself on its fresh food and great customer service. It’s in a competitive fast food industry and needs to have high efficiency to meet its targets.

You are the new OpEx manager for Salsa Squares and have been tasked with assessing and improving store operations. You decide to start with an understanding of the current customer fulfillment value stream from customer entry to order payment. You plan to use eVSM in support of the steps in the improvement cycle.

Case Study	Improvement Cycle
 The illustration shows a restaurant layout with four service stations: 'RICE & BEANS', 'SALSA & GUAC', 'GREENS / CHEESE', and 'PAY'. Staff members are shown at each station. A sign above the stations reads 'GREAT FOOD + FRESH SALSA! SALSA SQUARES'. Various food items like tomatoes, avocados, and cheese are depicted on the counter.	 The diagram illustrates the Improvement Cycle as a continuous loop of seven steps: Wall Map, Capture, Analyze, Brainstorm, Prioritize, Future State, and Implement. Each step is represented by a small icon: Wall Map (a network diagram), Capture (a value stream map), Analyze (a cycle time chart), Brainstorm (a mind map), Prioritize (a Kaizen Impact Matrix), Future State (a detailed value stream map), and Implement (a Gantt chart). Arrows connect the steps in a clockwise direction, with a large arrow pointing from the Future State back to the Wall Map, completing the cycle.
Assess and improve the Customer Fulfillment value stream of Salsa Squares.	Run the project using the steps in the improvement cycle.

Working with eLearner



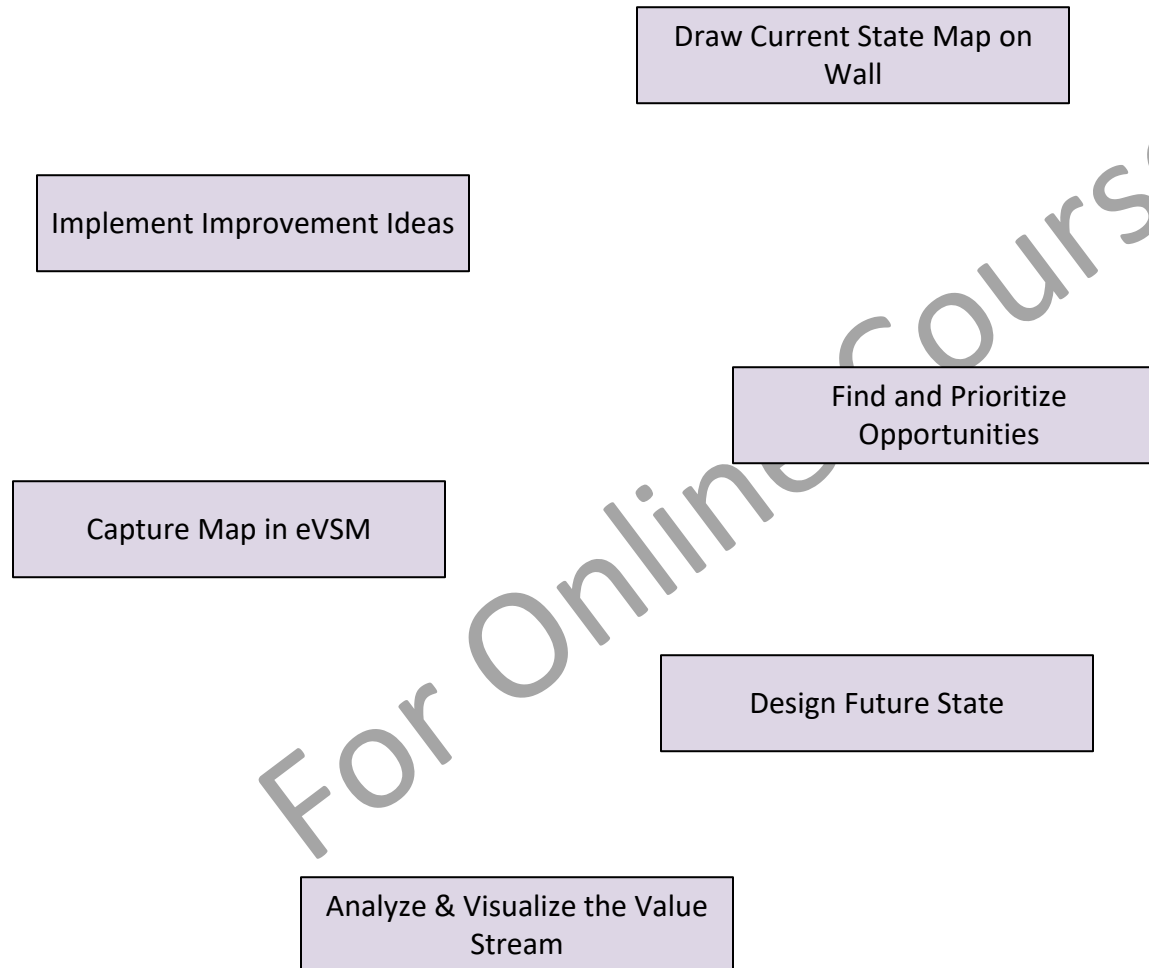
Important Notes

1. Make sure you have a good eLearner environment: large screen PC, 1280x720 resolution minimum, physical mouse with scroll wheel.
2. A PDF copy of the course is included in the downloaded fileset. It is very helpful to view this on a second while going through the training.
3. When you complete an exercise, you **MUST** click the "Grade It" button
4. You WILL lose points if you get an exercise wrong the first time
5. If you are stuck on an exercise, check the Hint. If that does not help, go back and review the preceding Readme pages. If you are still unsure, click the Feedback button in the eLearner panel and ask your question.

Continuous Improvement



You will need to watch the video (click the video button in the eLearner panel) to understand this sequence.



1

2

3

4

5

6

Value Stream Types

There are different value stream map types (plant, supply chain, etc.) and as you would expect, they differ a little in the metrics they use to visualize the waste. You should read through the map types (below), so you can select the appropriate type for each of your Salsa Squares value streams.

In eVSM, each value stream type is called a Quick Stencil.

Quick Stencils



Manufacturing

Plant level value stream maps for discrete parts and assemblies



Processing

Plant level value stream mapping of chemicals and food processes



Supply Network

Supply chain maps



Transactional

Office, services, and healthcare maps



Time

Simplest stencil that can be used for any value stream



Warehousing

Warehouse maps

Quick Stencil Components

Each value stream type or (Quick Stencil) has automation and standardized components built-in. These include:



VSM Icons

Complete Icon Set : The most comprehensive icon set for value stream mapping.



Variables

Standard Variables : Standardized variable names and meaning for each value stream type.



Macro Shapes

Macro shapes : Shapes with attached variables specific to each value stream type.



Equations

Built-In Equations : Equations to compute aspects of lead time, capacity, and cost.



Charts

Built-In Charts

Q. Which one of these Quick Stencil benefits is NOT true?

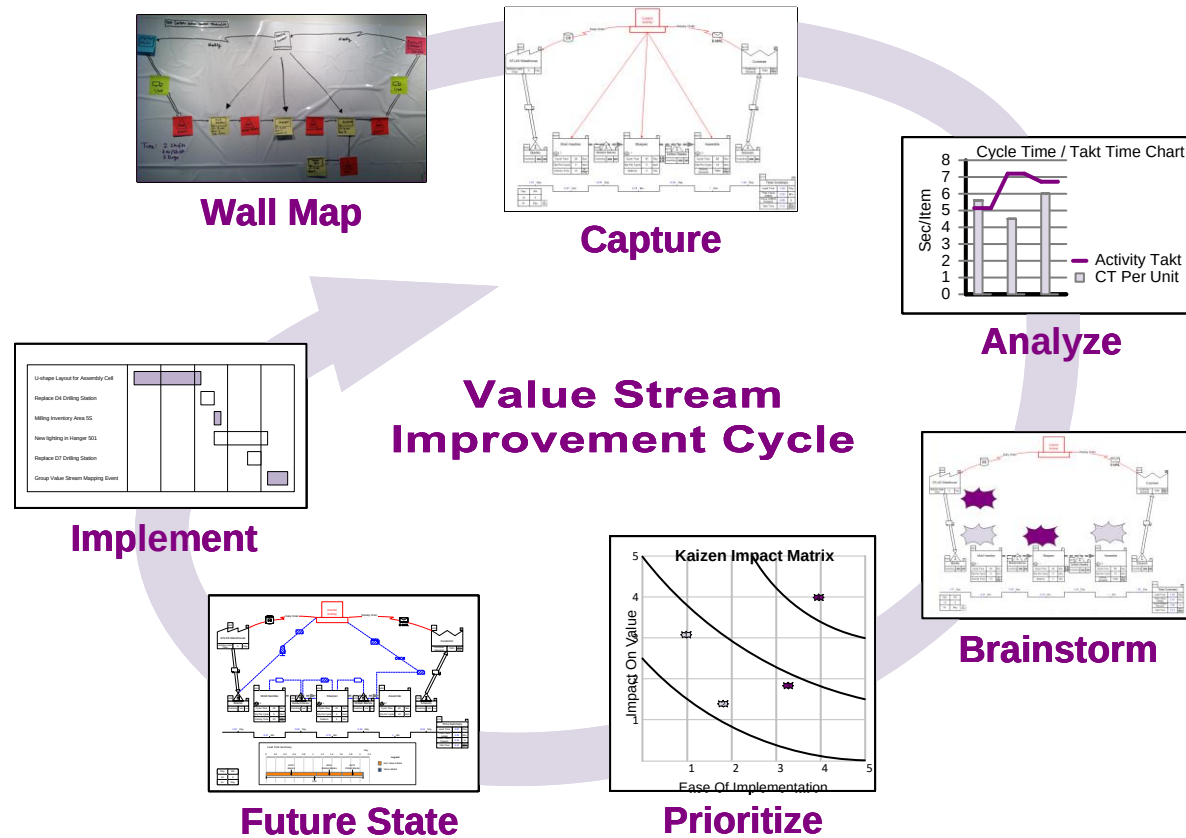
- ☐ Quick Stencils will tell you what improvements to make in what order
- ☐ The variables in the Quick Stencil will be pre-defined as a standard
- ☐ The calculation of Lead Time and other basic lean metrics will be automated
- ☐ Automated charts will be available
- ☐ Mapping shapes will come out as “macros” complete with data blocks



VSM Icons

You learned that:

- eVSM is designed to support the continuous improvement cycle
- “Quick Stencils” in eVSM support each value stream type
- Quick Stencils include a complete icon set, standard variables, macro shapes, built-in equations and built-in charts



What's next:

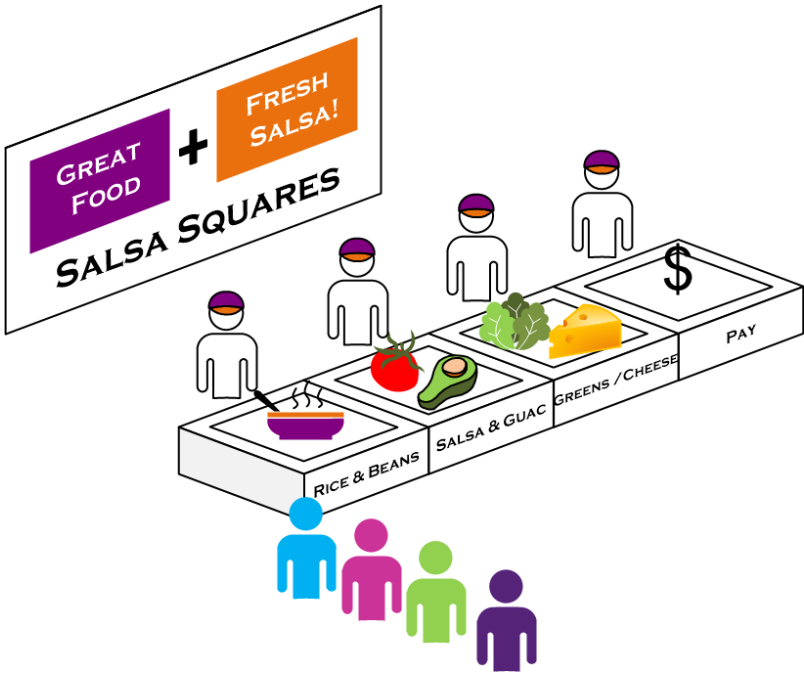
You are ready to apply the continuous improvement cycle to the Customer Fulfillment value stream, starting with the wall map and capture steps.

Capturing the Value Stream

You have established the improvement steps for the Salsa Squares Customer Fulfillment value stream and are ready to begin by creating a current state wall map, and then capturing it in eVSM. Let’s begin...

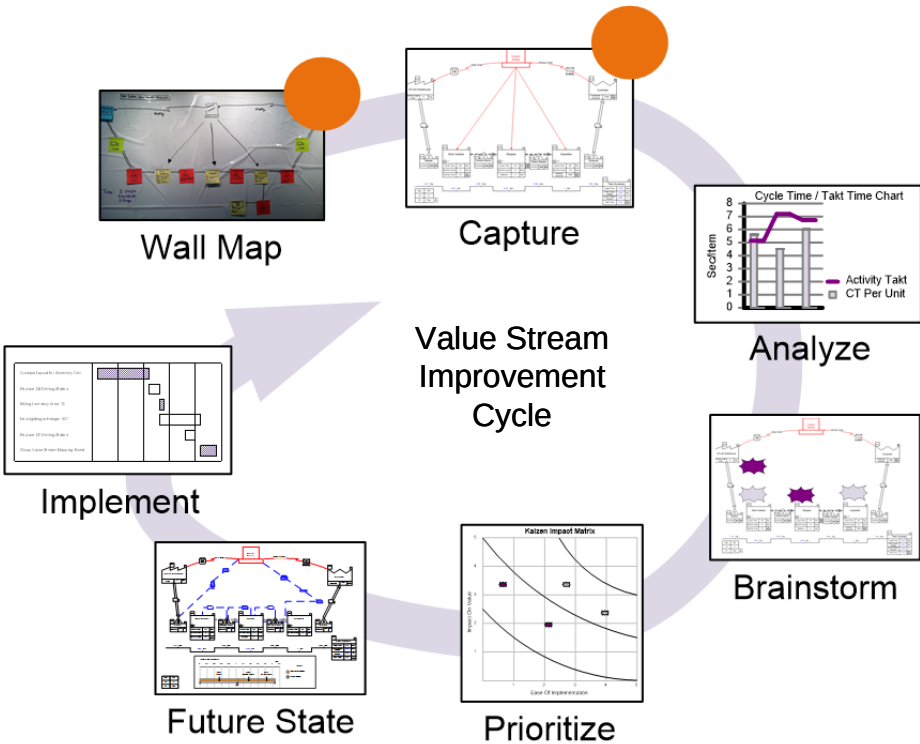
Capturing the Value Stream

Case Study



Work with the store team to develop a wall map of operations with average time measurements.

Improvement Cycle

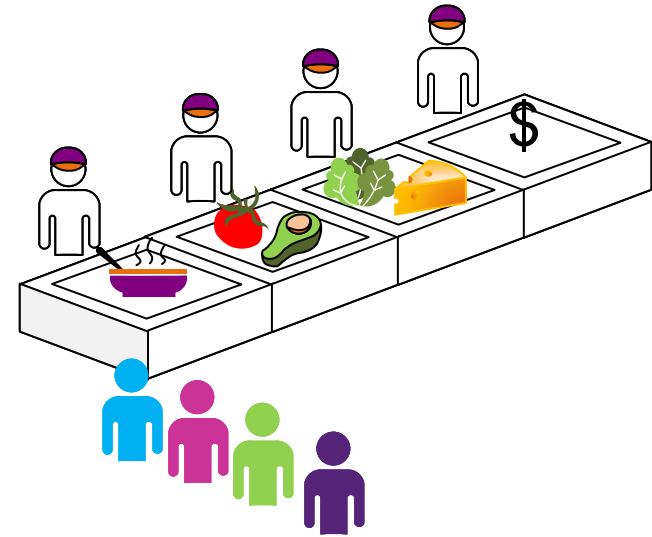


Accomplish the first two steps of the improvement cycle and with wall map capture in eVSM.

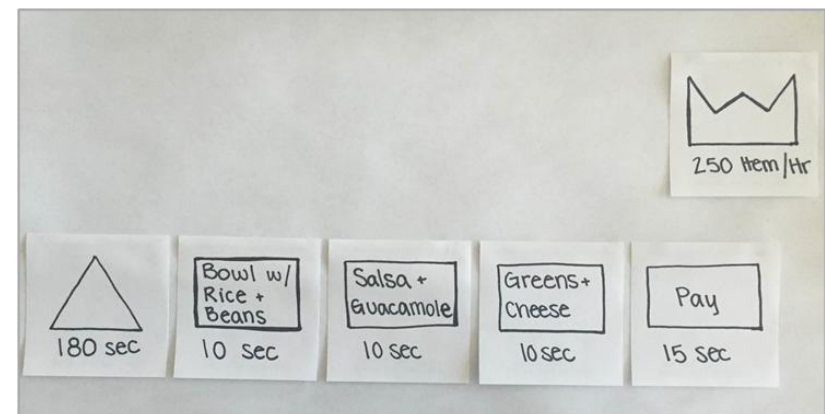
Creating a Wall Map

Important steps ahead of creating a wall map are in defining which value stream is being mapped, and depending on it's complexity, formally organizing a mapping event.

In our simple example, let's say you have already done this and then taken time measurements over an hour for the Customer Fulfillment process below, and over the lunch period which you want to analyze.

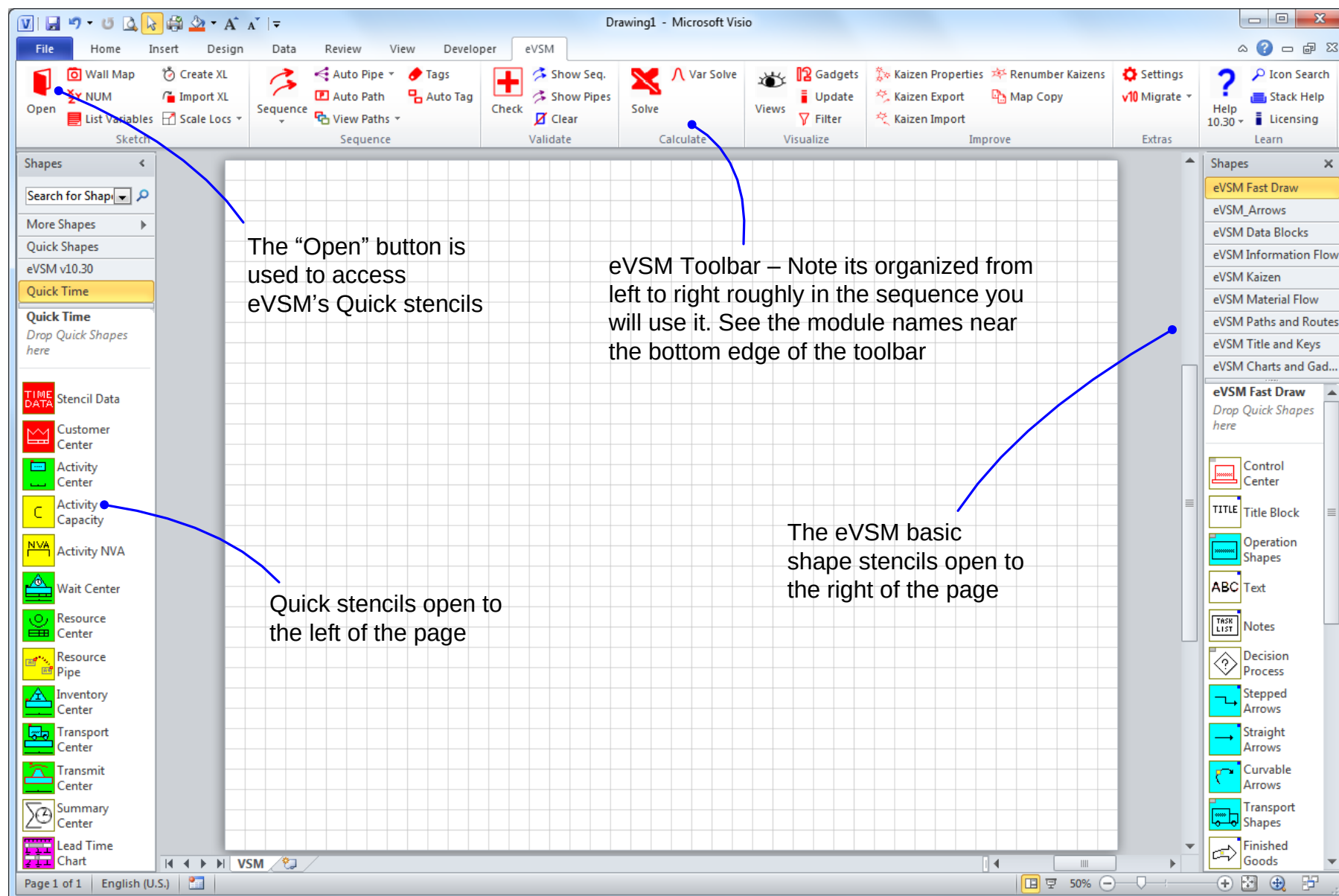


From averages of the measurements, you create the “wall map” on the right.

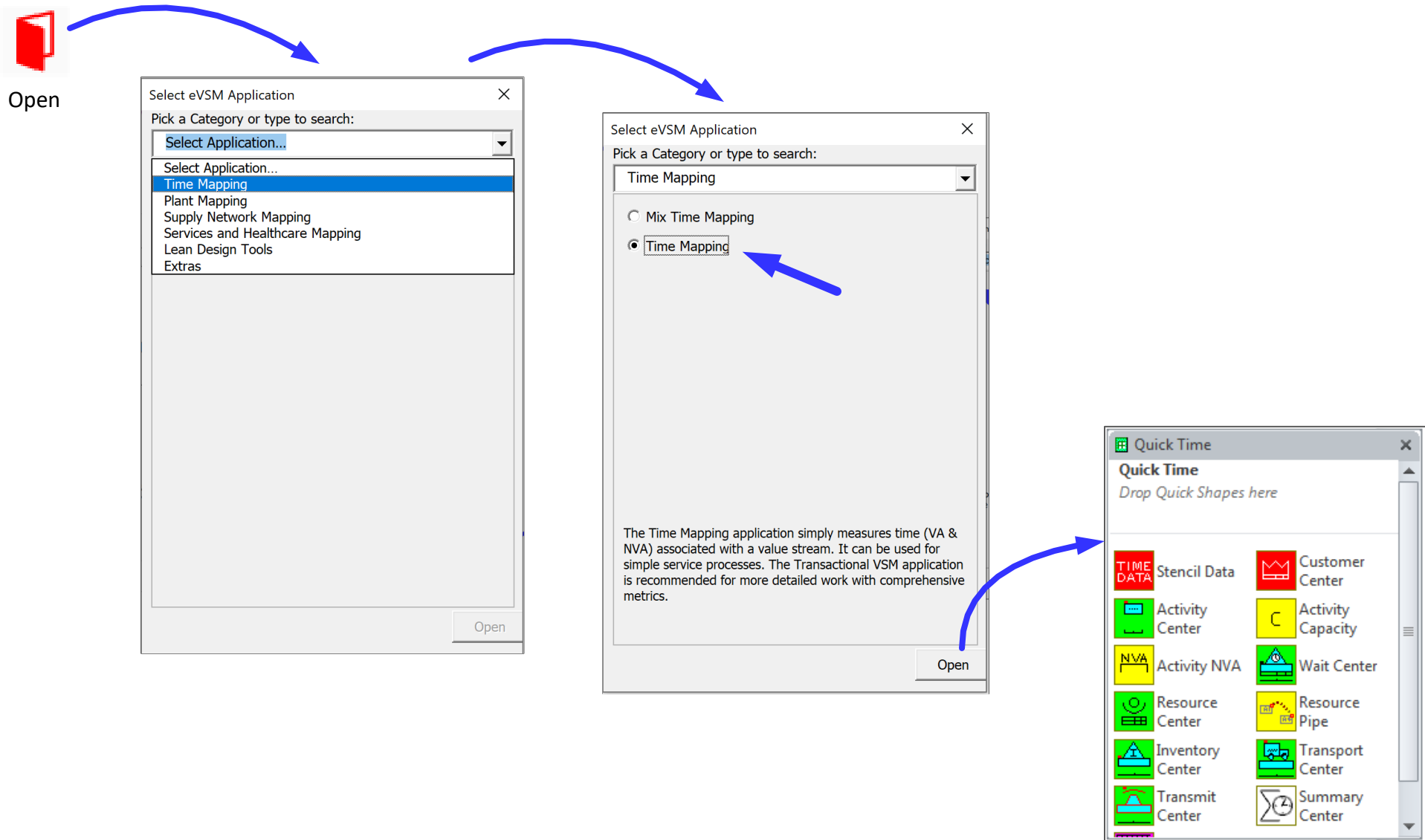


Capturing the Wall Map in eVSM

You have a wall map, and the next step is to capture it in eVSM. You review (below) the components of the eVSM environment.



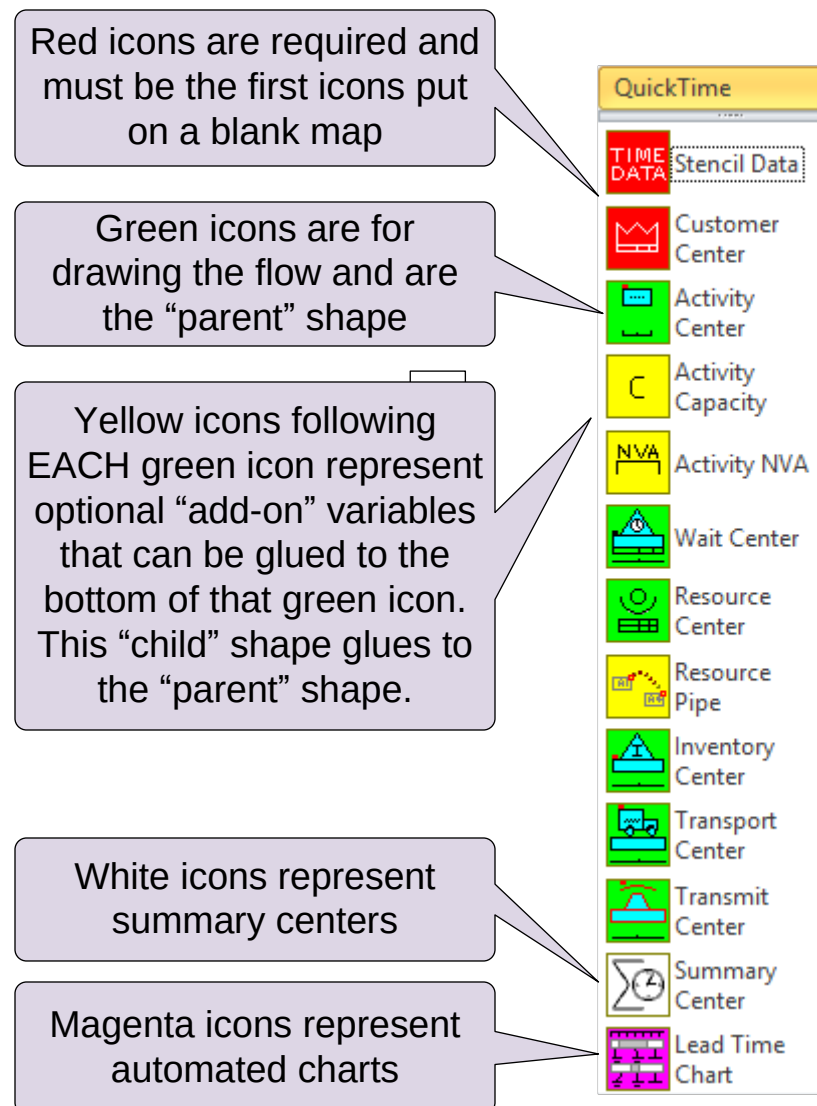
Opening the Quick Time Stencil



For Online Course Only

Icon Color Codes in Quick Stencils

Before you capture the wall map using eVSM's Quick Time stencil, you review how the icons in such stencils are color-coded and organized.



Q. Quick Stencil Icon Colors: Tick (double-click) all the following that are true.

- ☐ Magenta icons represent charts
- ☐ Red icons are the first mandatory quick stencil shapes on a map
- ☐ Green icons represent parent shapes for drawing the flow
- ☐ Yellow icons can be used instead of green icons



Customer
Center



Activity
Center



Activity
Capacity



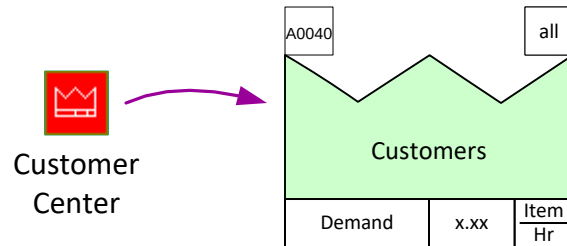
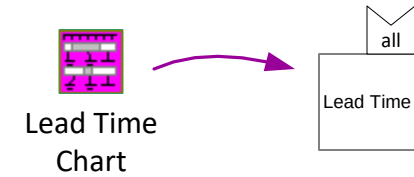
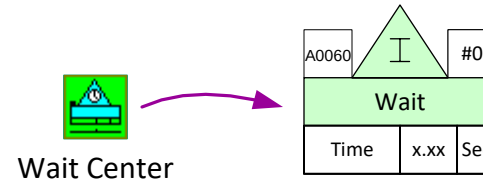
Lead Time
Chart



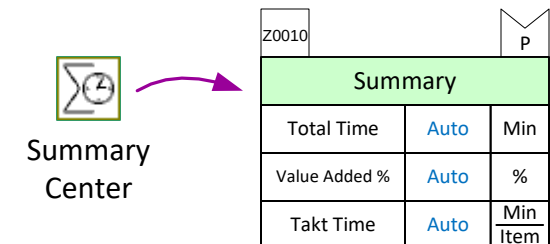
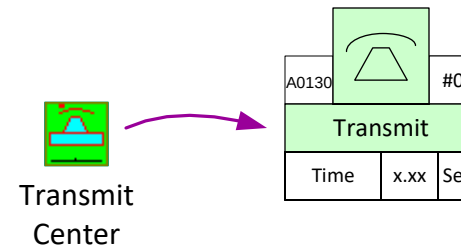
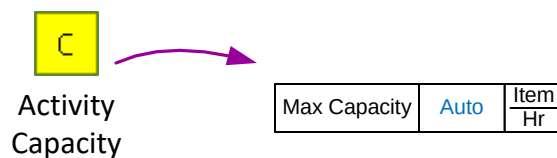
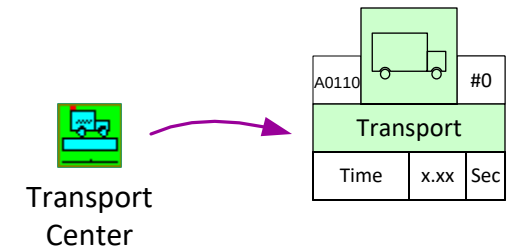
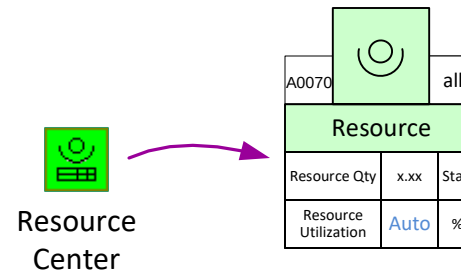
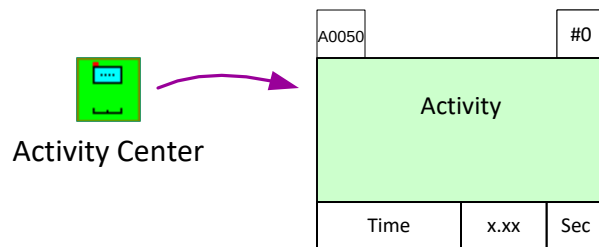
Summary
Center

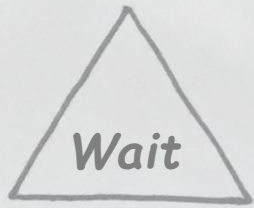
Quick Time Shapes

The Quick Time stencil has macro shapes that you can drag out and drop. You will have to use these to capture the wall map of the Customer Fulfillment value stream and review the contents of each macro shape (below).



Auto Sec





180 sec

Bowl w/
Rice +
Beans

10 sec

Salsa +
Guacamole

10 sec

Greens +
Cheese

10 sec

Pay

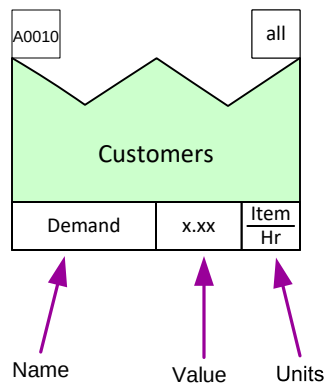
15 sec



250 Item/Hr

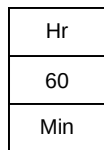
Moving Shapes Around in eVSM

We are going to be looking at changes in the Customer Fulfillment value stream and this will require editing the map and moving shapes around. Let's review some tips that make modification easy.

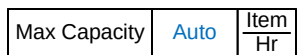


The green shapes dragged out from Quick Stencils are “parent” shapes. If you move the parent, then all the connected child shapes move also.

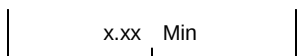
Data shapes consist of a Name field and/or a Value field, and/or a Units field. To move or delete a data shape, you MUST click on the Value field.



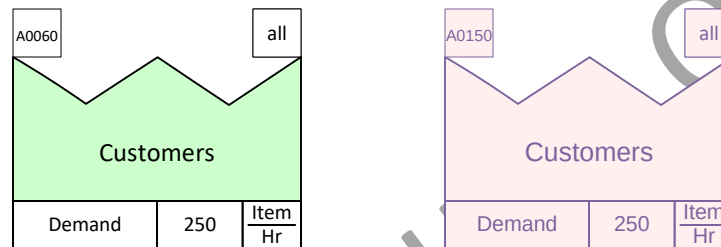
For unit converter shapes, the middle block is the parent.



For data shapes, the middle block (the value field) is the parent shape also. So, if you want to move or delete the data shape, you must click on the value field.



For VA or NVA shapes, the left block is the parent.



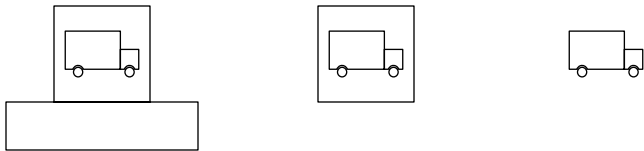
How are the eVSM Base Stencils Organized?

The eVSM Base Stencils are on the right of the screen and contain individual eVSM drawing shapes. The Quick Stencils on the left of the screen are actually macro combinations of shapes.

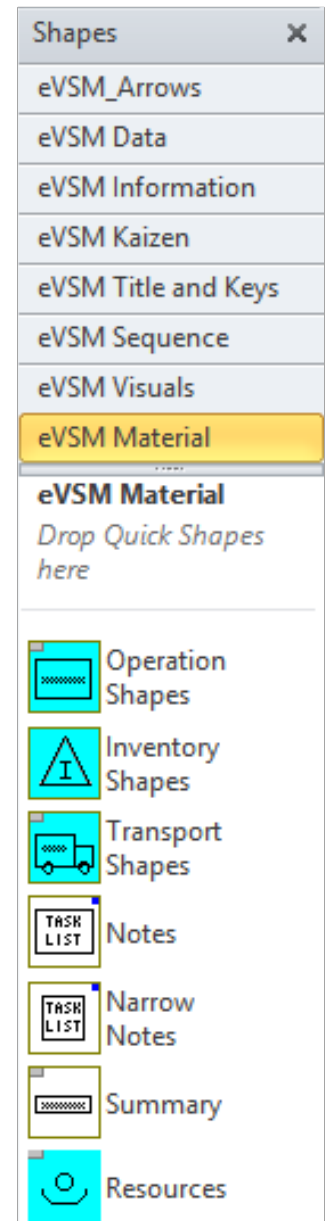
Base stencils have blue and white icons. The blue icons are actually FAMILIES of shapes. To access members of the family, you:

- Drag out the icon
- Right-click the shape on the page
- Use “Change Shape” to select a different member

There are also right-click options to change the framing around the shape.

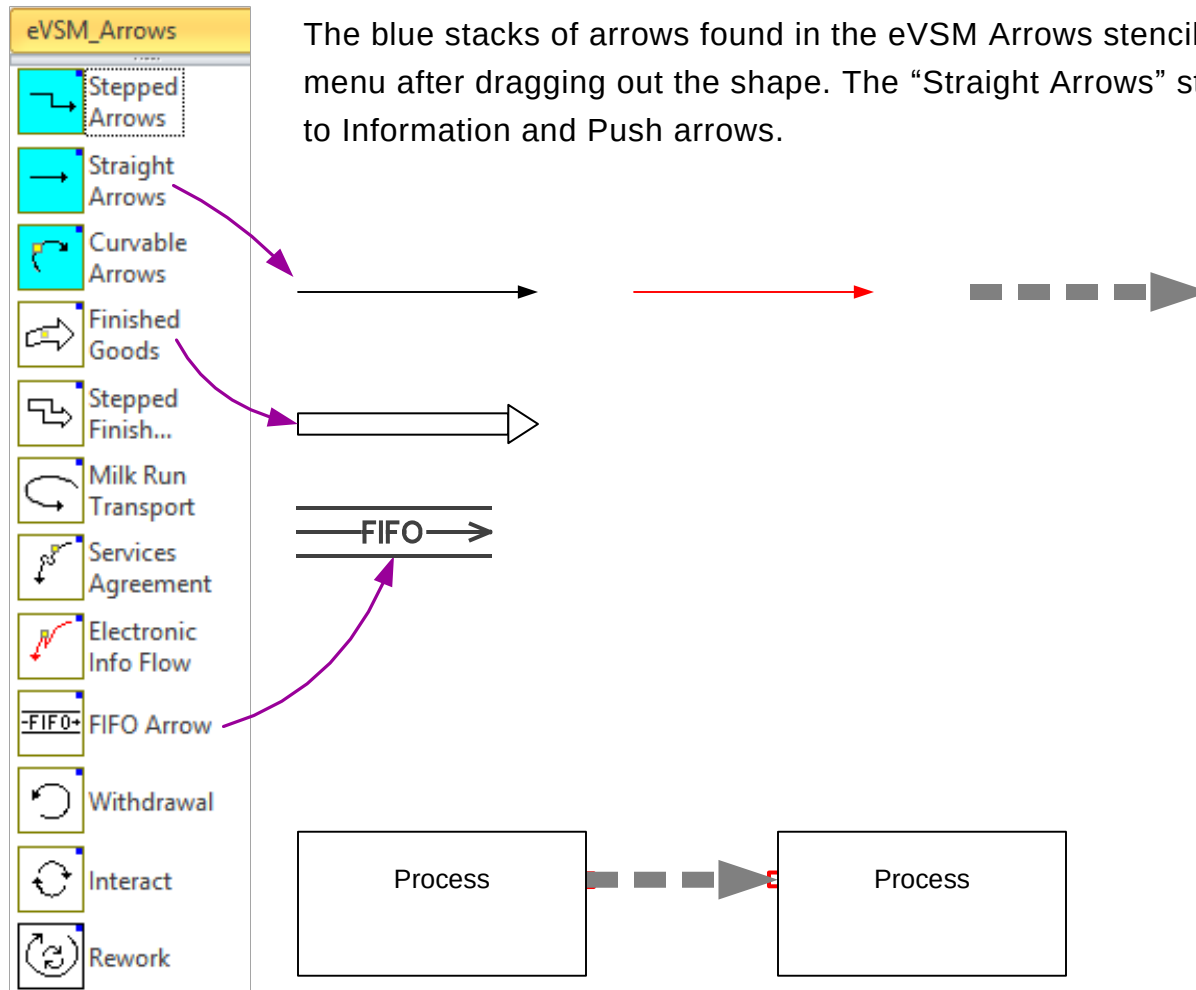


Base stencils are organized by categories like eVSM Arrows, eVSM Data, etc.



Arrows on Value Stream Maps

You remember that the arrows between the VSM shapes have specific meaning on a map. For example, push arrow, information arrow, etc. You review the different arrow types before you add the arrows to the captured model.



The blue stacks of arrows found in the eVSM Arrows stencil have a variety of arrows in the right-mouse menu after dragging out the shape. The “Straight Arrows” stack below has been dragged out and changed to Information and Push arrows.

Make sure to glue each end of the arrow (by making sure a glue connection appears) to the center it is associated with so that if the center is moved, the arrow will move with it.

A0010				1
Salsa / Guacamole				
Time		10	Sec	

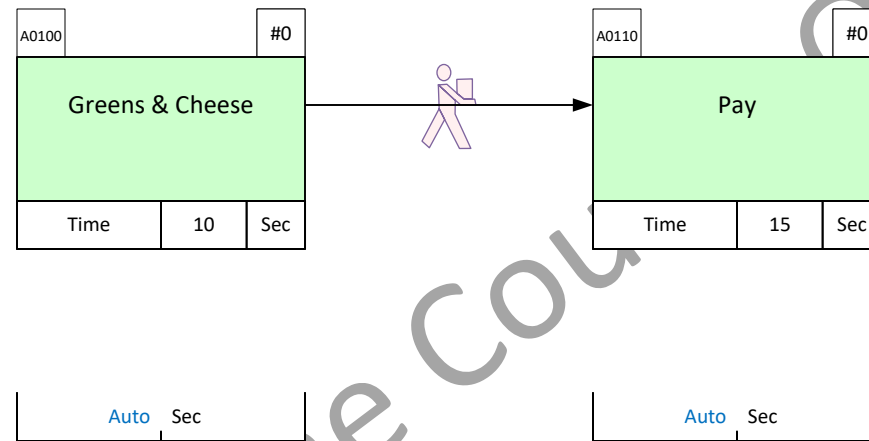
Auto	Sec
------	-----

A0020				1
Greens & Cheese				
Time	10	Sec		

Auto	Sec
------	-----

A0030			1
Pay			
Time	10	Sec	

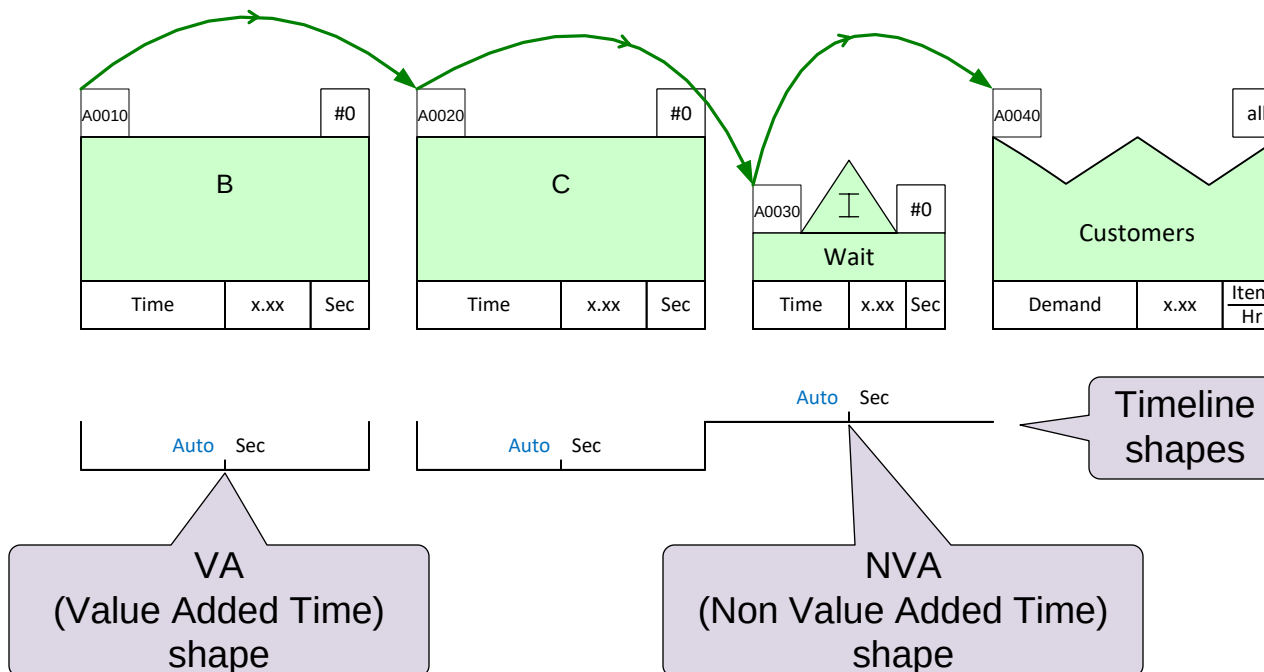
Auto	Sec
------	-----

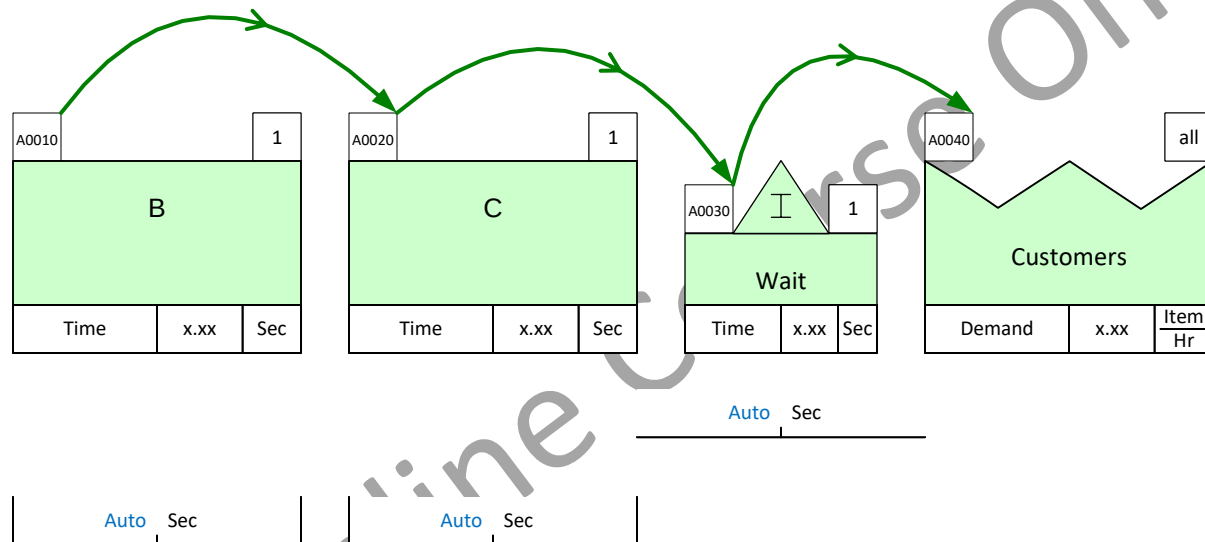


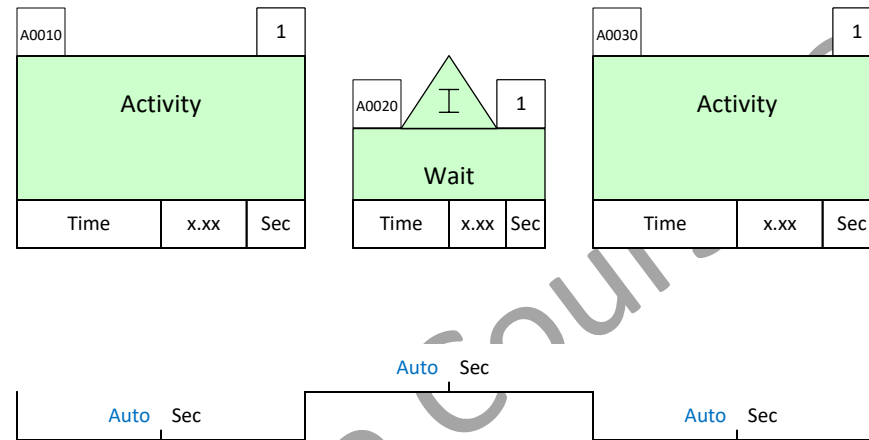
Working with Timeline Shapes

As you draw the value stream map, you notice that the timeline shapes are part of the macro shapes. Aligning these can be painful so there are a few help functions. You review these below.

1. The timeline shapes that appear below many of the center's (shown in green below), denote VA and NVA times for the map.
2. In general if you line up the base of the green shapes, you will find that the timeline aligns correctly also.
3. If you want to align a timeline because the shapes are misaligned, then select any one of the timeline shapes, and right-click to access the "Align Timeline" function.
4. If you want to move a whole timeline, then select any one of the timeline shapes and right-click to access the "Select Timeline" function. Once you have selected the whole timeline, just drag it down to move it (typically to make space for additional data blocks for the centers).



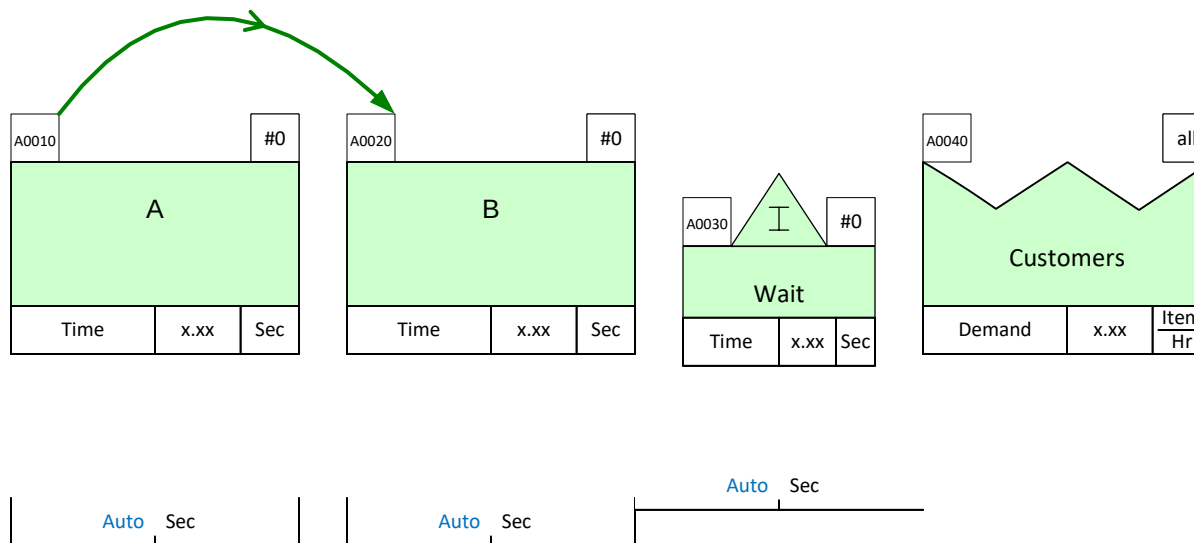




Establishing Sequences on the Map

You will want to know the Total Customer Time through the process, and this requires some knowledge of the process sequence. This is established as shown below.

1. Sequence arrows allow you to perform upstream and downstream calculations.
2. To connect a single arrow, select the upstream center for example, “A”, then shift-select the downstream center, for example, “B” and finally click the “Sequence” button in the eVSM toolbar.  Sequence
3. You can create arrows individually or as a chain. To create the chain, keep selecting more downstream centers in the correct order using shift-select. Then click the sequence button in the toolbar.



A0010			#0
A			
Time	x.xx	Sec	

Auto	Sec
------	-----

A0020				#0
B				
Time		x.xx	Sec	

Auto	Sec
------	-----

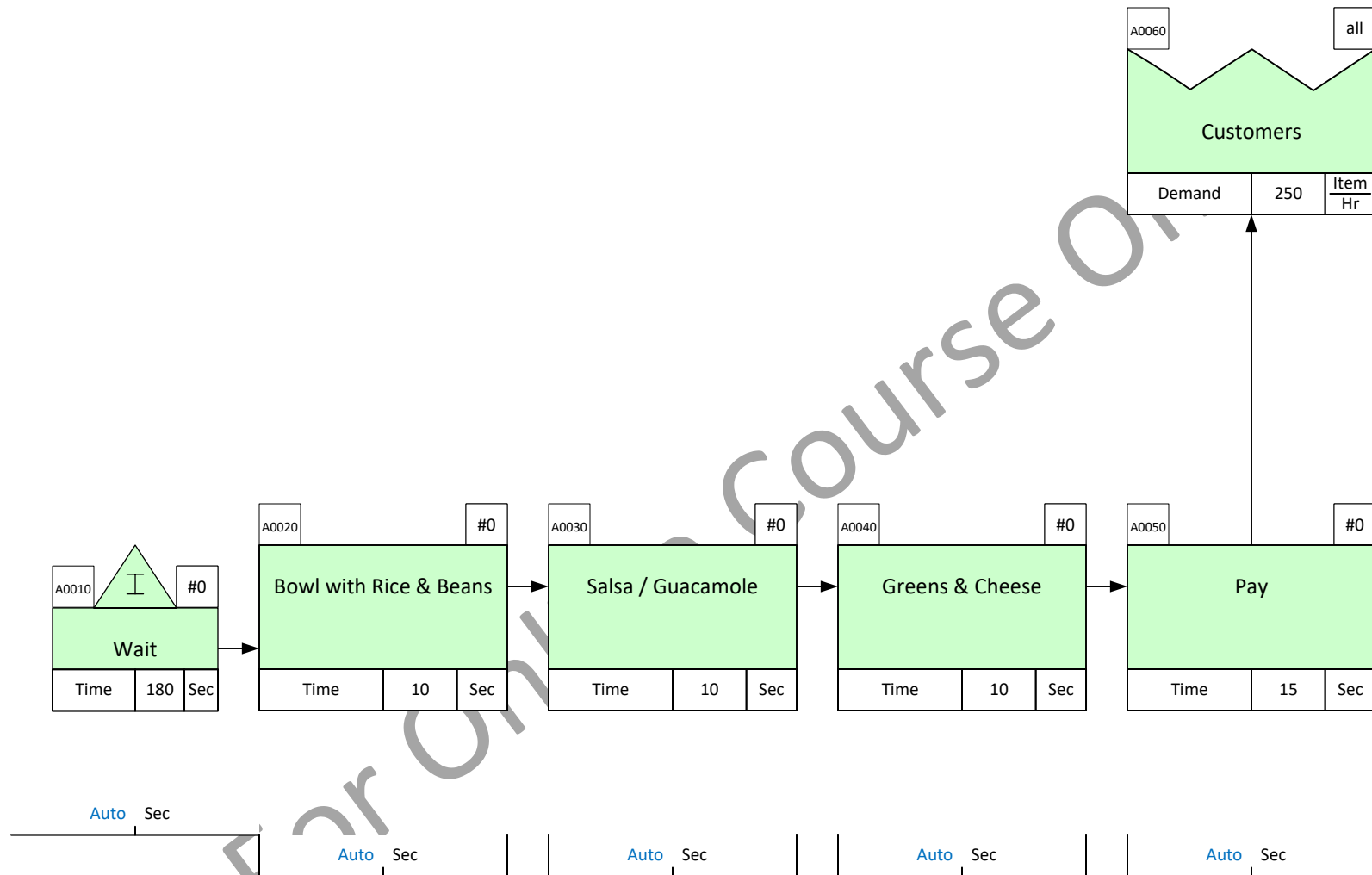
A0030				#0
C				
Time		x.xx		Sec

Auto	Sec
------	-----

A0040	I	#0
D		
Time	x.xx	Sec

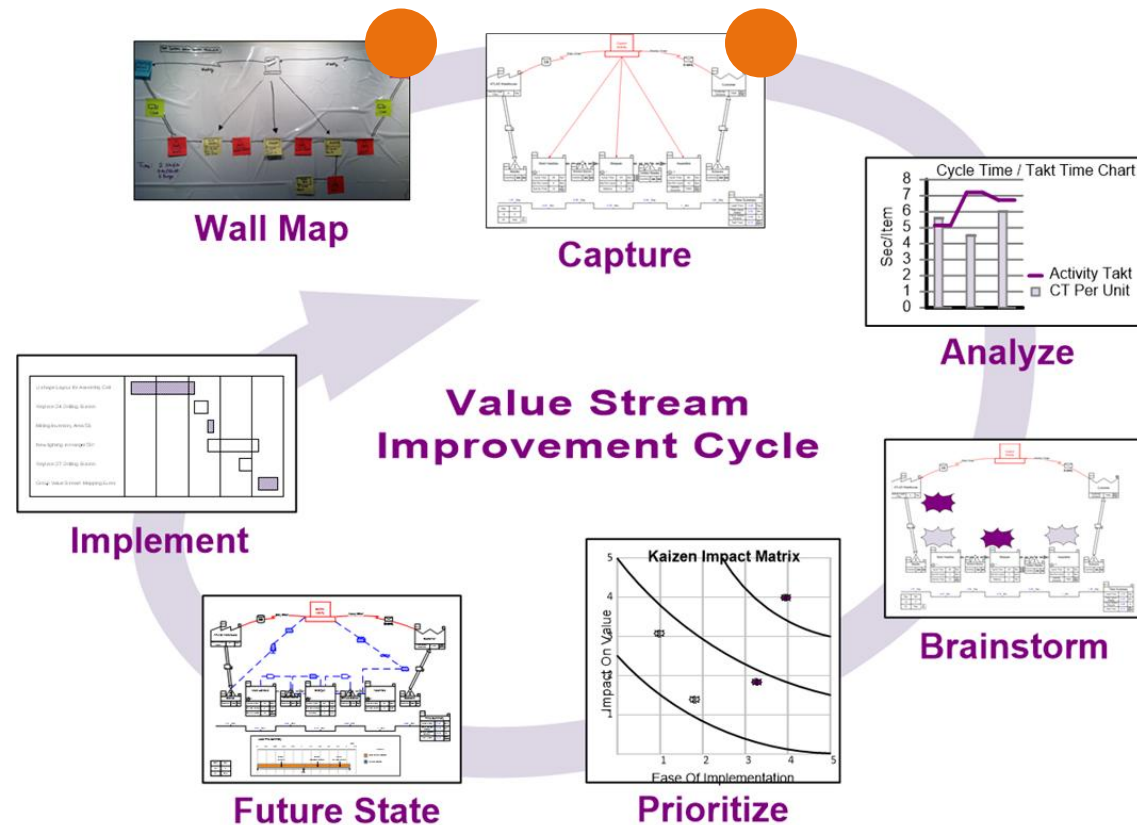
Auto	Sec
------	-----

A0050	all	
E		
Demand	x.xx	Item Hr



You learned that:

You can capture the wall map electronically in eVSM, complete with data values and process sequence. You can see it's fast and simple using the Quick Time stencil.



What's next:

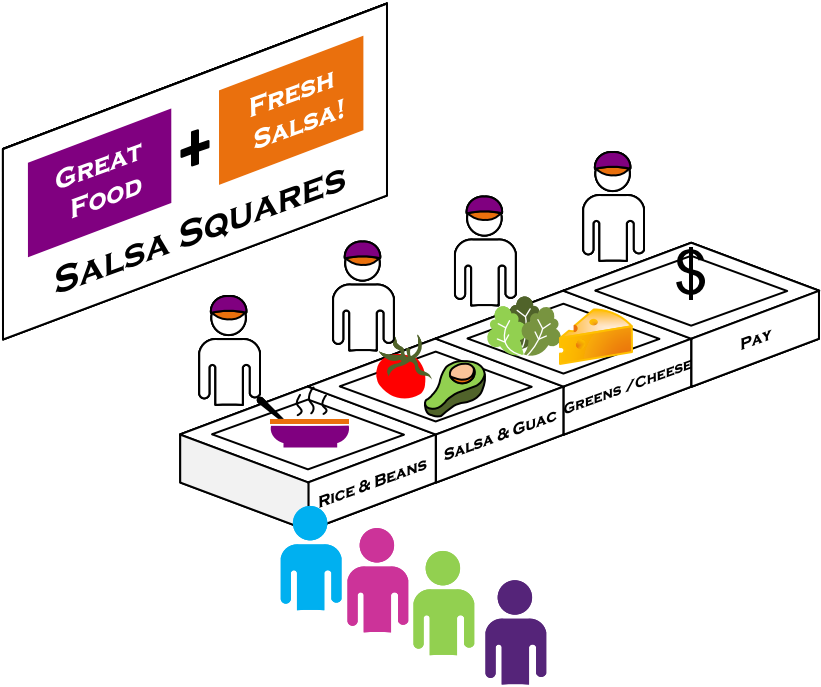
This map can be shared with others, but how do you use it to analyze and visualize the waste in the value stream?

Analyzing and Visualizing the Value Stream

You have captured the map for the Customer Fulfillment value stream and your next step is to analyze the map to help you see the waste in terms of metrics like VA, NVA, Lead Time, and Capacity. Let’s learn how.

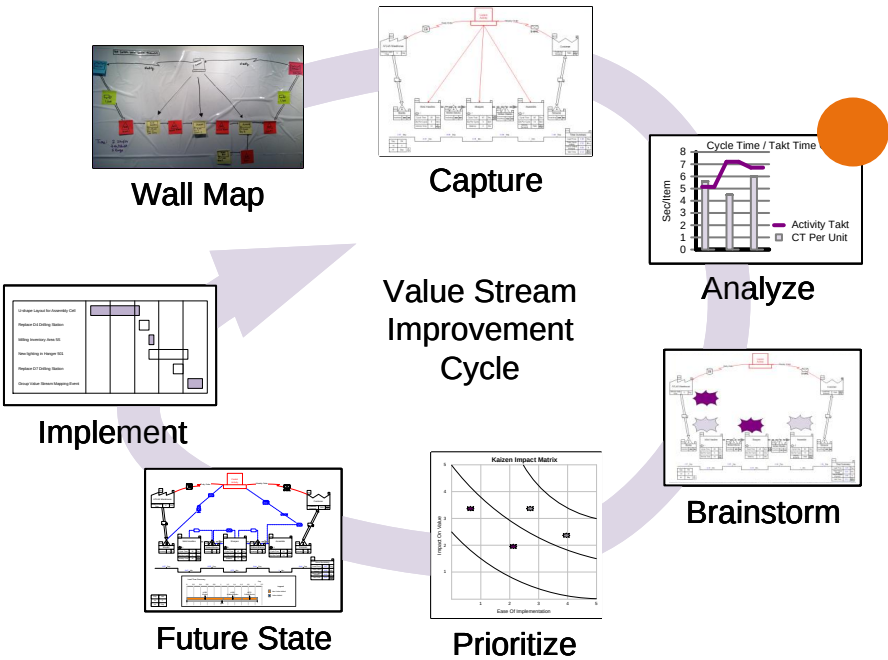
Analyzing and Visualizing the Value Stream

Case Study



By analyzing the map, it will be easier to “see” the waste in the Customer Fulfillment value stream.

Improvement Cycle



The analyze phase will allow waste to be looked at in terms of key metrics and associated charts. The goal is to provide a clear visual understanding that will lead to better ideas in the “Brainstorm” step.

Checking for Common Mapping Mistakes

Before you Solve the map, just check it for common mapping errors. The Check button  Check identifies common user mistakes on the map. In the map below for example, the user has forgotten to fill in the two input data fields (x.xx)

The “Check” button Check

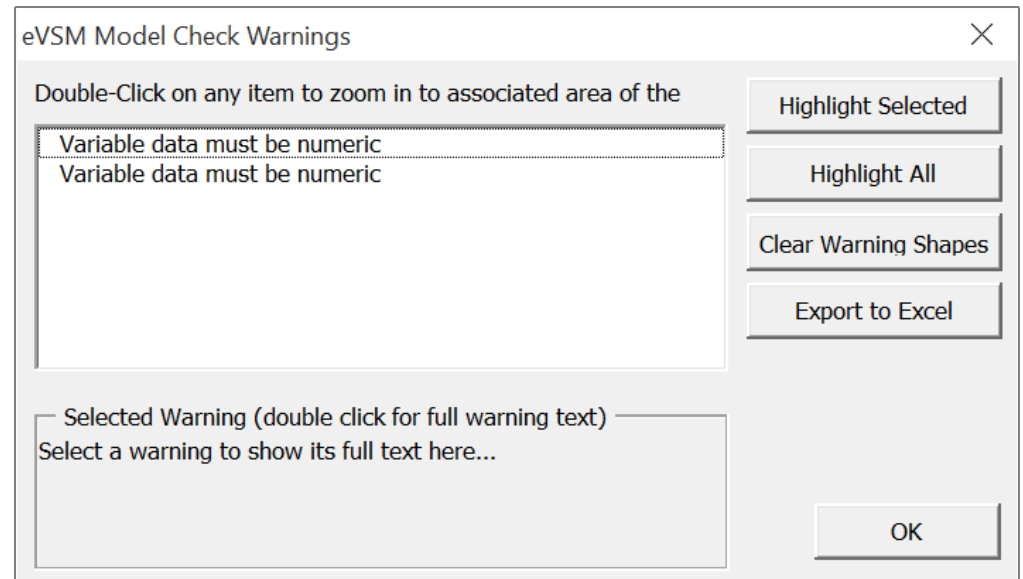
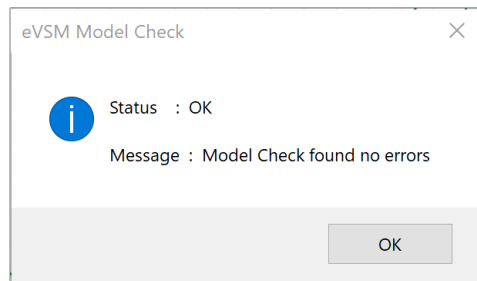
The “Check” button is designed to catch some of these problems and will display this form if it finds any.

Example problems are:

- Unconnected Sequence arrow
- Missing mandatory data
- Missing units convertors
- Unglued datashapes
- Duplicate Operation tags

You can select the error and use the Highlight buttons in the form to highlight and zoom into the problem area on the map.

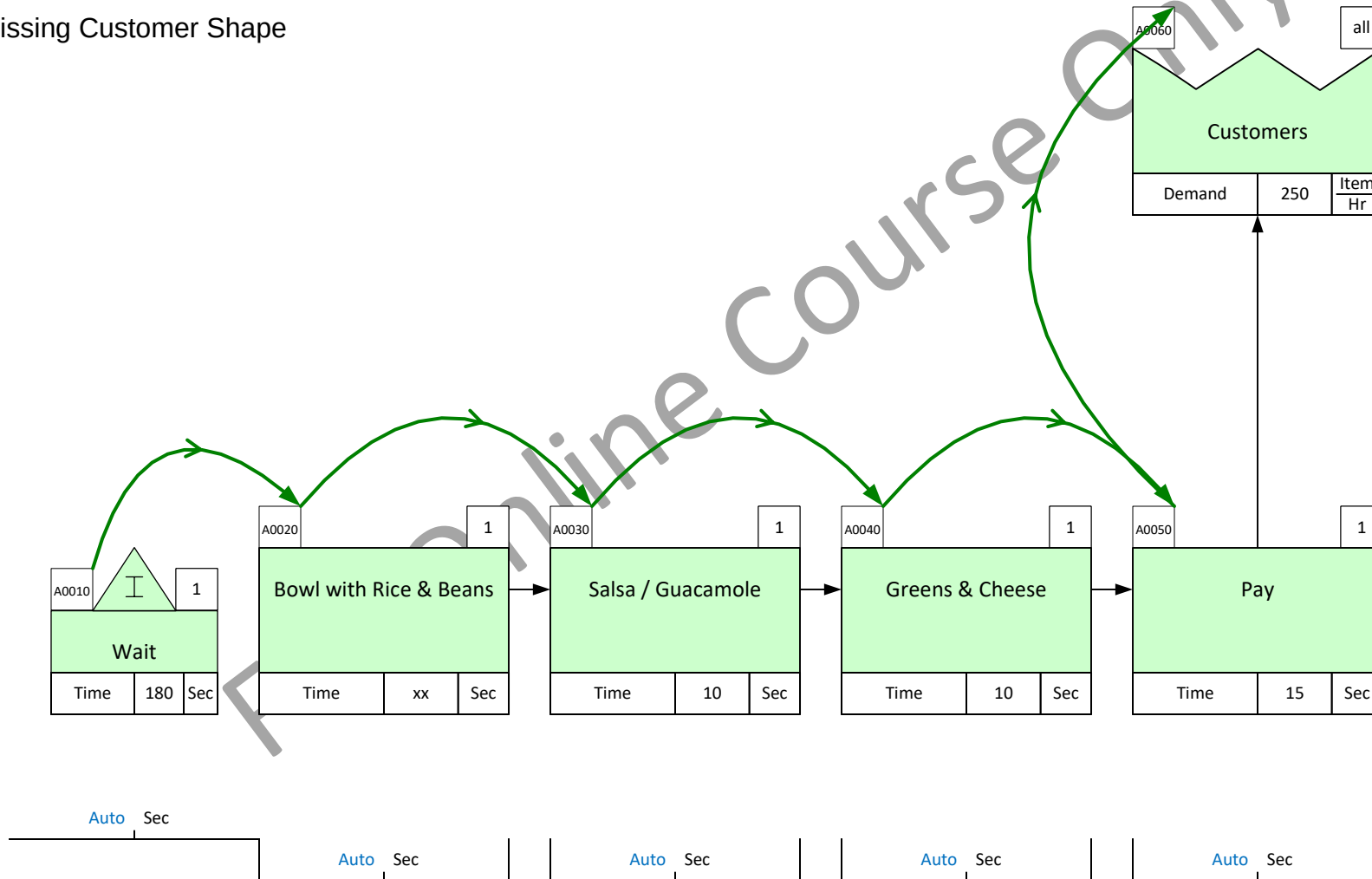
The problems will be reported incrementally. So, fix the problems reported and then run the Check function again. Repeat until you see a message like this:



Which of the following problems were reported by the Check button?

Tick all that apply.

- ☐ Missing data value
- ☐ Disconnected Sequence Arrow
- ☐ Missing Customer Shape



Solving the Map

The Quick Time stencil you are using includes equations that do the basic lean calculations for VA time, NVA time, Takt Time, and Total Customer Time.

Once you have checked the map, you can use the “Solve” button to apply the equations. Values on the map that are a result of the equations are shown in blue, (see below).



Before Solve

A0010			#0
Bowl with Rice & Beans			
Time	10	Sec	
Max Capacity	Auto	Item Hr	

Auto	Sec
------	-----

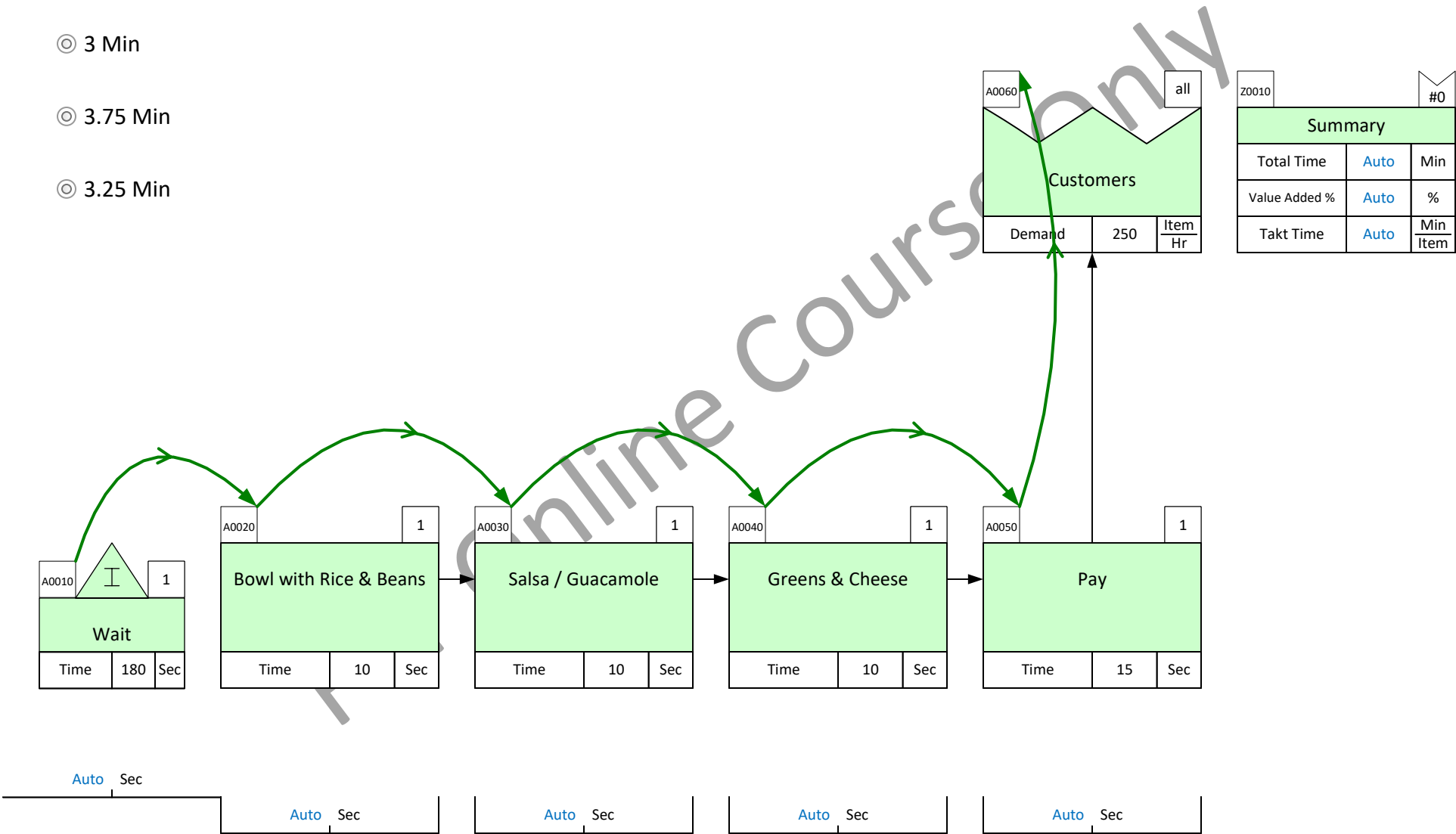
After Solve

A0020			1
Bowl with Rice & Beans			
Time	10	Sec	
Max Capacity	360.00	Item Hr	

10.00	Sec
-------	-----

“Solve” the map to find the “Total Time” for the value stream, and select the correct value below.

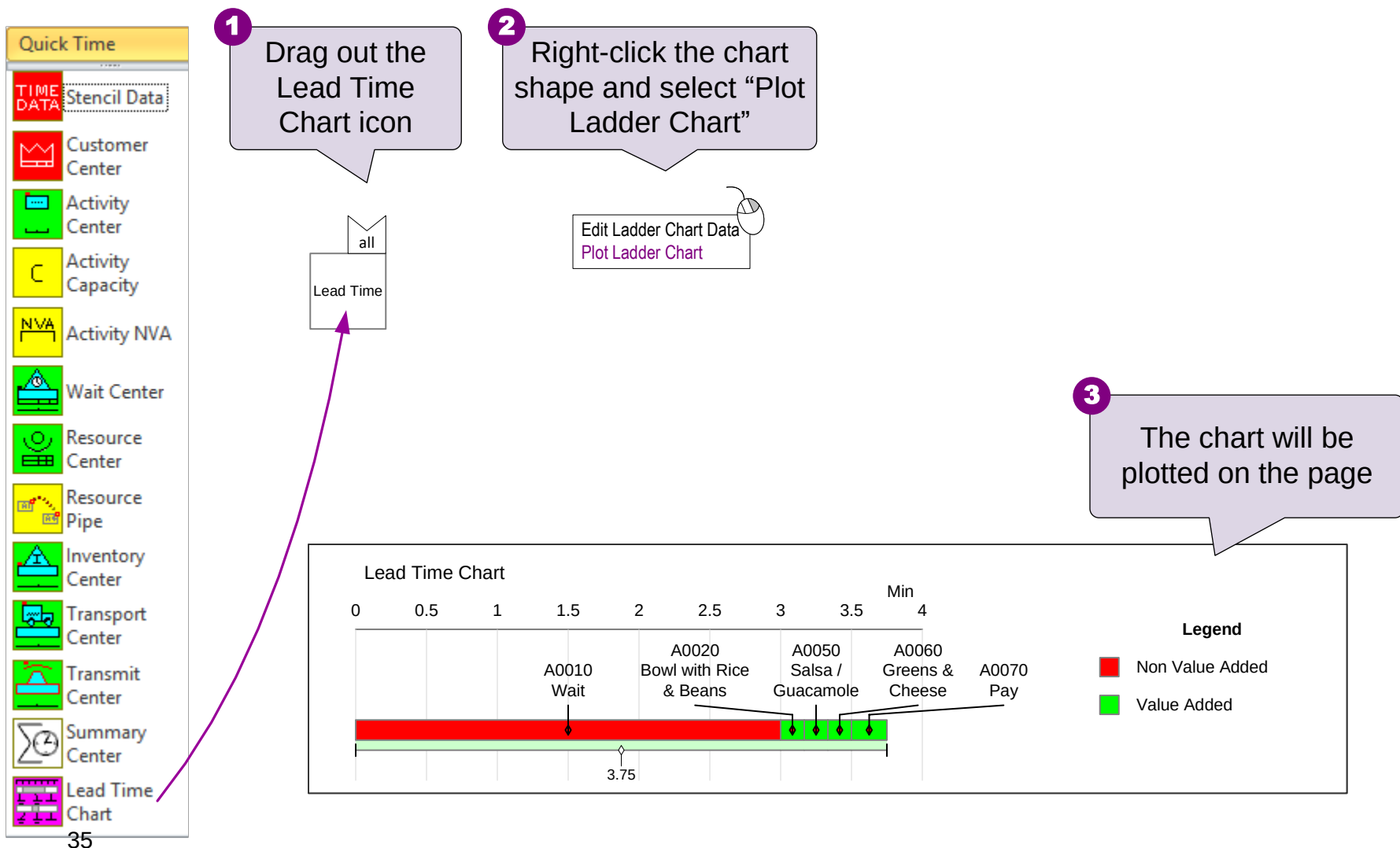
- ☐ 4 Min
- ☐ 3 Min
- ☐ 3.75 Min
- ☐ 3.25 Min

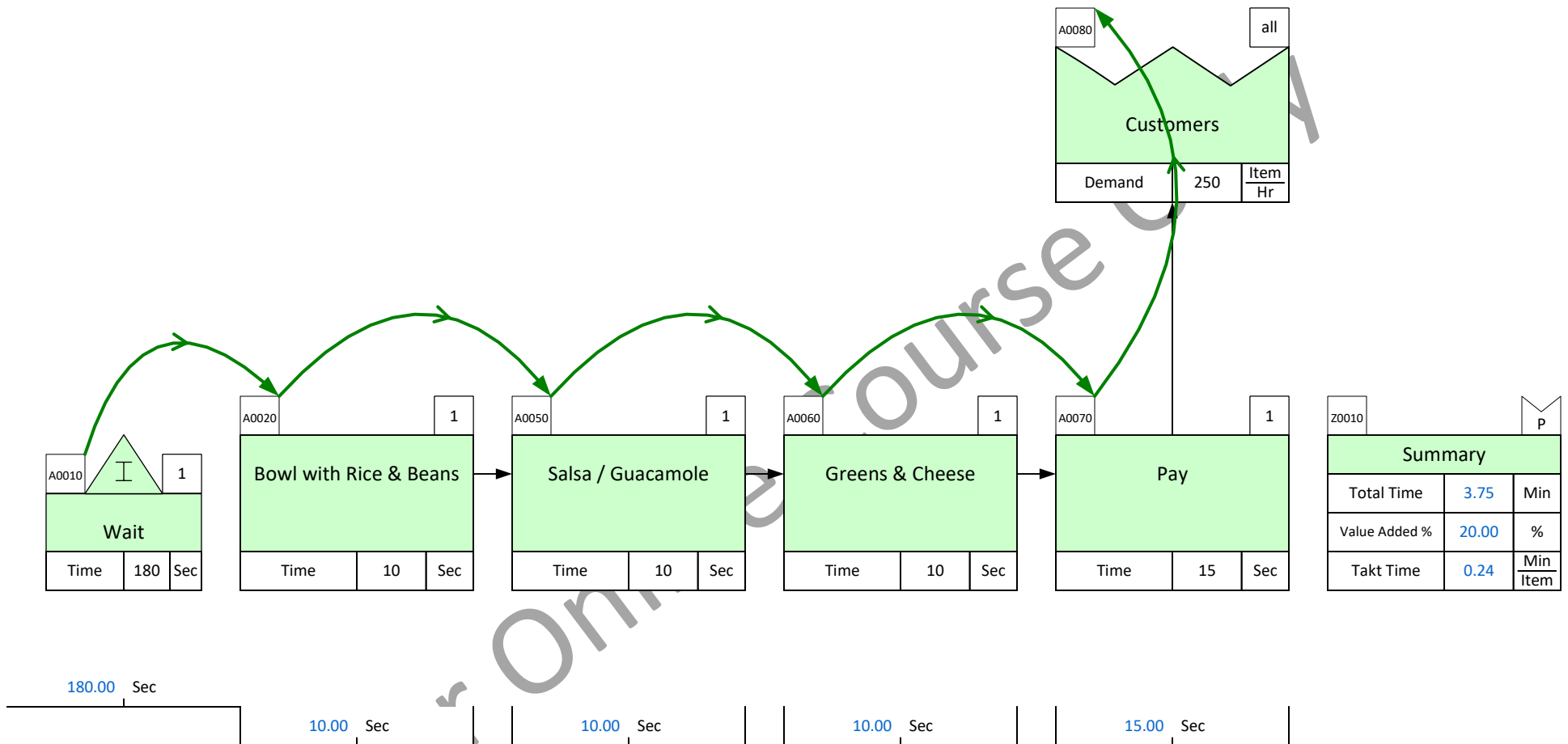


Using Charts to Visualize the Data

As maps get larger, it is hard to understand the data by just looking at the numbers. Each Quick Stencil comes with pre-built charts (like the Lead Time Chart), and it's also possible to create user-defined charts.

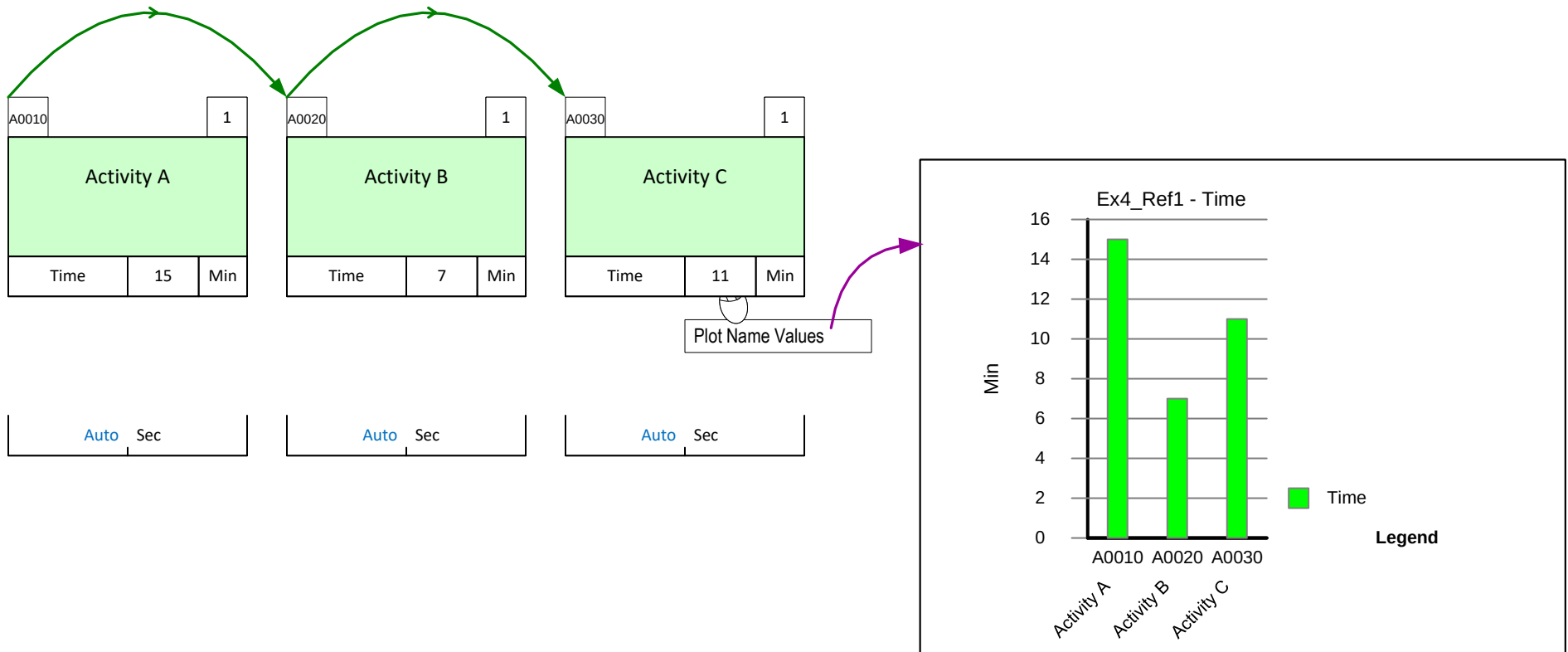
Pre-built charts are dragged out from the Quick Stencil, and then plotted via the right-click menu.

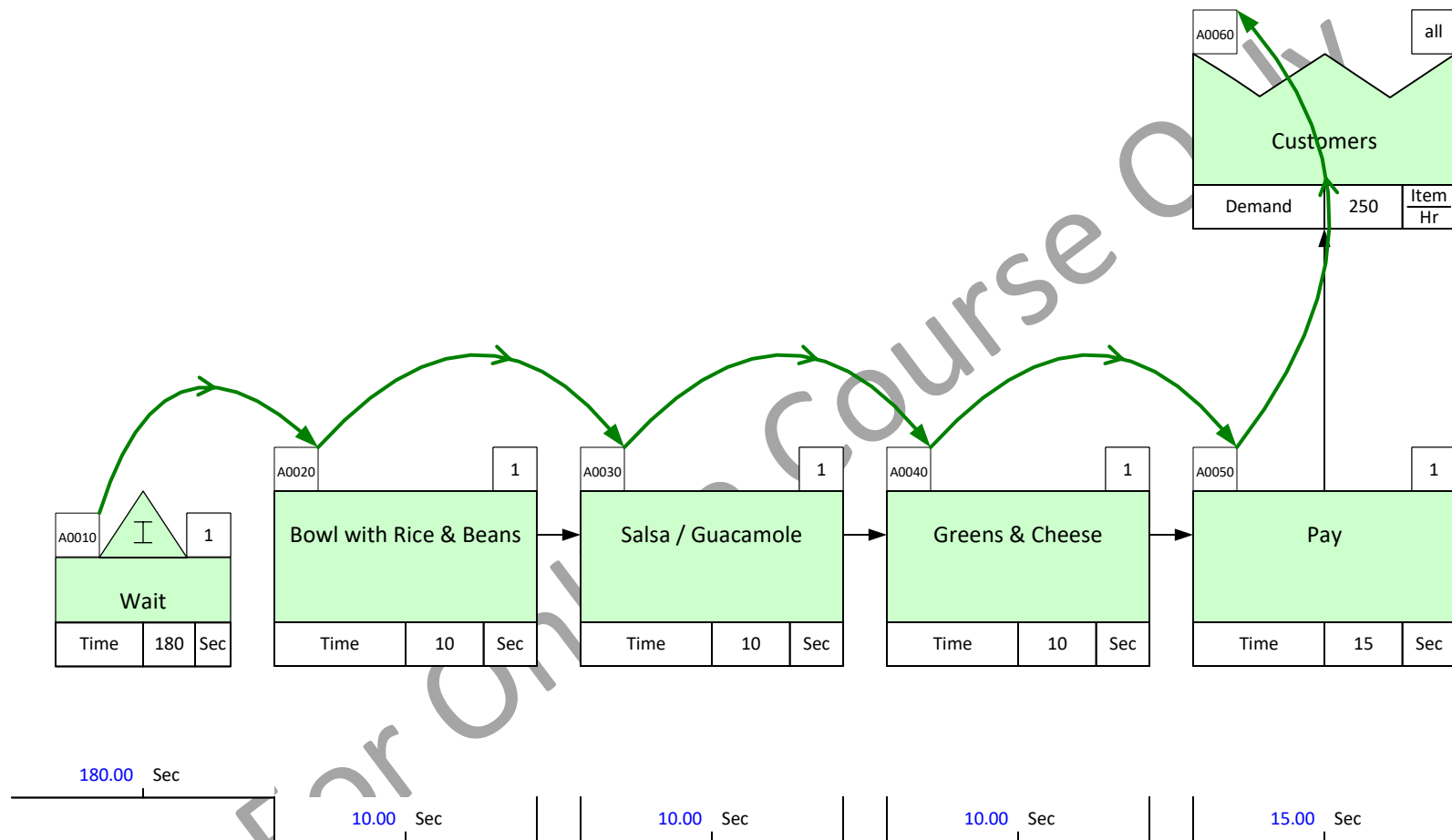




Plot Name Values Command

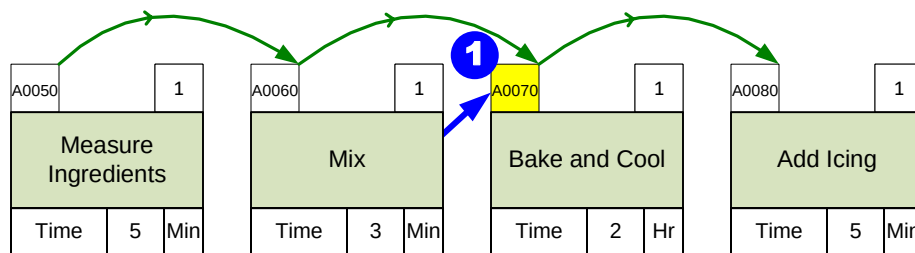
The “Plot Name Values” command generates a bar chart for any variable on the map.



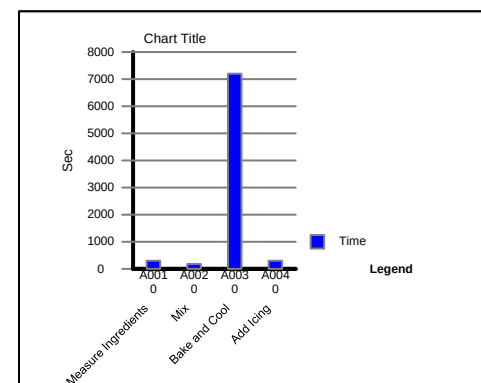


Filtering Data on Bar Charts

Sometimes, one or two activities can swamp out all the other activities on a chart. Example:



XY
Chart

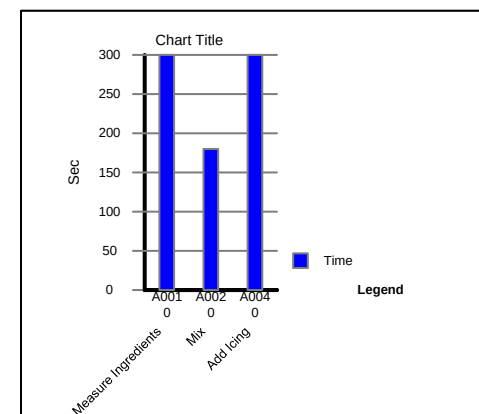


Unwanted data can be omitted from the chart with these steps:

1. Color the operation tags for activities to be omitted with any non-white color.
2. Open the Edit Chart dialog
3. Make sure the “Color Tags” check box is Off.
4. Update (re-plot) the chart.

XY
Chart

Edit Chart Data
Plot Chart
Create Spreadsheet
Add leader
Enable Leader Line
Where From



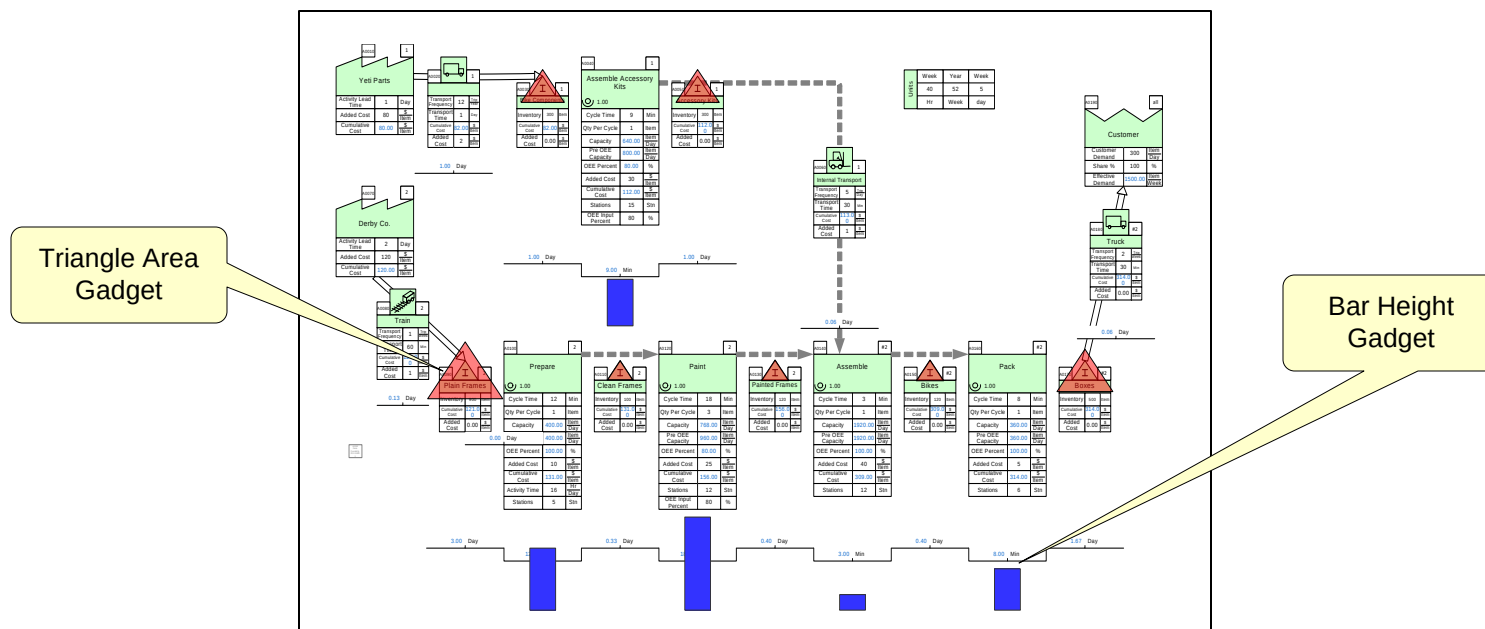
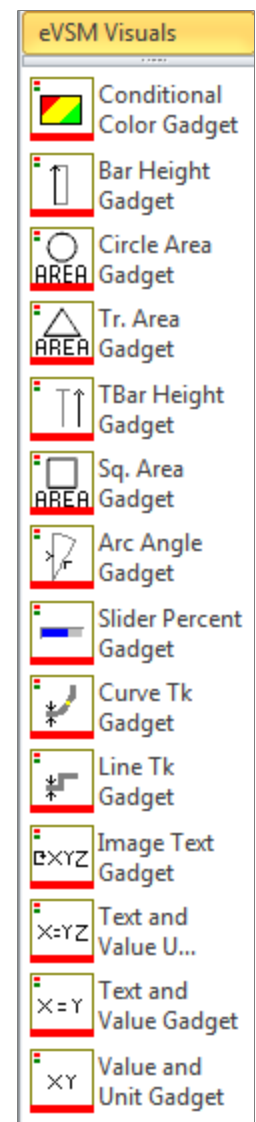
Using Gadgets to Visualize Data

In addition to charts, you can use geometric gadgets to visualize data with the gadget positioned near the data block they represent. Any NVU (data shape in Name-Value-Unit format) variables on the map can be linked to visual gadgets whose size and color changes in proportion to the value. This allows you to visualize values “in-place” on the map.

Gadgets are available in many shapes as shown on the right. Each gadget has exactly one parameter (eg. length, area, thickness, etc.) which can be tied to represent a data value. Different gadgets can be used to represent different variables on the same map. Gadgets size/color will update automatically for any changes in data when the “Solve” is run. Gadgets may also be updated with the Update



Gadget properties (scale, colors, visibility) can be managed with the Gadgets button in the eVSM ribbon.



Using Gadgets to Visualize Data (continued)

To apply, drag out the “Bar Height” gadget from the “eVSM Charts and Gadgets” stencil on the right and position it near the data value with which it will be associated. Apply a fill color to the gadget if you wish.

Select the gadget, glue the gadget’s yellow flying connector to the side of the NVU data shape.

Right-click on the gadget and use the “Create By Example” option to create similar gadgets near all “Time” NVU’s on the map as shown on Process B.

The “Create By Example” creates a similar gadget on the maps for every instance of the selected variable.

You can manage the gadget family using the “Gadget Manager” button in the eVSM ribbon. This allows you to scale the gadget and even apply color-coding to it based on variable value

The diagram illustrates the process of applying a gadget to NVU data shapes. Three NVU data shapes are shown, each with a 'Time' column and a value (2, 1, and 6). A 'Bar Height' gadget is placed next to the first shape, and its yellow flying connector is attached to the '2' in the 'Time' column. A context menu is shown with the 'Create Gadgets By Example' option highlighted. To the right is a stencil of various gadgets.

Activity

Time	2	Min
------	---	-----

Activity

Time	2	Min
------	---	-----

Activity

Time	1	Min
------	---	-----

Activity

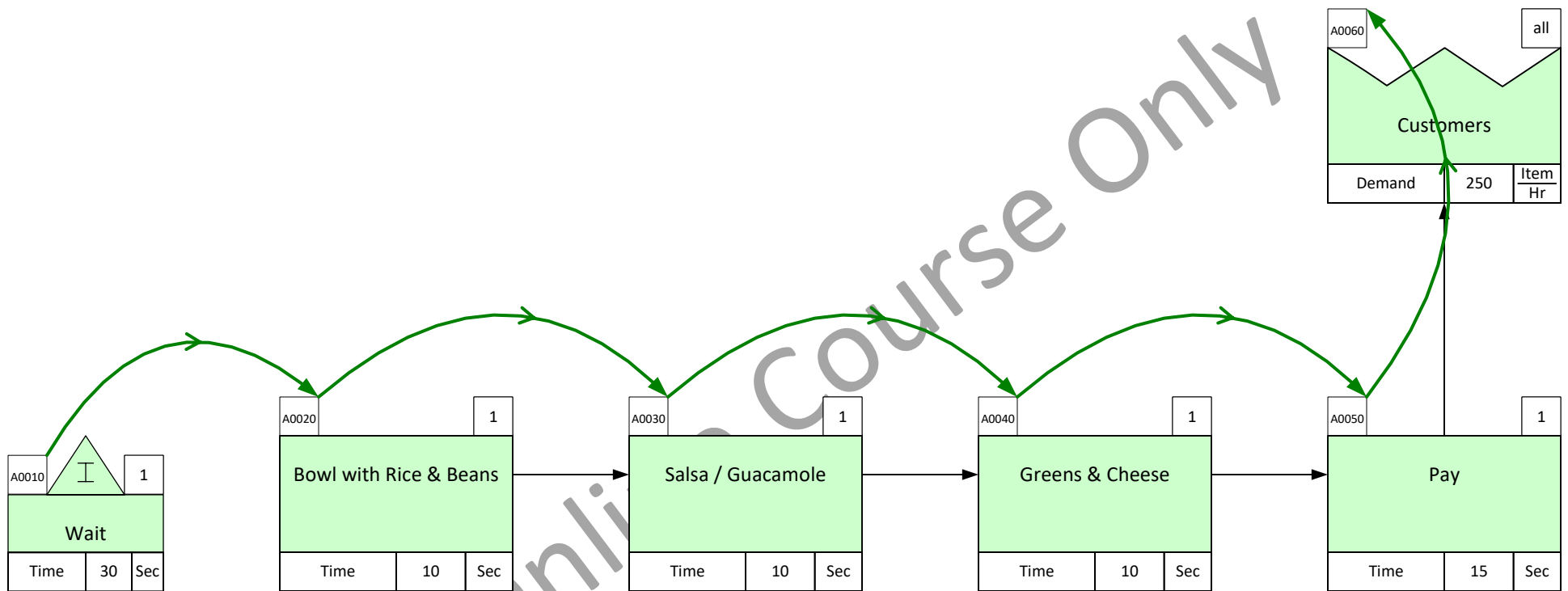
Time	6	Min
------	---	-----

Activate Gadget
Create Gadgets By Example
Show Gadget Collection
Manage Gadget Collections

Gadget Stencil:

- Kaizen Risk Matrix
- XL Report
- Tag Sequence Chart
- Conditional Color Gadget
- Bar Height Gadget
- Circle Area Gadget
- Tr. Area Gadget
- TBar Height Gadget
- Sq. Area Gadget
- Arc Angle Gadget
- Slider Percent Gadget
- Curve Tk Gadget
- Line Tk Gadget
- Image Text Gadget
- Text and Value U...
- Text and Value Gadget
- Value and Unit Gadget

For more help on eVSM Gadgets, see <http://www.evsm.com/toolbarguide>

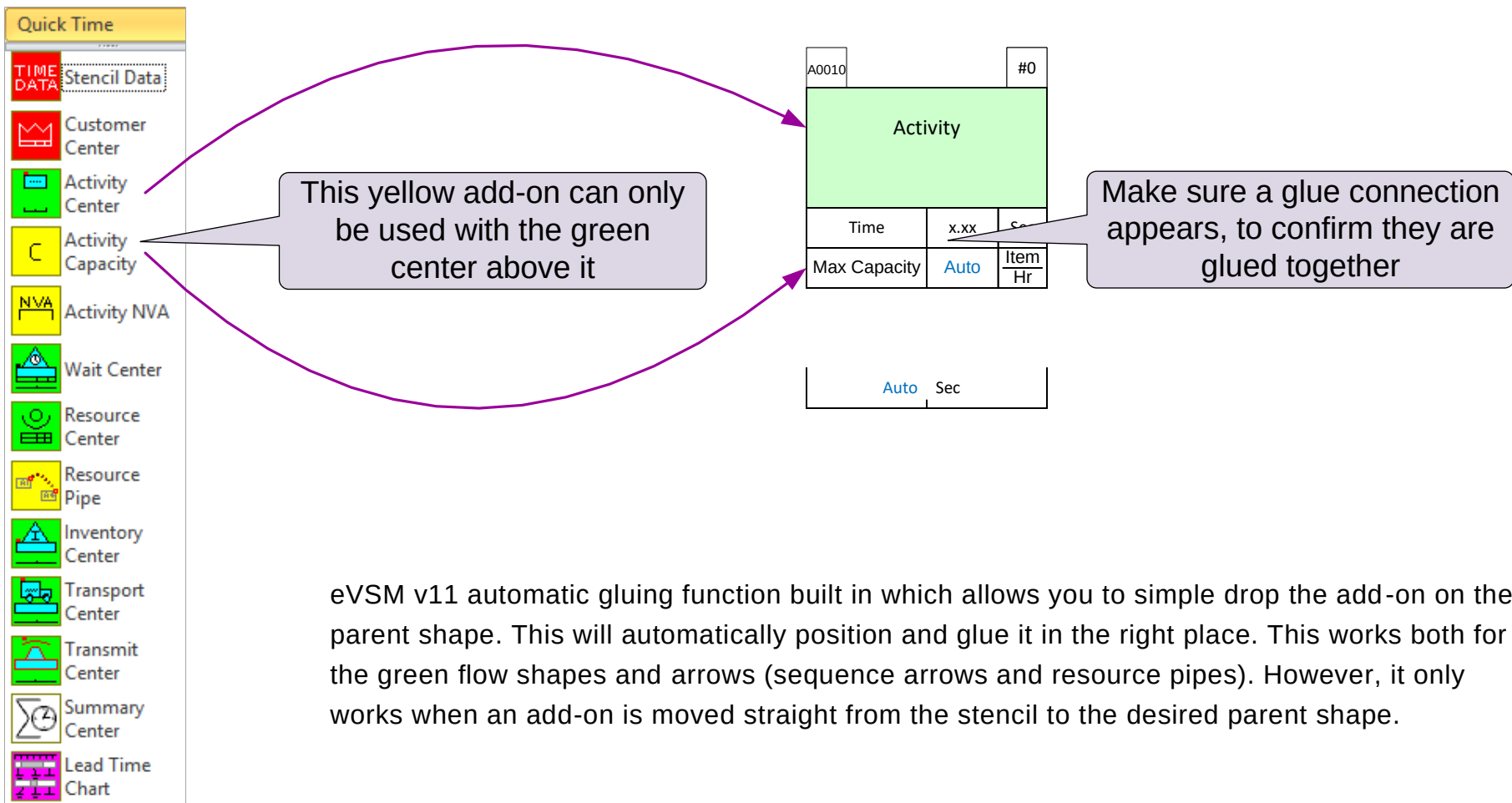


Using the Yellow Add-ons in Quick Stencils

In addition to the Customer Lead Time, you are also interested in understanding the capacity of the different stations and in making sure there is no overloading or excessive wasted capacity.

You recall that the yellow icons in the Quick Stencils represent optional input data and calculations. The shapes from the add-on are glued to the bottom of the existing data blocks on the map.

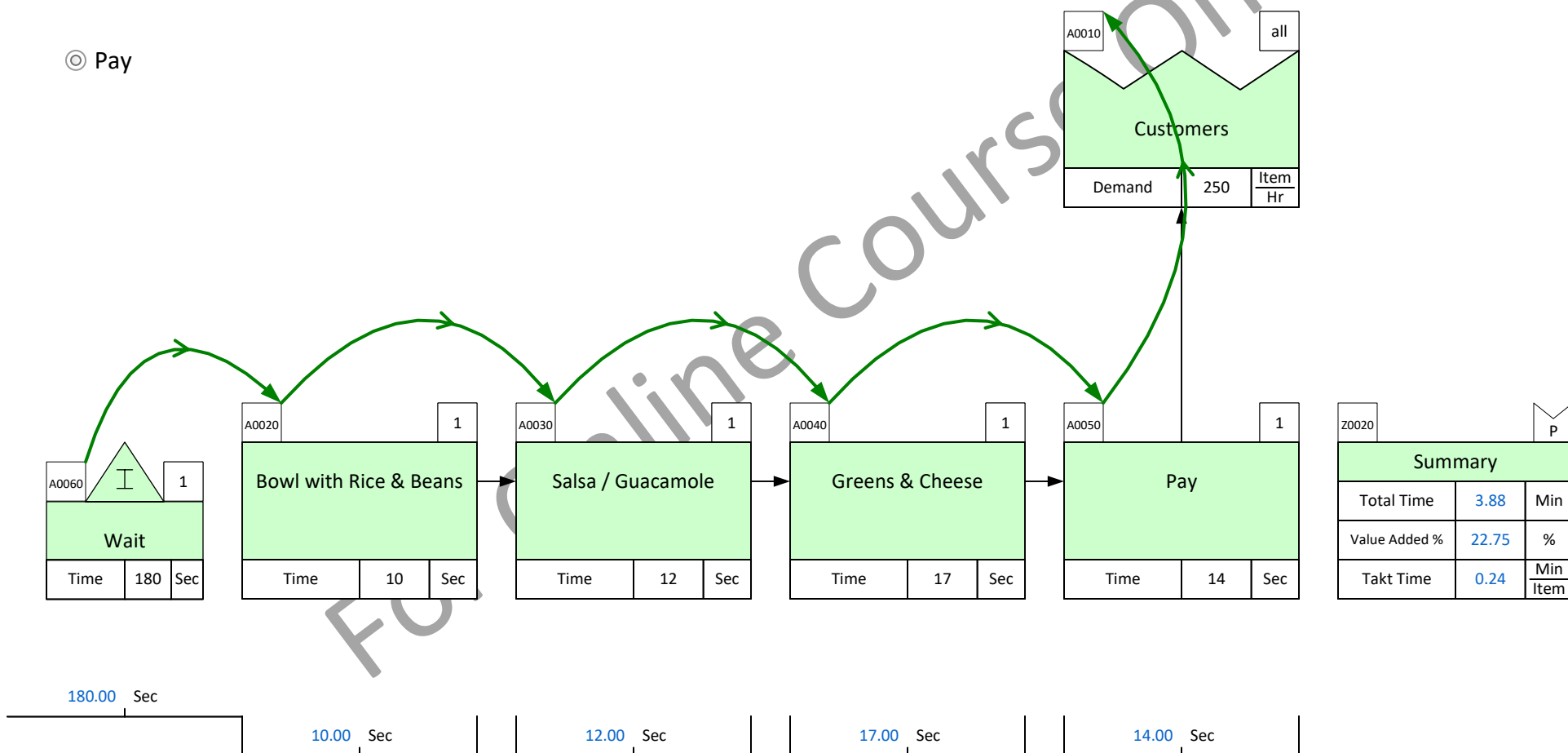
Note that there is a unique correspondence between green icons in the stencil and the yellow icons that immediately follow the green icon. So yellow icons can be used ONLY with the green icons they “belong” to.



eVSM v11 automatic gluing function built in which allows you to simple drop the add-on on the parent shape. This will automatically position and glue it in the right place. This works both for the green flow shapes and arrows (sequence arrows and resource pipes). However, it only works when an add-on is moved straight from the stencil to the desired parent shape.

Q. Which one of these activities is the most severe bottleneck process?

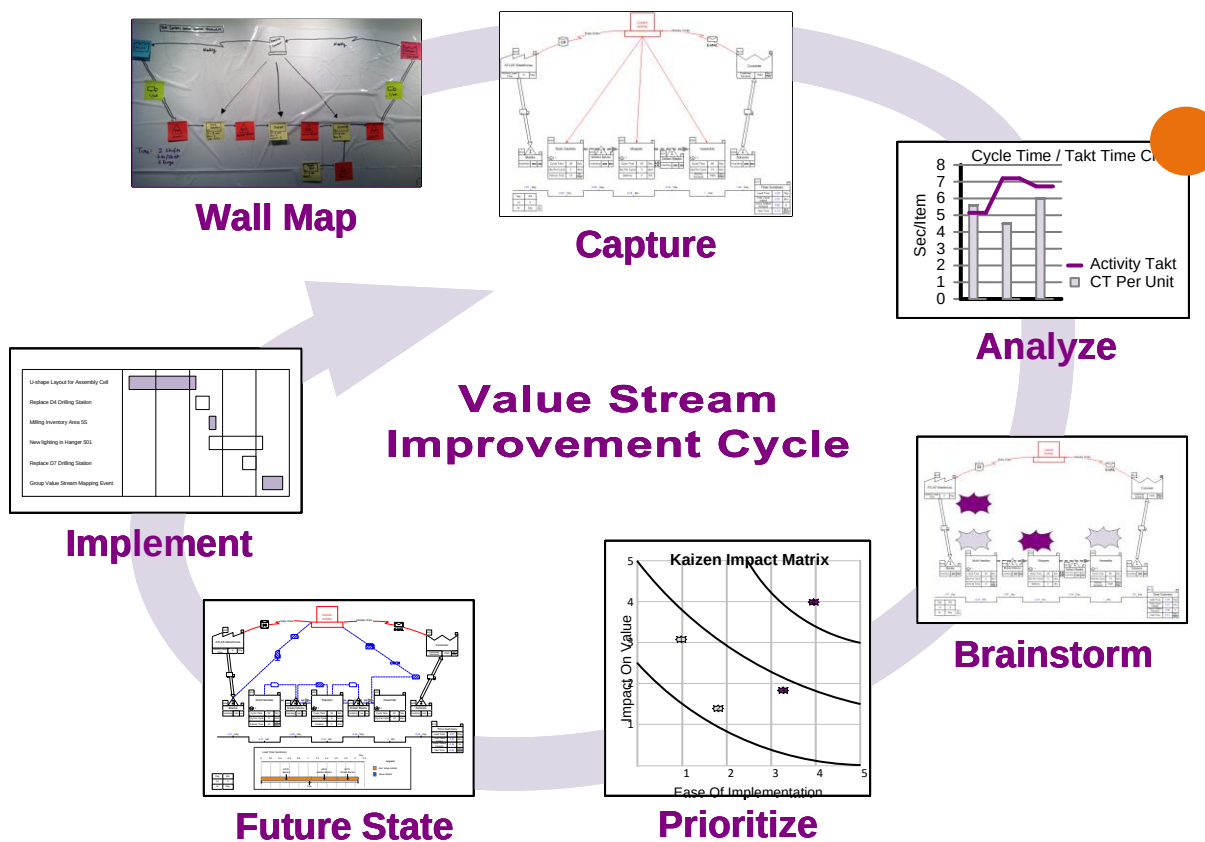
- ☐ Salsa / Guacamole
- ☐ Bowl with Rice & Beans
- ☐ Greens & Cheese
- ☐ Pay



You learned that:

Visuals can be pre-defined charts, value plots of any variable on the map, or geometric gadgets like bars that represent variable values and are positioned next to the variables they represent.

You have seen that visuals are easy to apply and are critical in prioritizing what improvements to make. They also make the map easier to understand by the whole team and this leads to better improvement ideas and prioritization.



What's next:

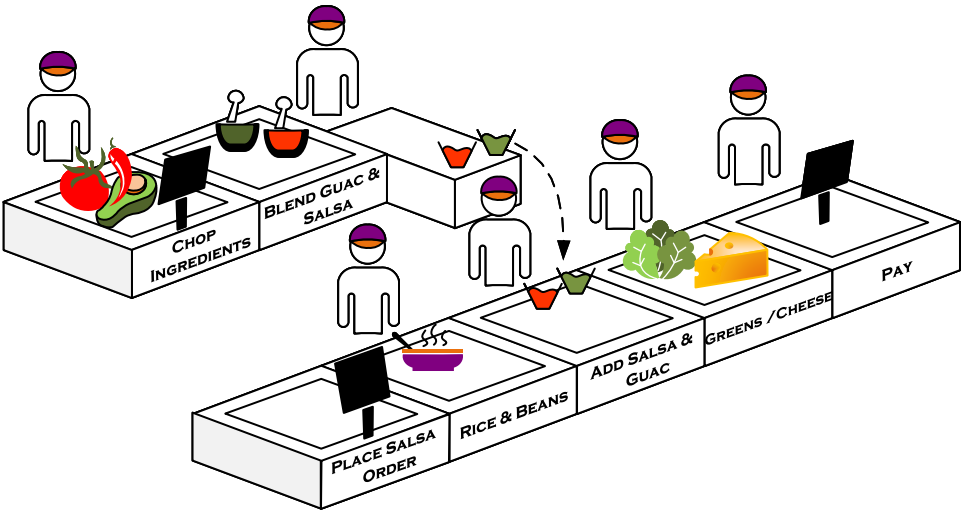
So far you have captured and analyzed a simple "linear" value stream map. But what if there are parallel operations?

Handling Complex Value Streams

Marketing wants to introduce a station to make fresh, custom, and on-demand salsa into the process. This is going to have to happen on a parallel lane in the value stream. Let’s modify the map to accommodate this.

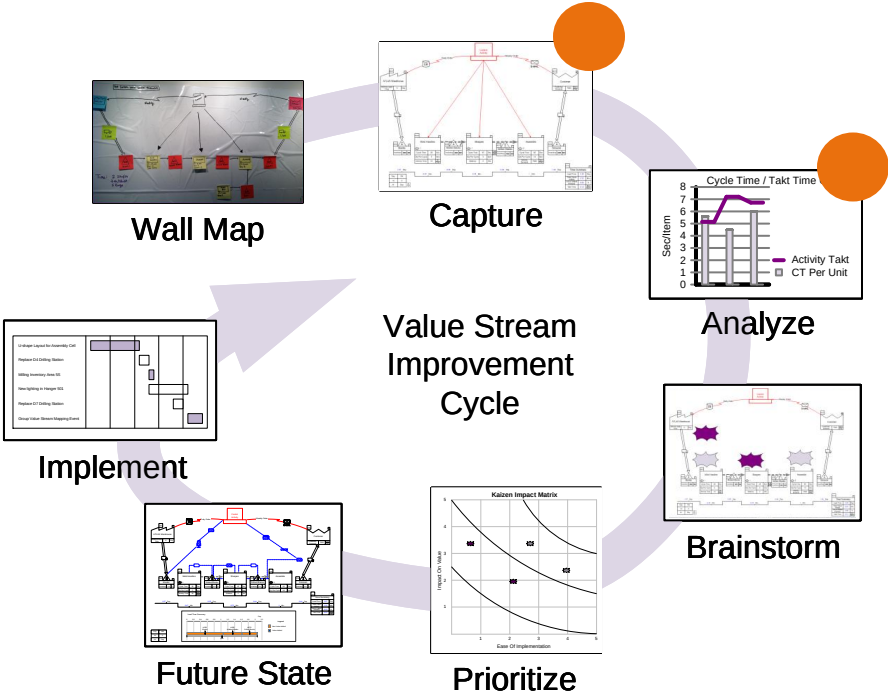
Handling Complex Value Streams

Case Study



Operational efficiency is going to face a stiff challenge with marketing’s decision to have a “Fresh” made to order salsa station.

Improvement Cycle

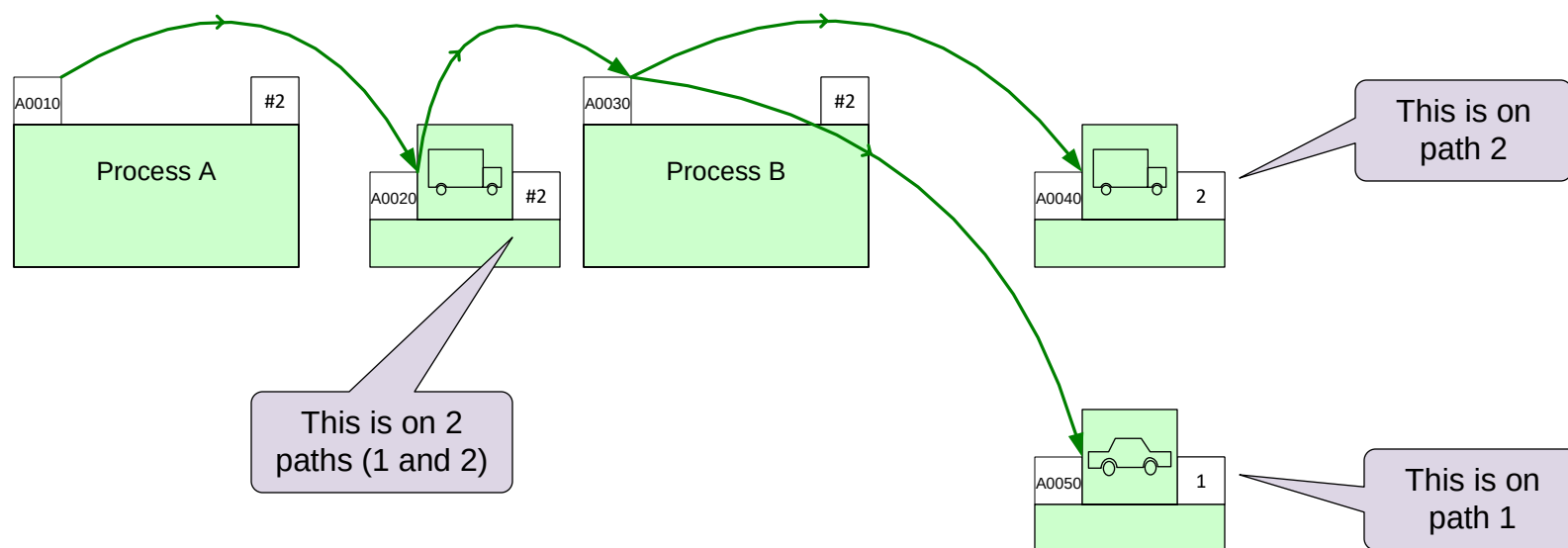


Modifying the map becomes a bit easier with an electronic model. You change the model and then re-analyze it to assess the value stream.

Parallel Processes

Parallel processes require some special handling on maps because key outputs like Lead Time are not simply a case of adding up all of the VA and NVA times.

To support parallel work, we will introduce the concept of value stream paths and you will generate path numbers for the map.

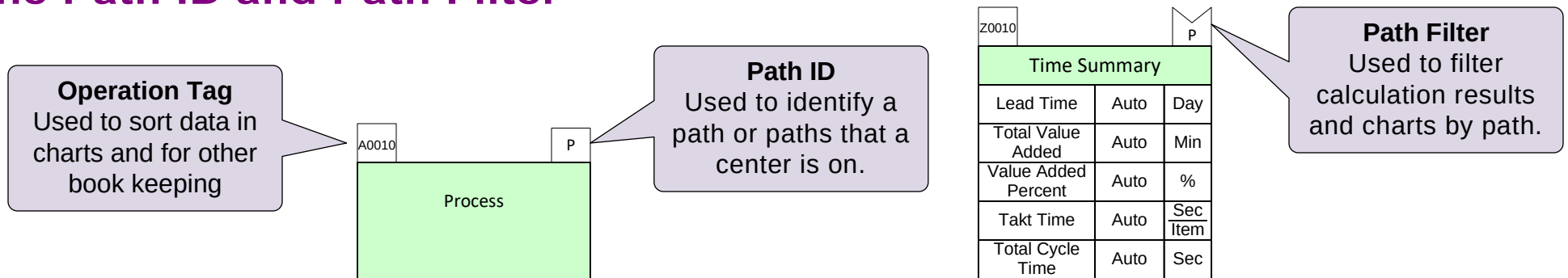


What is a Path?

Each unique sequence through the value stream is considered a separate path and is assigned a path ID. The Green Centers can each be on one or more paths.

In the example above, the left 3 green centers are on both paths 1 and 2. The right side shows two transportation options, either a truck or car. The truck option is on path 2 and the car option is on path 1.

The Path ID and Path Filter



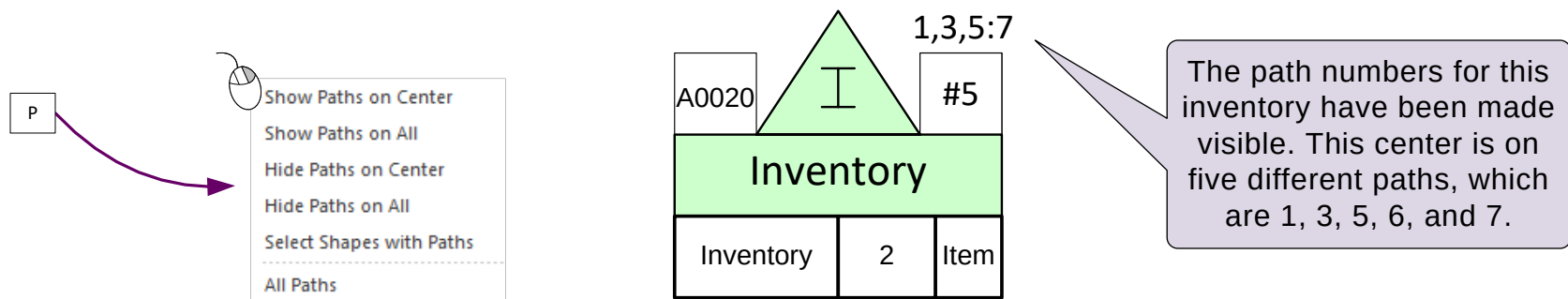
To generate Path ID's, add sequence arrows to the centers. Then click the  Auto Path button in the toolbar.

The Path ID shapes will get populated with path numbers. Examples:

-  The center is on 3 different paths
-  The center is on path 7

You can make the actual path numbers visible above the Path IDs through right-mouse menus.

To minimize the numbers displayed, a colon is used to show any continuous ranges of numbers. For example 7:10 represents paths 7,8,9,10.



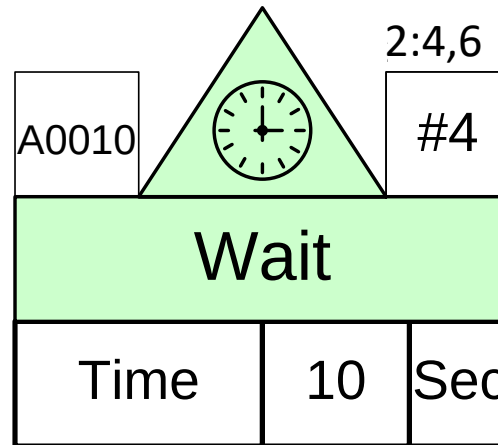
Q. Which of these paths is this Wait Center below on?

☐ 2, 4, 5, 6

☐ 1, 2, 3

☐ 2, 3, 4, 6

☐ 1, 2, 4, 6



Auto Unit

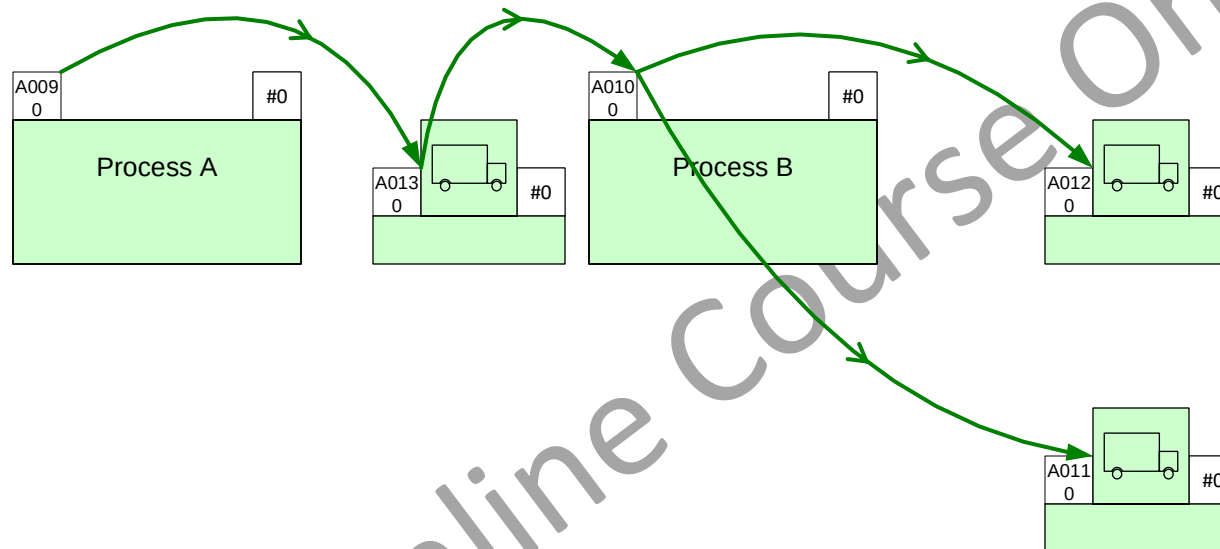
Auto Path Auto Path

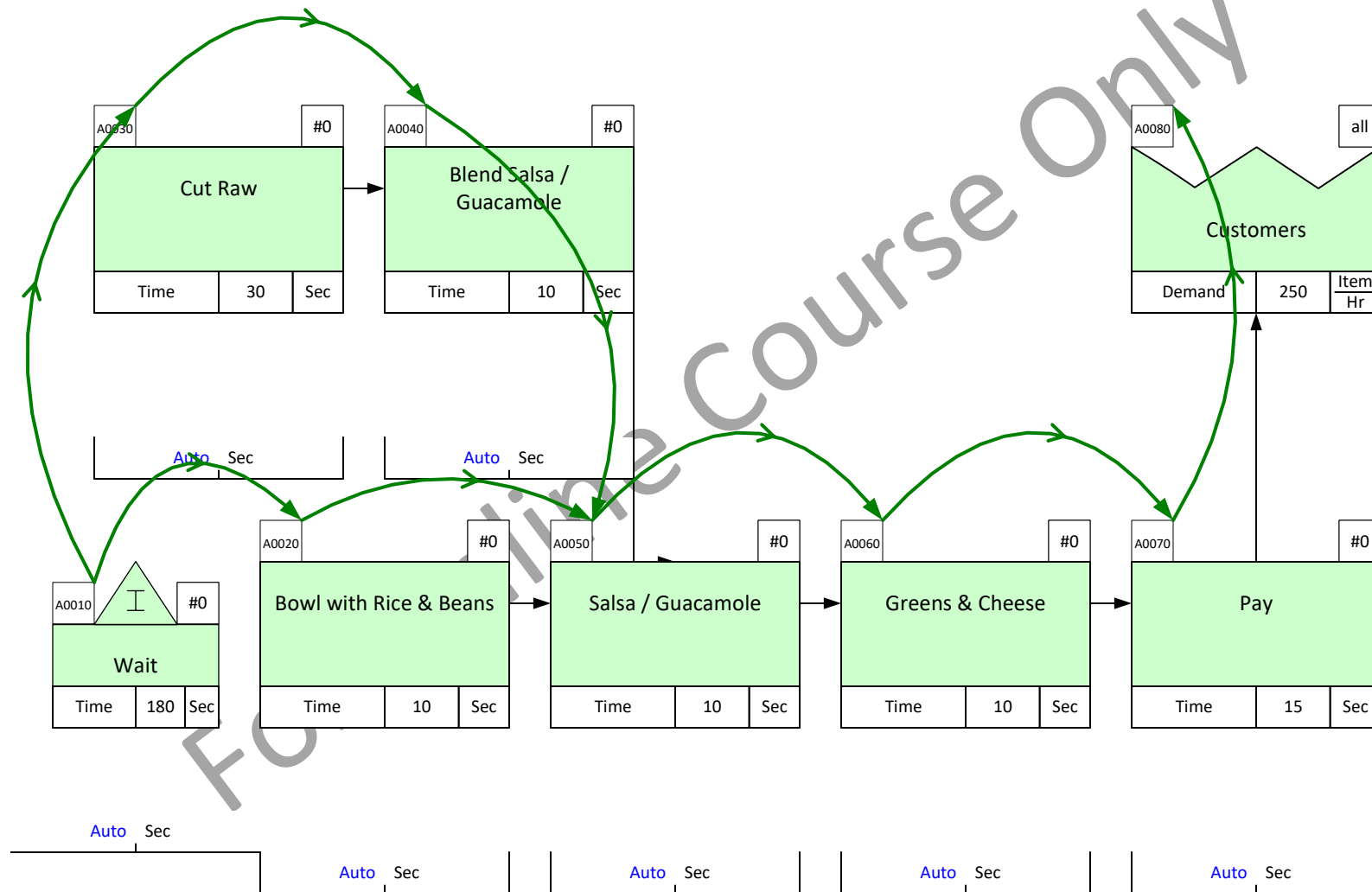
After adding the Sequence arrows, the Auto Path button can be clicked to automatically generate the path IDs for all the possible paths on the map based on the sequence arrows.

AutoTag Auto Tag

A quick way to tag activities in order of the sequence arrows is to use the eVSM Auto Tag button.

Charts are plotted in alphanumeric tag sequence, and the AutoTag button does a great job to help charts plot in upstream to downstream sequence.

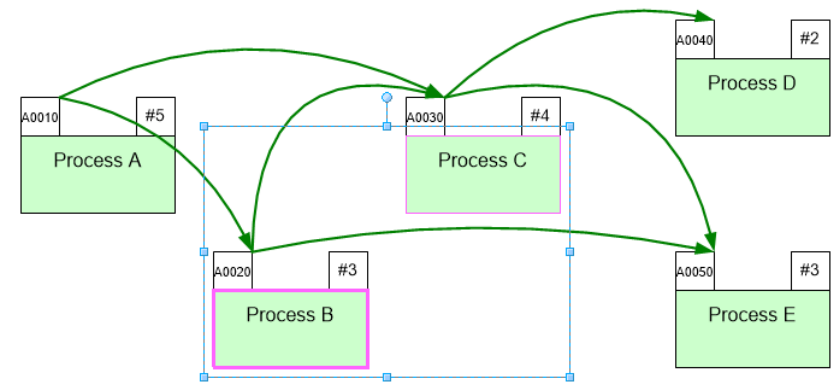




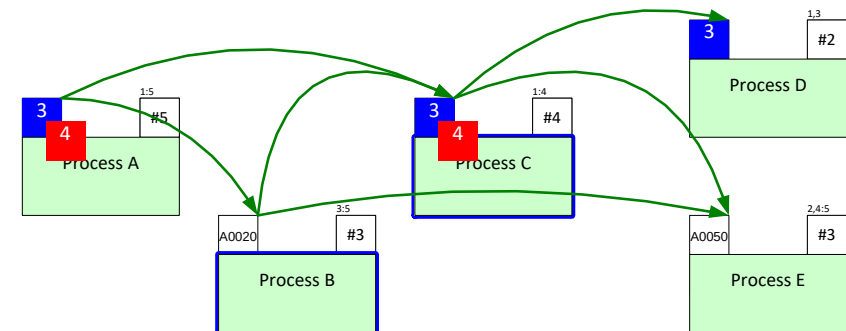
Highlighting and Selecting Paths Same Paths

If you want to see the common paths that go through centers of interest, then use the Same Paths button in the toolbar.

First select which centers on the map you'd like to view the paths for. Hold down the SHIFT key if you want to select more than one center. In this example, Process B and Process C are selected.



After the Same Paths button was clicked, Process B and Process C are highlighted, reminding you which centers were selected. Only the paths that go through both centers are highlighted, which are paths 3 and 4 in this case, shown in blue and red.



You can clear the highlighting on the page at any time by clicking the Clear button in the toolbar.  Clear

The Path Filter shape  can be populated in two ways:

1. **Manually:** Just double-click and a dialog will appear.
2. **Automatically:** While paths are highlighted on the map, click the “Set Path from View” command in the right mouse menu of the Path Filter.

Z0010

P

Summary		
Total Time	Auto	Min
Value Added %	Auto	%
Takt Time	Auto	Min Item



Show Paths on Center

Show Paths on All

Hide Paths on Center

Hide Paths on All

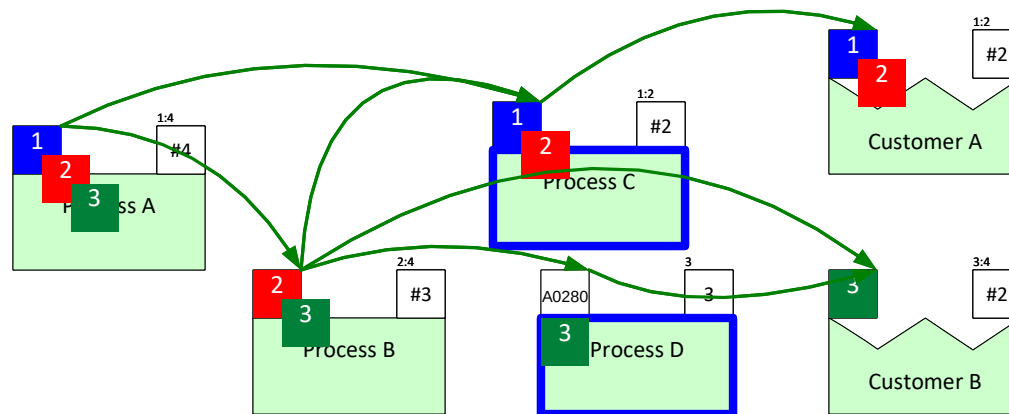
All Paths

Select Paths

Set Path From View

All Paths All Paths

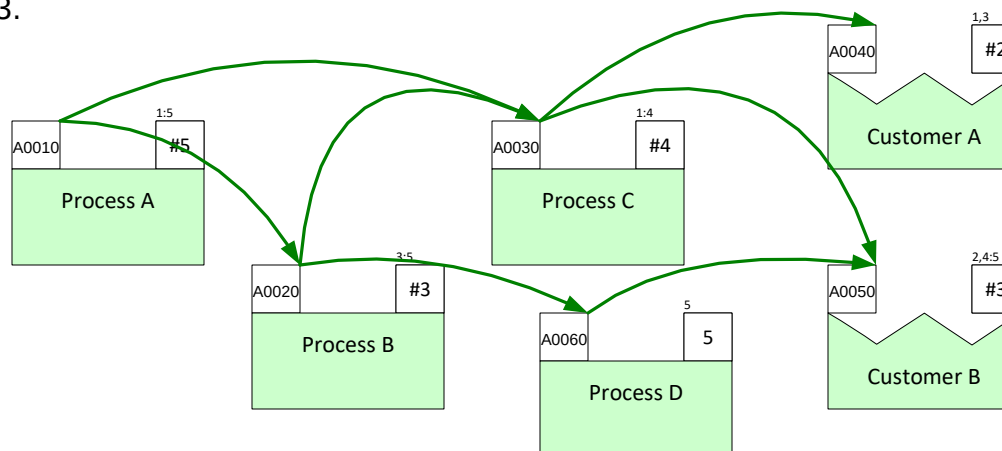
If you have various paths on the map, and want to see all of the paths that go through multiple centers, then use the All Paths button in the toolbar. In the example below, Process C and Process D were selected using the SHIFT key, and then the All Paths button was pressed. All of the paths are highlighted from Process C (1 and 2), and Process D (3).



The right-mouse option on the Path Filter of “Set Path from View” will work for this function as well.

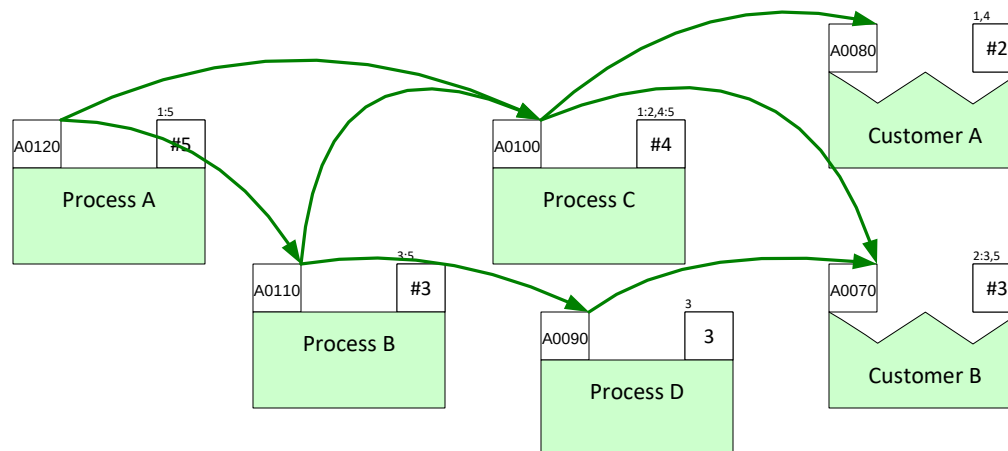
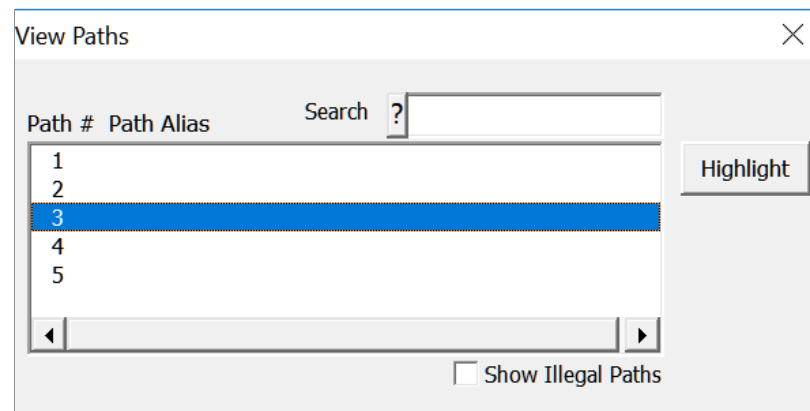
Highlight Path Arrows

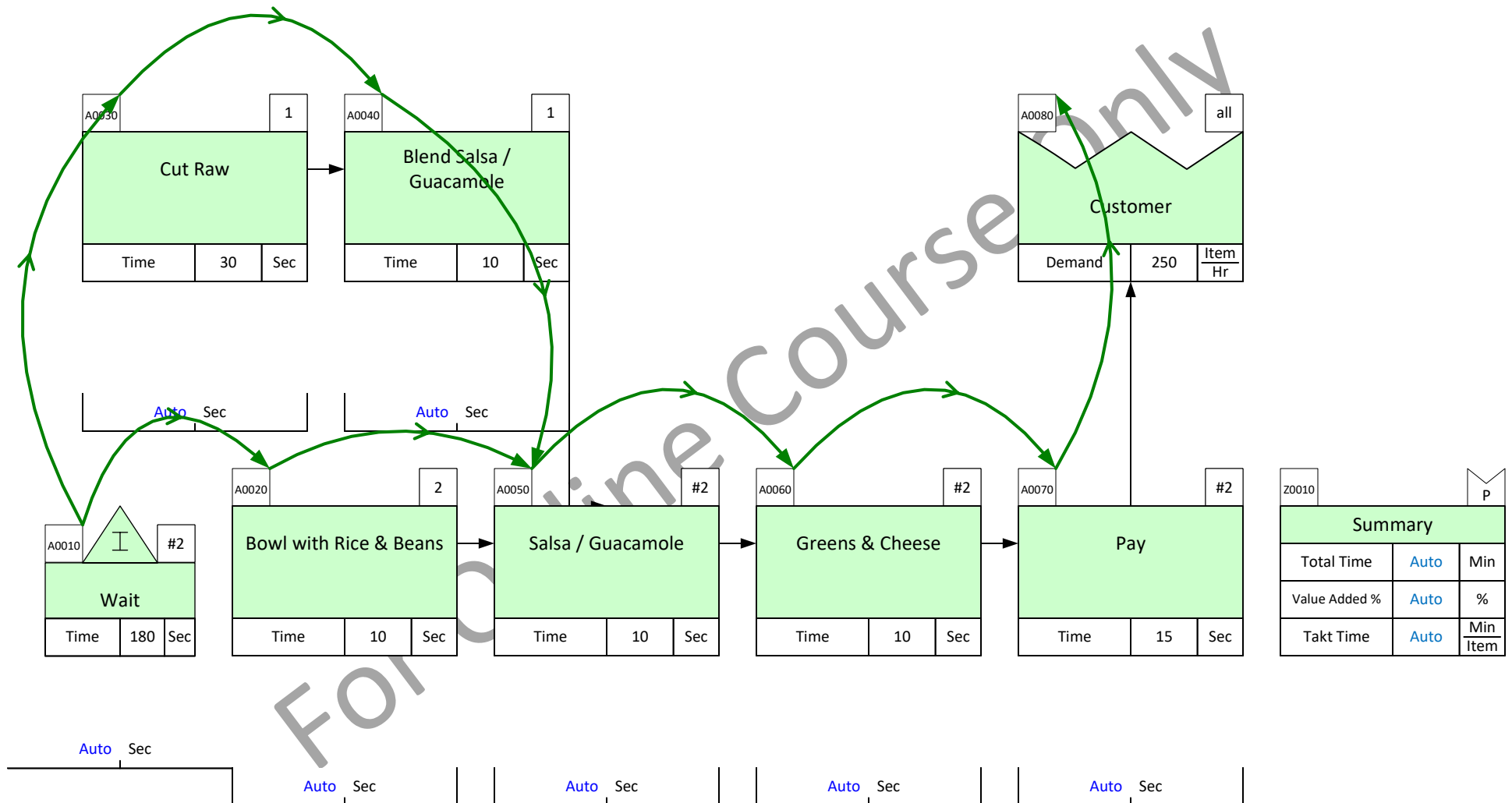
If you have multiple paths highlighted, like the red, green, or blue boxes, you can view their path by highlighting the sequence arrows in the matching color. If you right-click on any of the highlighted path boxes, there is an option to “Highlight Path Arrows”. This has been done for path 3.

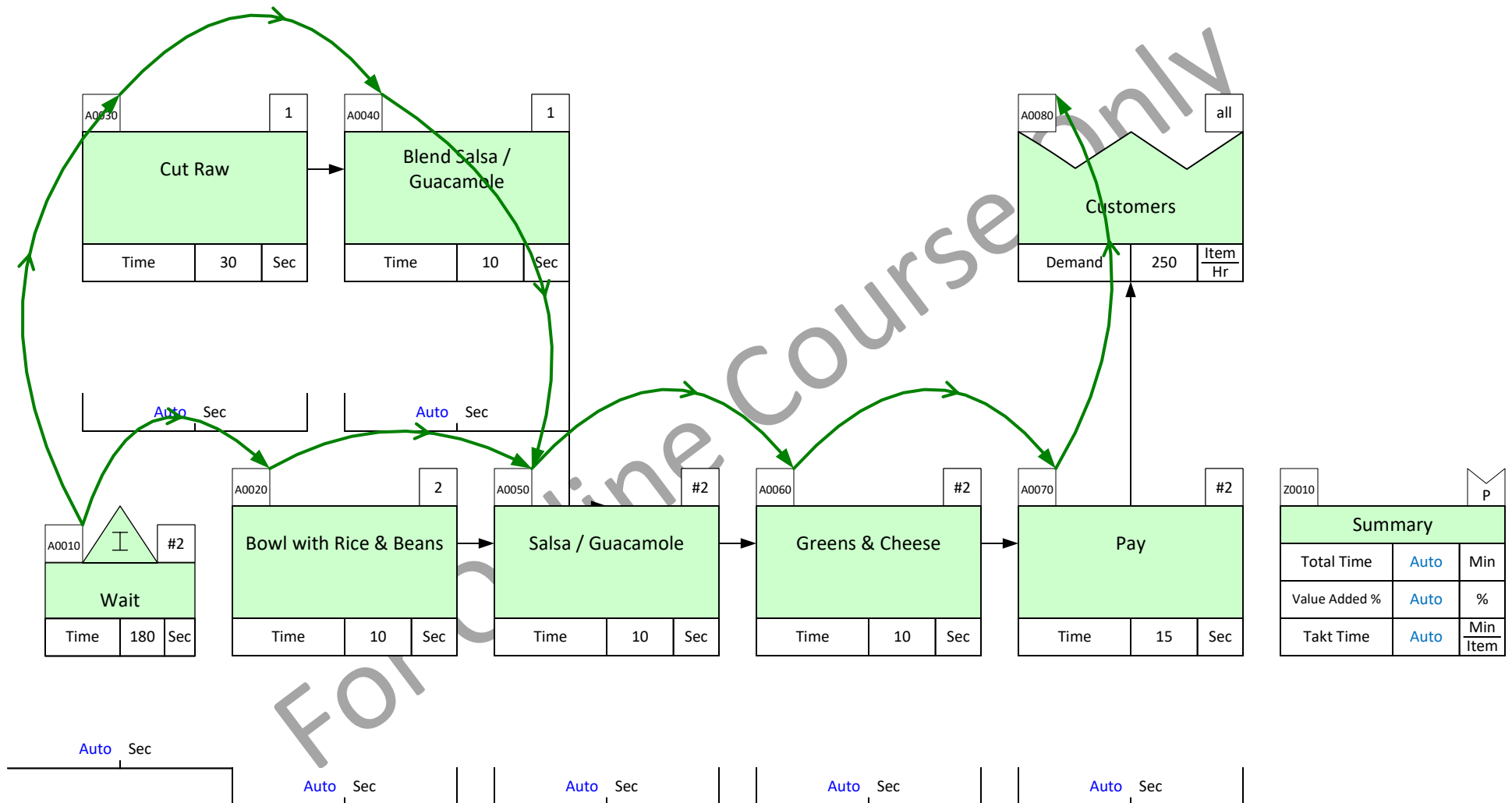


View Paths View Paths

This toolbar button brings up a list of the paths in the model so that one or more can be selected and highlighted. The example shows highlighting of path 3 below.

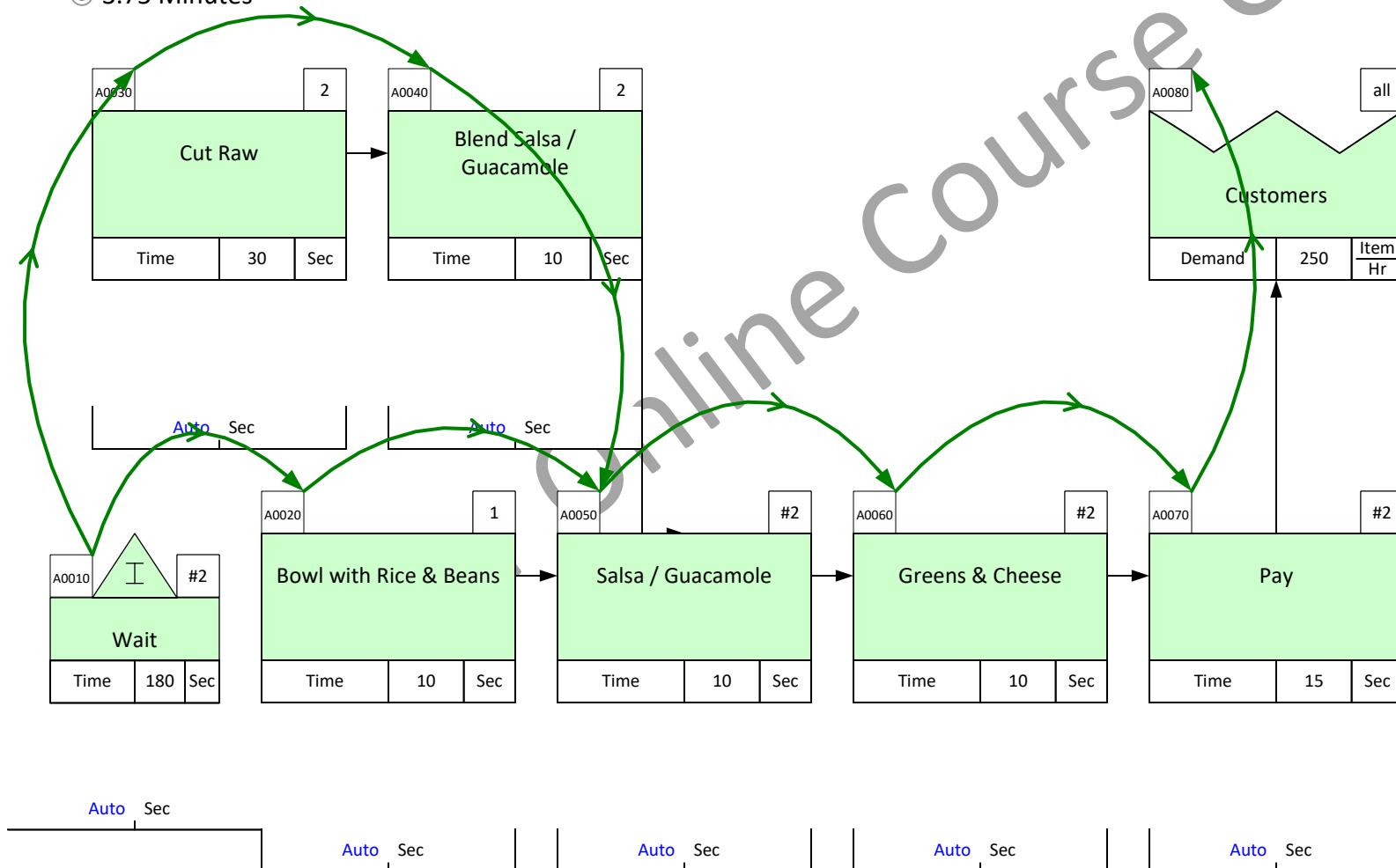






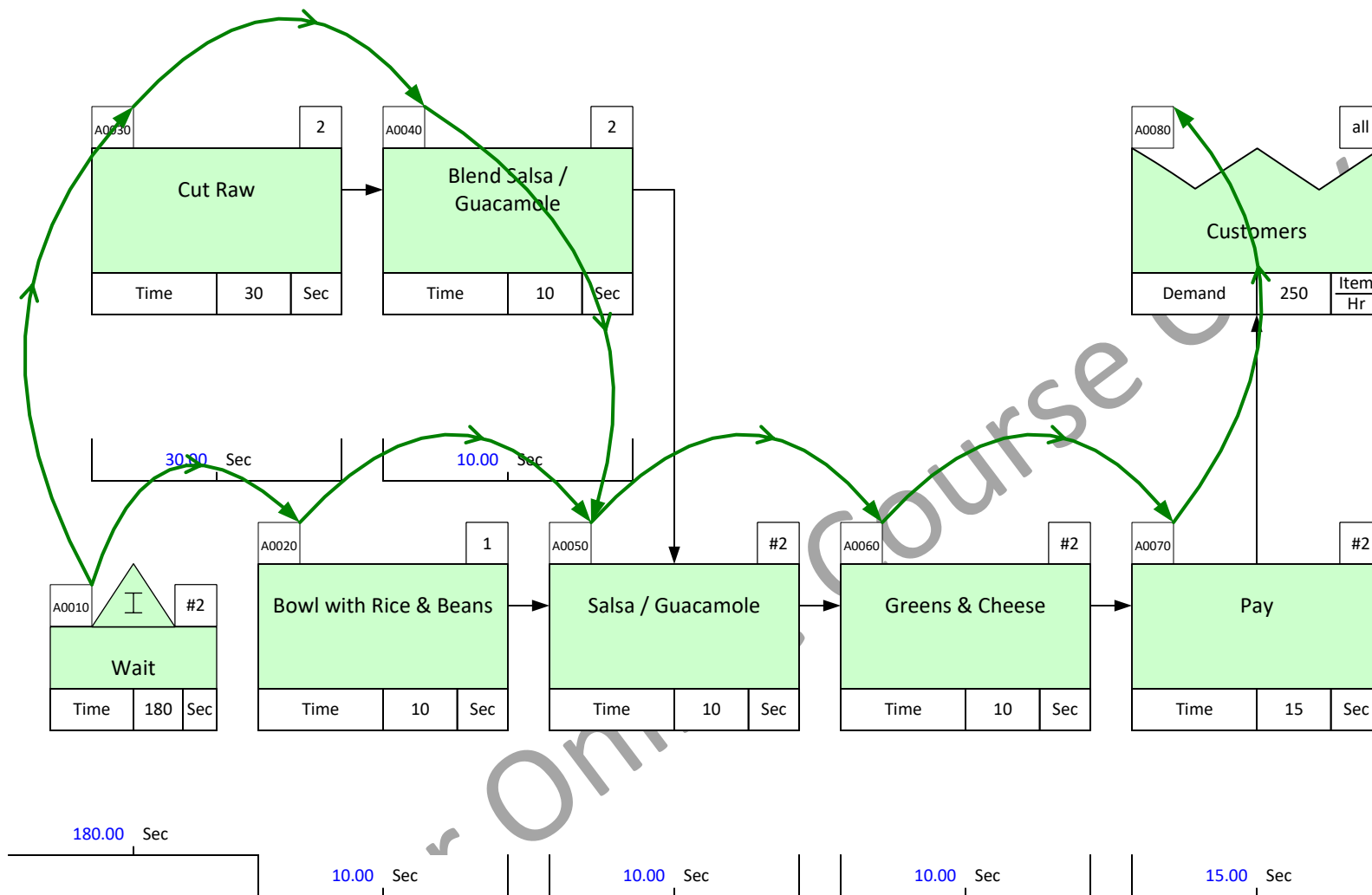
Q. Which one of these is the “Total Time” for the longest path?

- ☐ 8 Minutes
- ☐ 5 Minutes
- ☐ 4.25 Minutes
- ☒ 3.75 Minutes



Summary		
Total Time	Auto	Min
Value Added %	Auto	%
Takt Time	Auto	Min Item

Summary		
Total Time	Auto	Min
Value Added %	Auto	%
Takt Time	Auto	Min Item

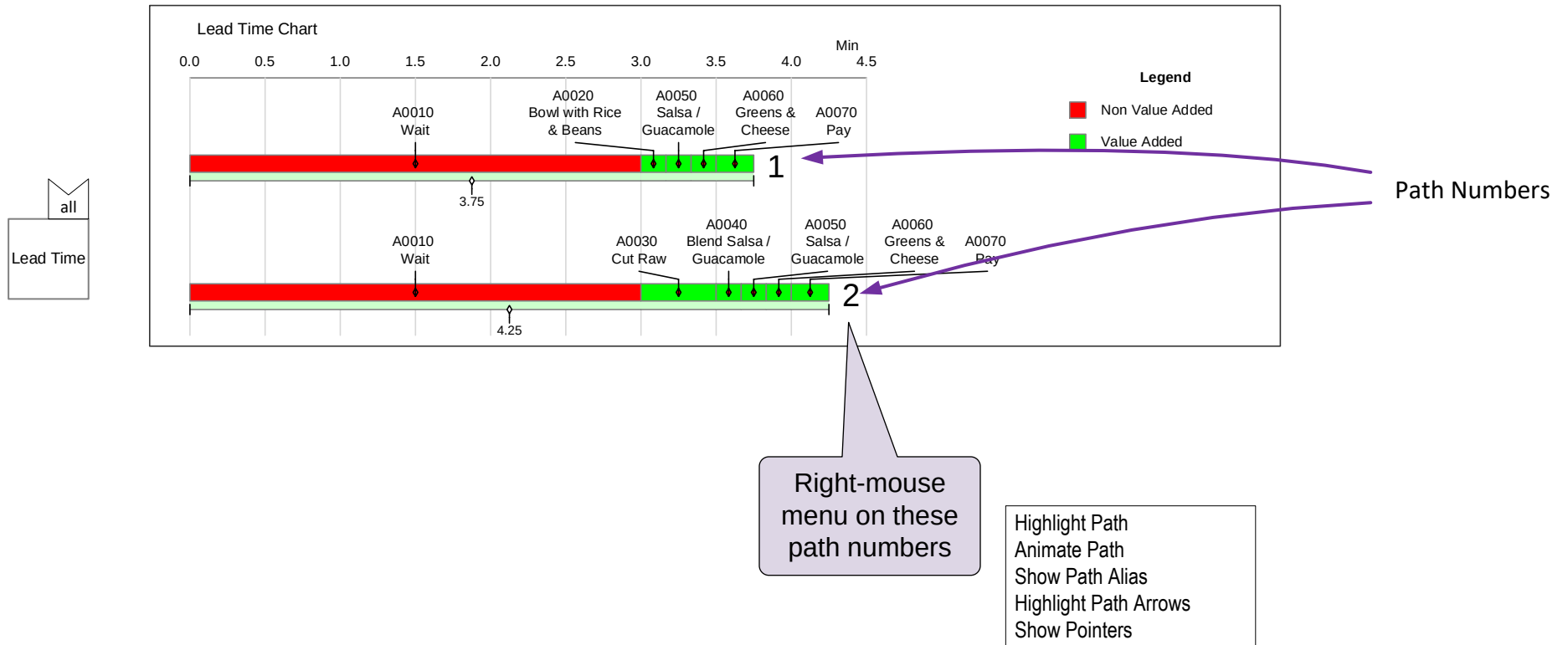


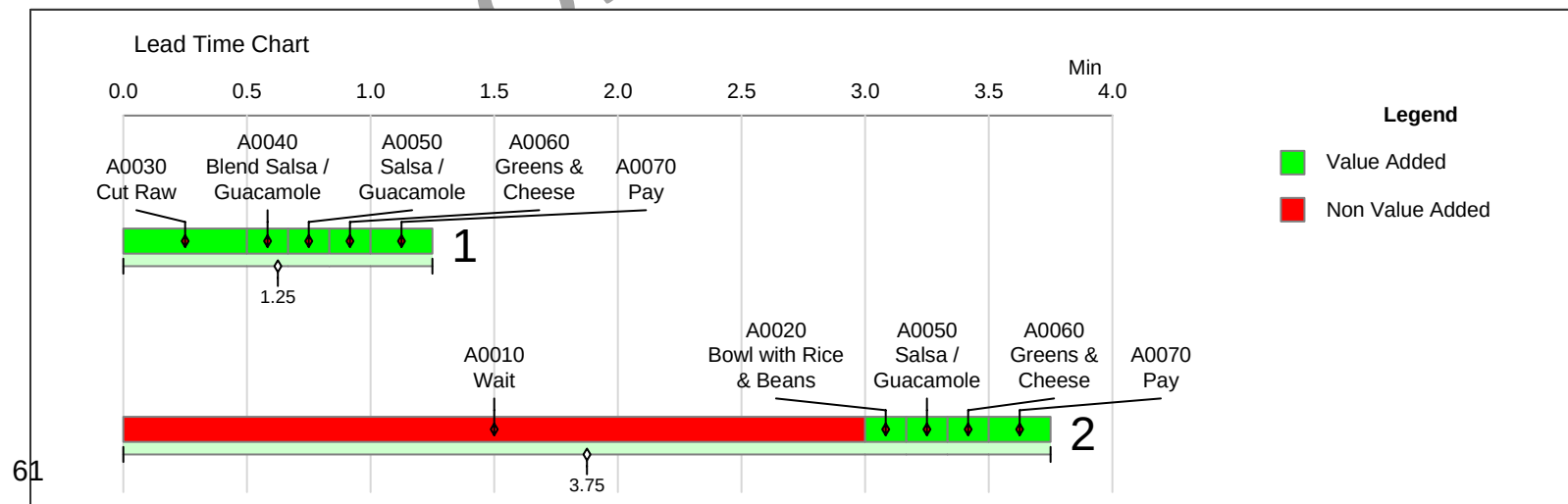
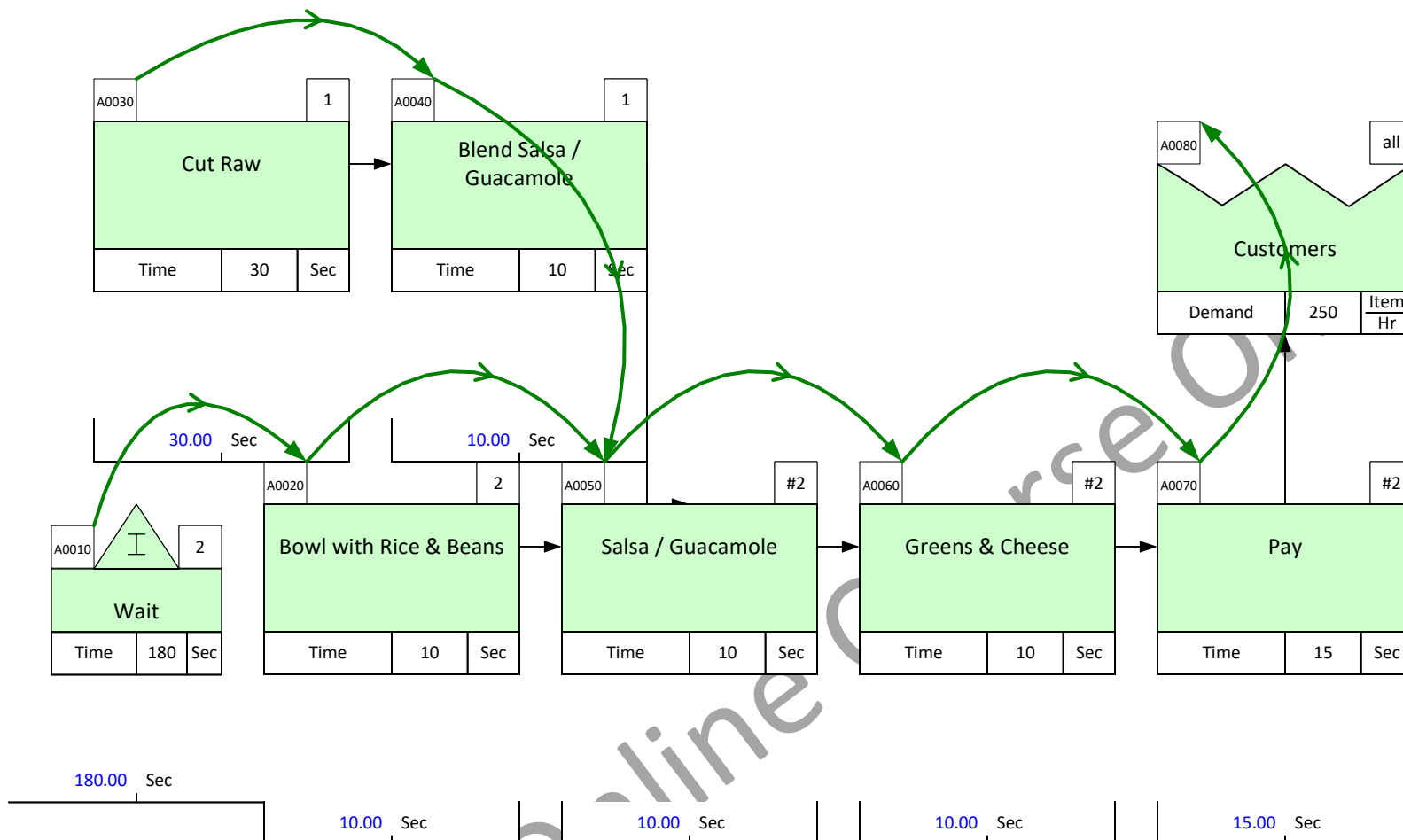
Z0020			#0
Summary			
Total Time	3.75	Min	
Value Added %	20.00	%	
Takt Time	0.24	Min Item	

Z0010			#0
Summary			
Total Time	4.25	Min	
Value Added %	29.41	%	
Takt Time	0.24	Min Item	

Path Identification on Ladder Charts

The lead time chart can display the timeline for all or a subset of the paths on the current map. Each timeline has a path number to the right. These path numbers have a right-mouse menu which allows you to highlight or animate that path on the map.

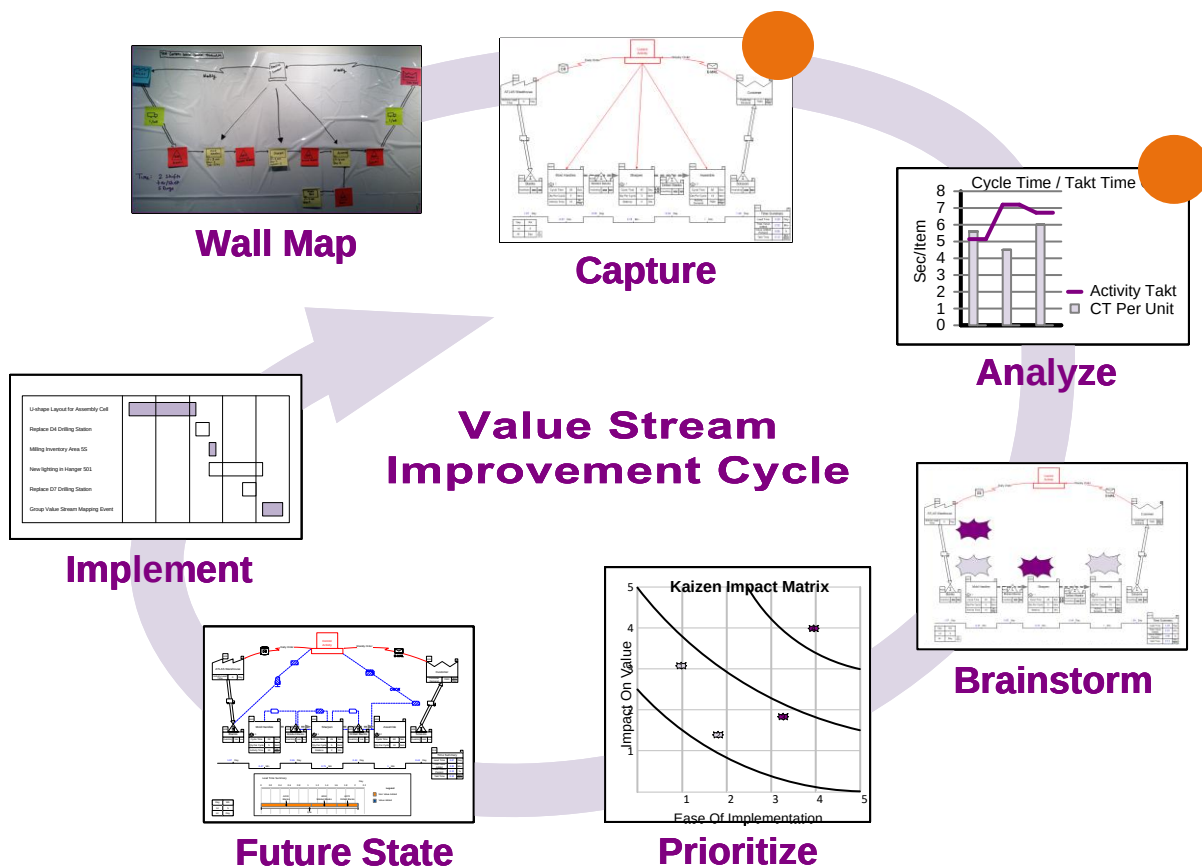




You learned that:

Using the concept of sequence arrows and value stream paths, you were able to modify the map and analyze it to support marketing's concept of an additional fresh salsa station.

As you can imagine, this all has to operate very efficiently to maintain good margins. Later, you and the team are going to look for opportunities to remove waste in the value stream, thus reducing cost and improving customer throughput.



What's next:

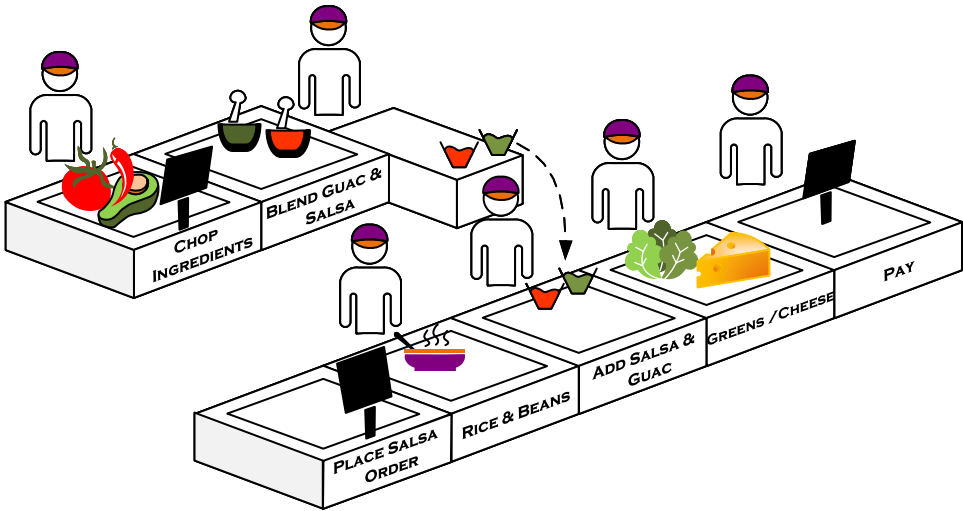
As the map gets more complex and includes multiple products, it becomes important to be able to work in convenient units that are easy to understand. You will look at units and unit converters that make this possible in eVSM.

Units Handling

The team needs to visualize the map easily so they can identify opportunities. One thing that makes visualization difficult, is use of inappropriate units for time and for product quantity. You are going to take a quick look at units and how they can be managed for both easy data input and output.

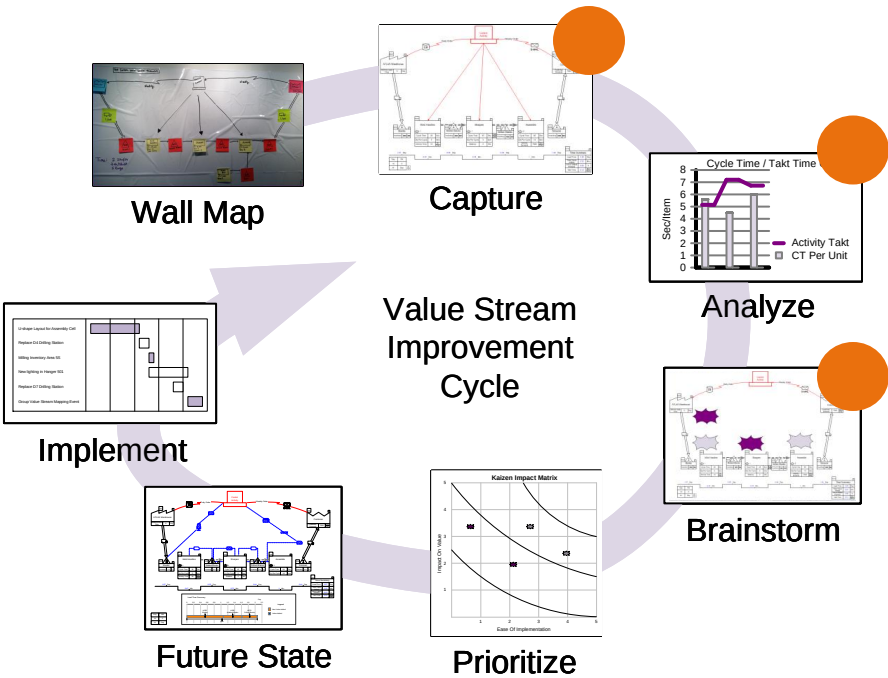
Units Handling

Case Study



You plan to put the data and any charts in the appropriate units so mistakes are easy to spot, and the map easier to understand when you show it to the team.

Improvement Cycle



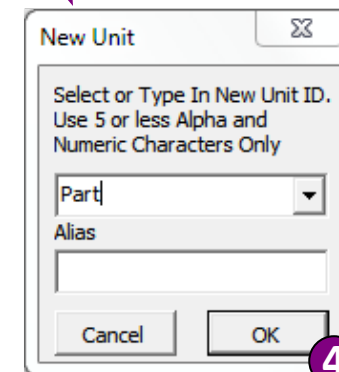
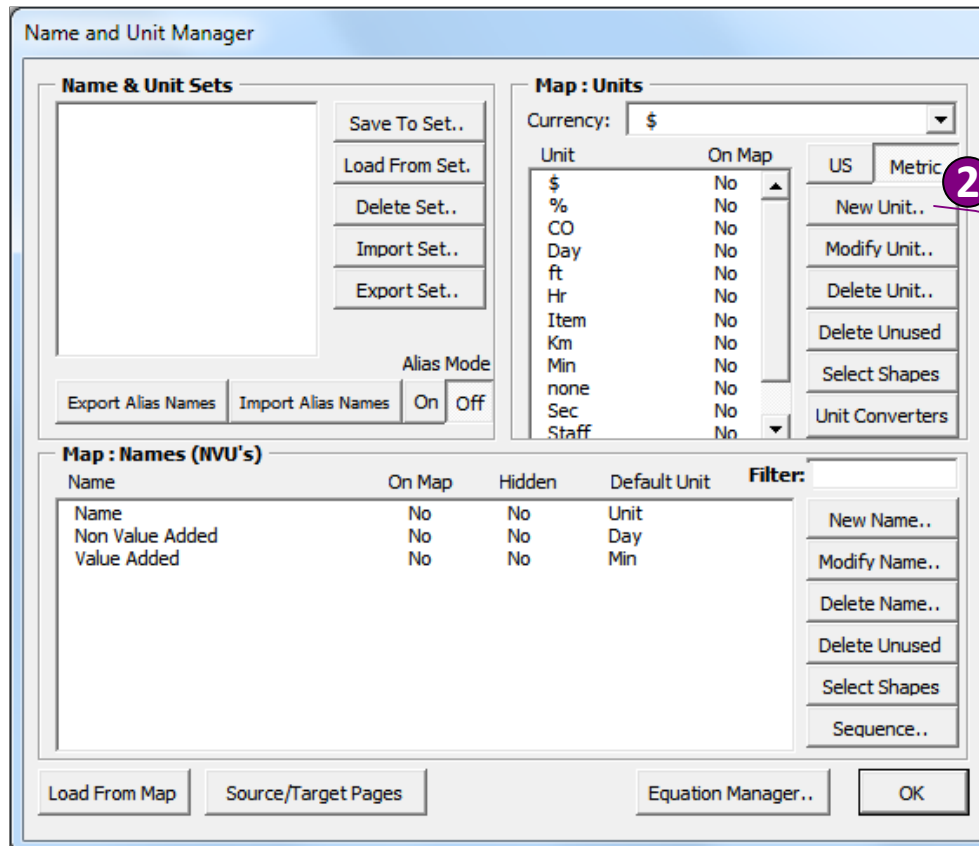
Using appropriate units will help the capture, analyze, and brainstorming phases of the improvement cycle.

Working with Units

You may want to work in different time or quantity units in different parts of the value stream such as boxes, trays, bowls, items, etc. Some of these units are already built into eVSM. Custom units may be added

Creating a Custom Unit

- 1 Click the Name and Unit Manager button in the eVSM toolbar.  NUM



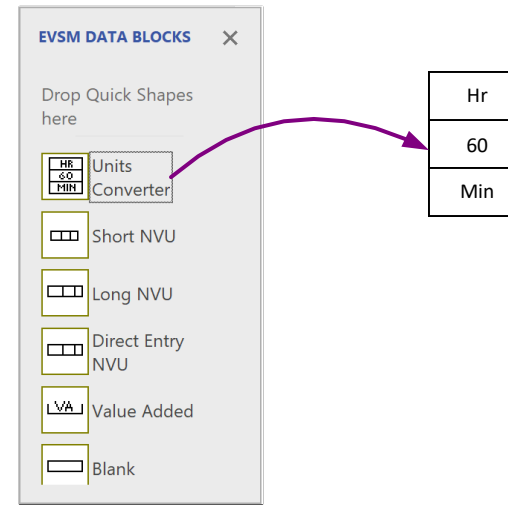
3 Enter new unit name

4 Click OK

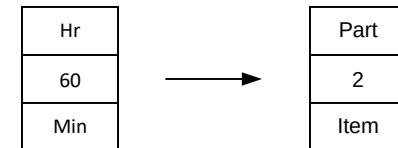
Converters Between New and Existing Units

Unit Converters are necessary to provide conversion factors between compatible units. To create a Unit Converter:

1. Drag out a Units Converter shape from the eVSM Data Blocks stencil on the right of the screen.

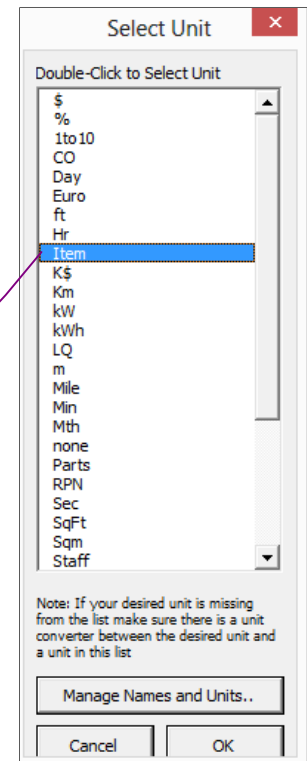


2. Double click on the default unit “Hr” to change the top unit to “Part” and the default unit “Min” to change the bottom unit to “Item.” This allows the calculator to account for the new unit. The top unit is the original unit, while the number specifies how many of the bottom unit makes up one of the top unit.



3. If you have more than one custom unit, repeat steps 1 and 2.

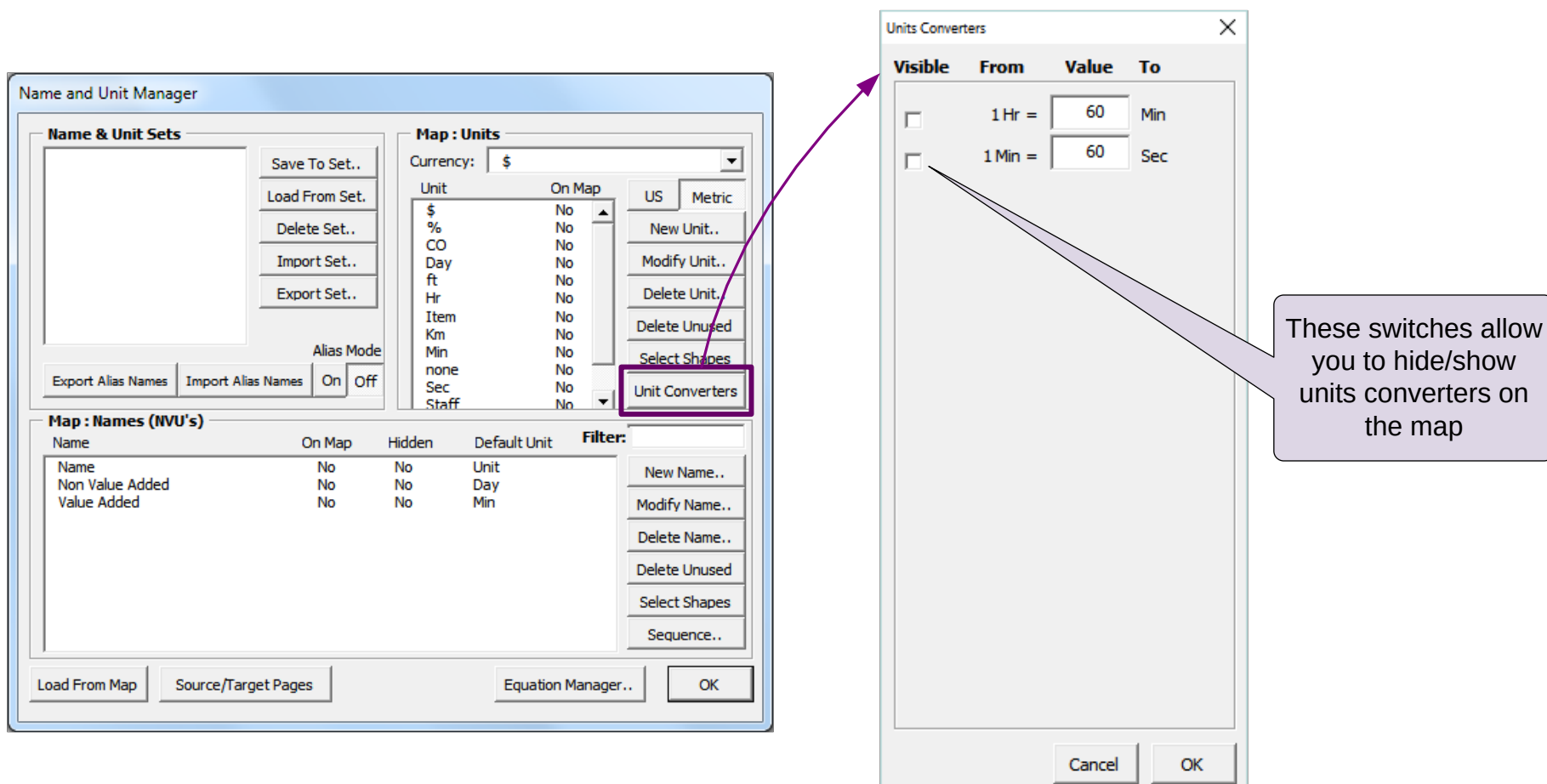
Part	Box
2	60
Item	Part



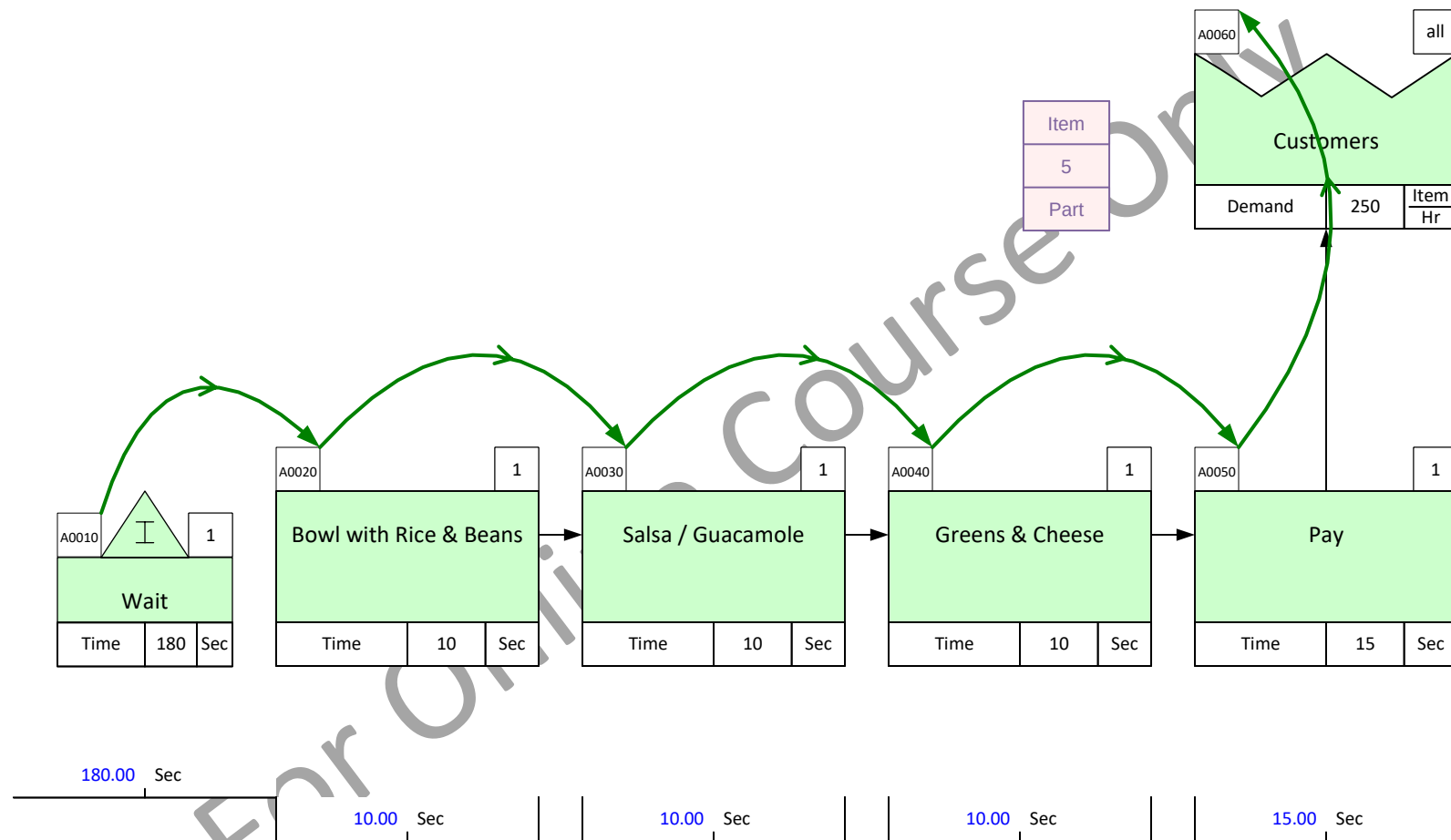
Note: Only the units that are connected via unit converter(s) will appear in the form when trying to change a variable unit on a center. This is to avoid (for example) people converting quantity units to time units for a center, like the Inventory Center. To ensure that the units you want to change appear in the form, create the converters first between any units you plan to use.

Hidden Unit Converters for “Well-known” Conversion Values

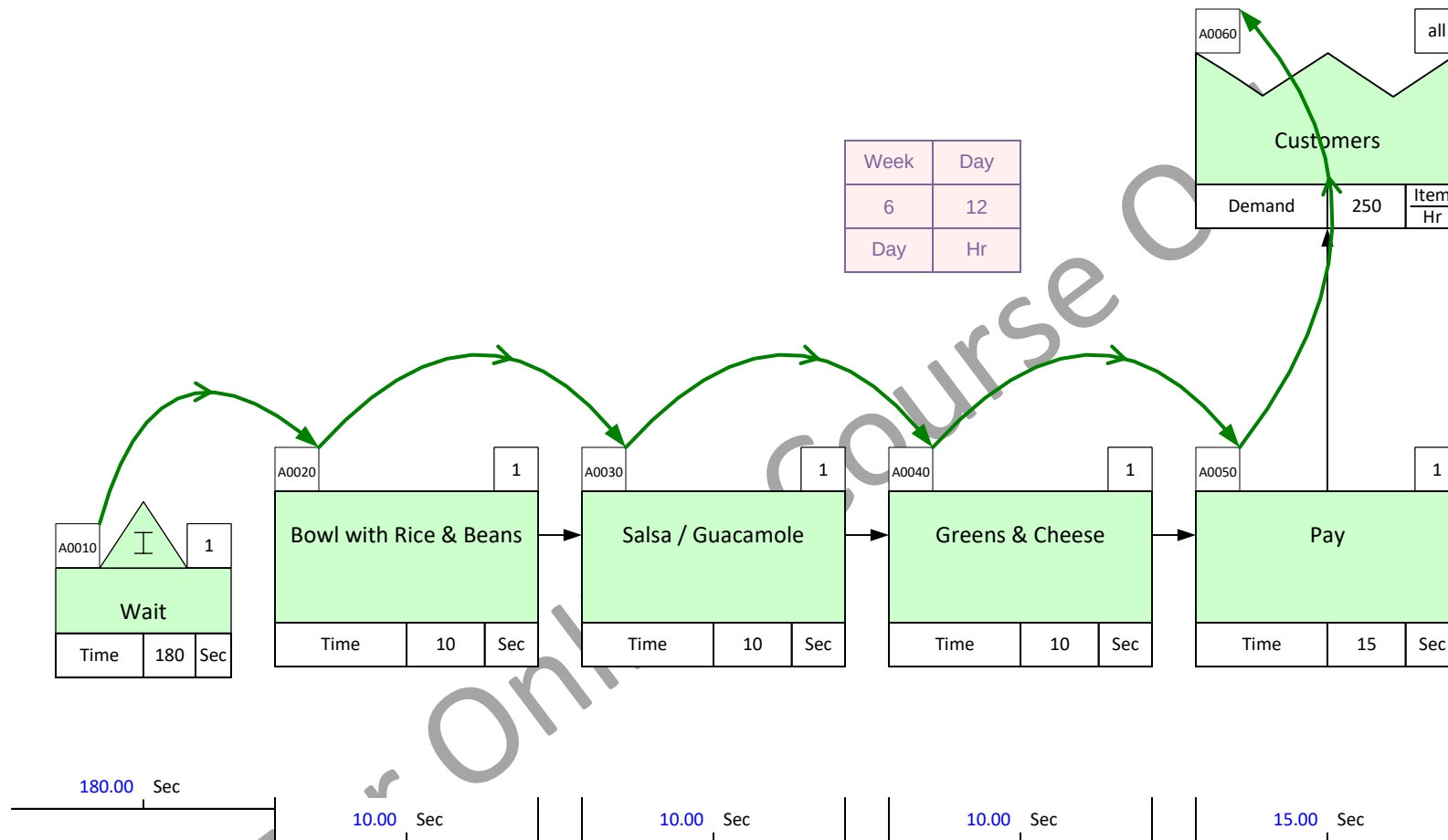
eVSM comes with hidden unit converters for well-known conversion values (Wks to Hrs to Mins). To check and see which unit conversions are included, open the NUM and click “Unit Converters.”



If the conversion you need isn't listed, you will need to include a Units Converter on your map in order to use your units. If the conversion you need is listed then you will not need to include a Units Converter. An error may appear if you add an additional Units Converter for a conversion that is included in eVSM, such as hours to minutes.

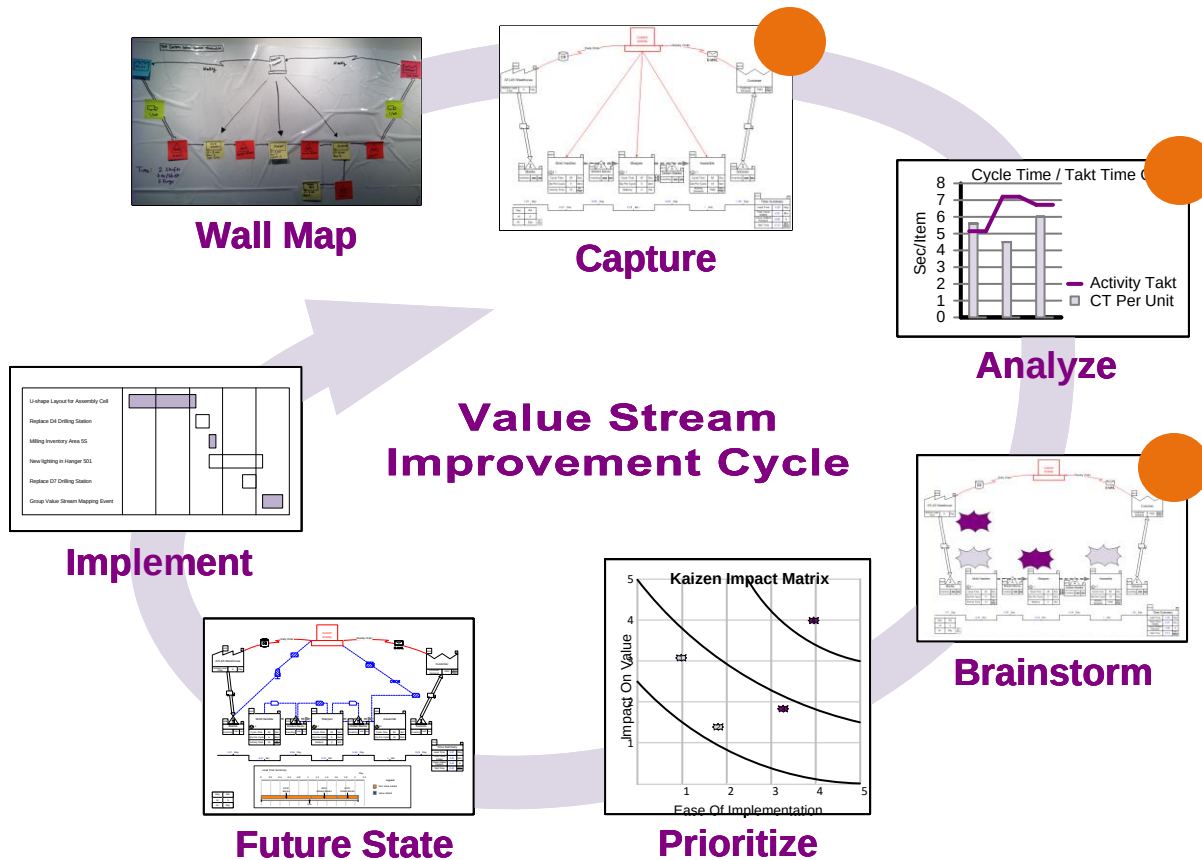






You learned that:

You can input and output data in convenient units on the map. This will let you make the Customer Fulfillment map easier to read and understand. Next let's go back to the Salsa Squares and the revised value stream intended by marketing. One of the major costs is labor so we will add that to the model and see how it can be optimized.



What's next:

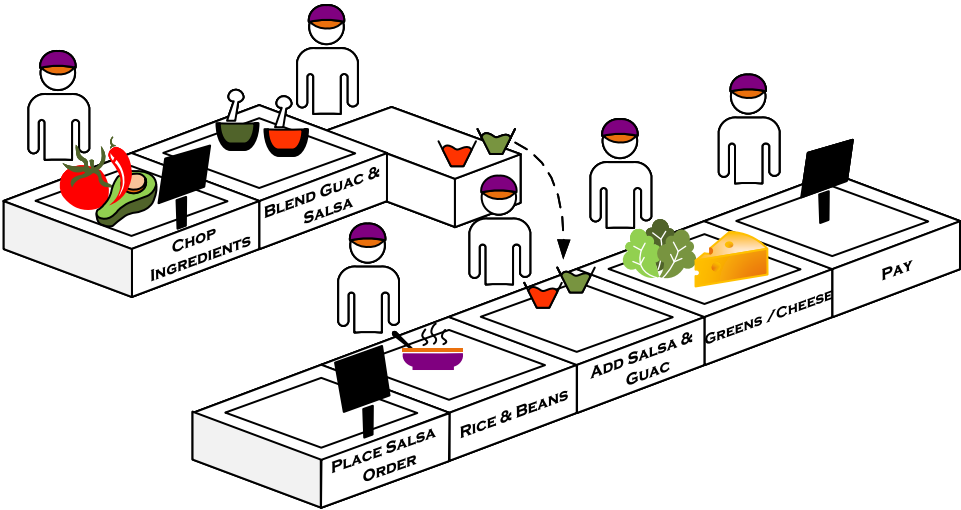
You have created a map to include marketing's concept of fresh salsa and it shows the staffing operation and hence cost has gone up. Next you review the map with the team and see what operational improvements can be made.

Resource Balancing

Marketing’s concept of a fresh salsa station has also introduced the need for additional staffing, and this is increasing the operational cost. You plan to look at resource usage efficiency for the operation, and if possible, directly on the map...

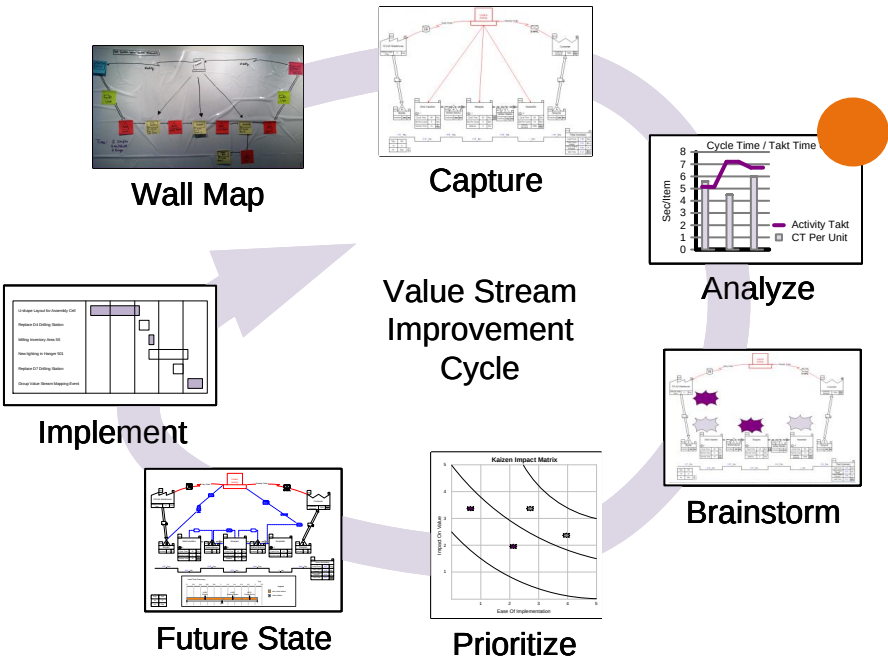
Resource Balancing

Case Study



The new salsa station needs to be introduced in a way that minimizes additional head count.

Improvement Cycle



You can analyze the map for key metrics like resource utilization in support of the marketing initiative.

Modeling Resources

It is useful on maps to:

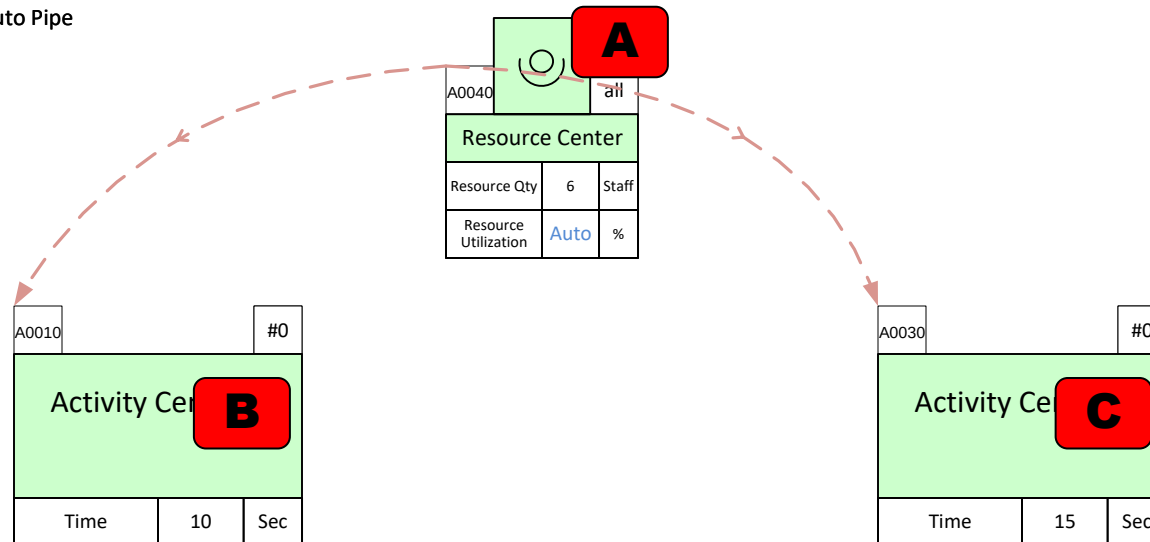
- See how resources are used for activities.
- Make sure the work is balanced across the resources.
- Ensure there is no over-burdening which would lead to bottlenecks and queues.

In eVSM, the Resource Center is put on the map and data is added for the quantity of resource available. This resource is then connected by “Pipe” arrows to the activities that need it. Pipe arrows are used for resource allocation only. The steps are as follows:

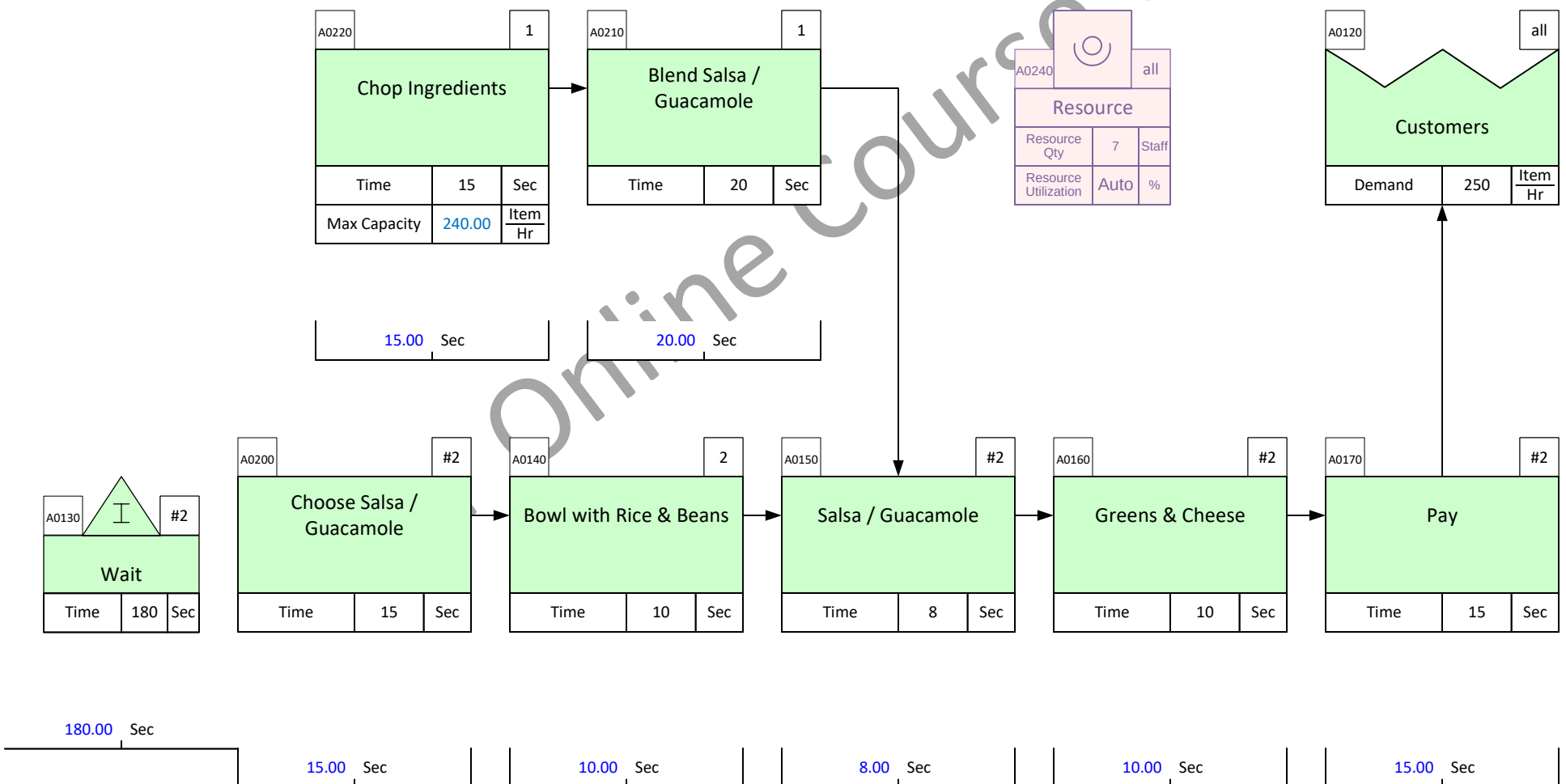
1. Drag the Resource Center from the stencil onto the map.
2. Select the Resource Center (A), then while holding down the shift key, select the activities (B & C) the resource serves.
3. Click the “Auto Pipe” button in the ribbon to automatically create pipe arrows from the Resource Center to each of the activities.

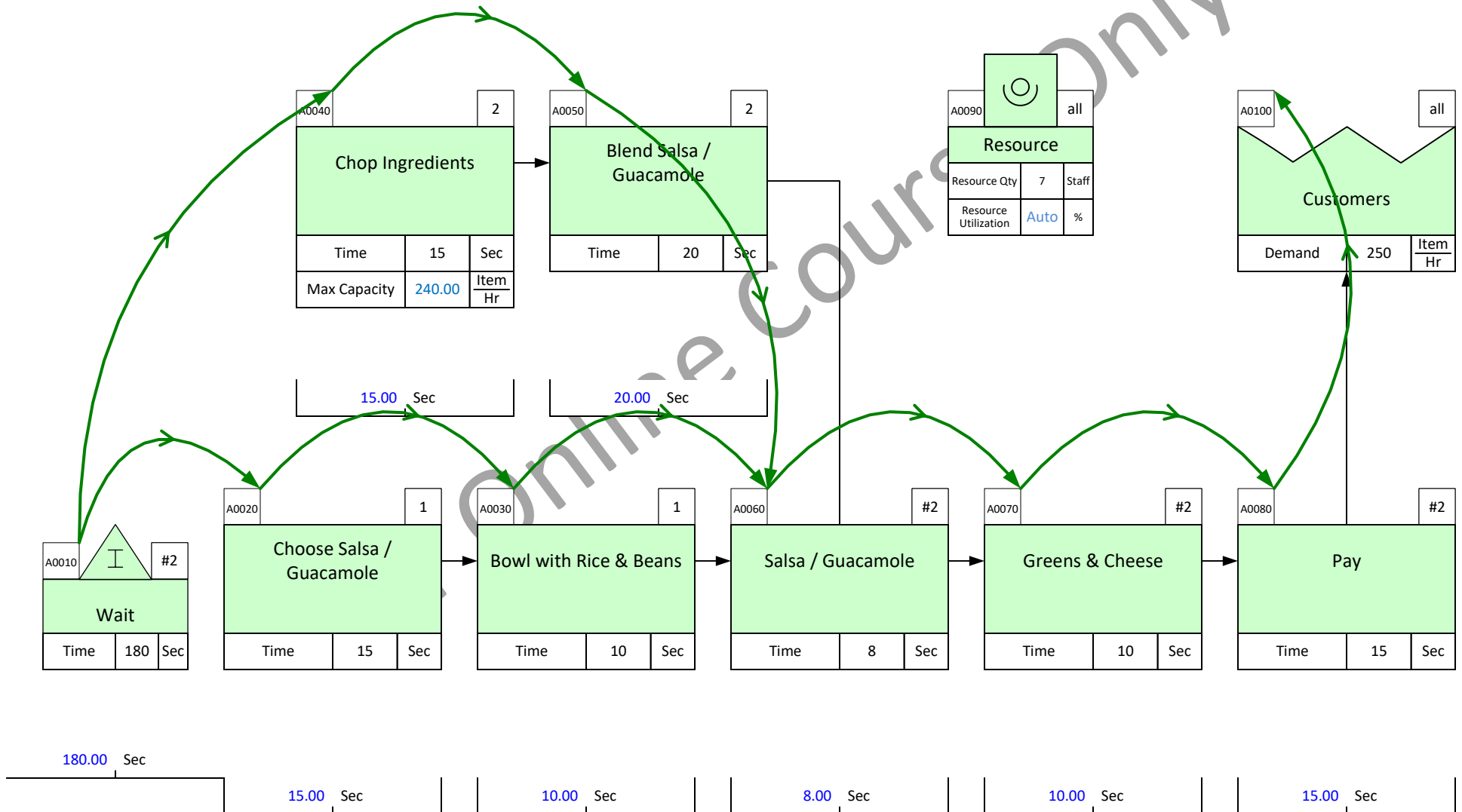


Auto Pipe



4. Fill in any required data (denoted by xx) on the Resource Center.
5. When the map is solved, the Resource Utilization numbers will be calculated.





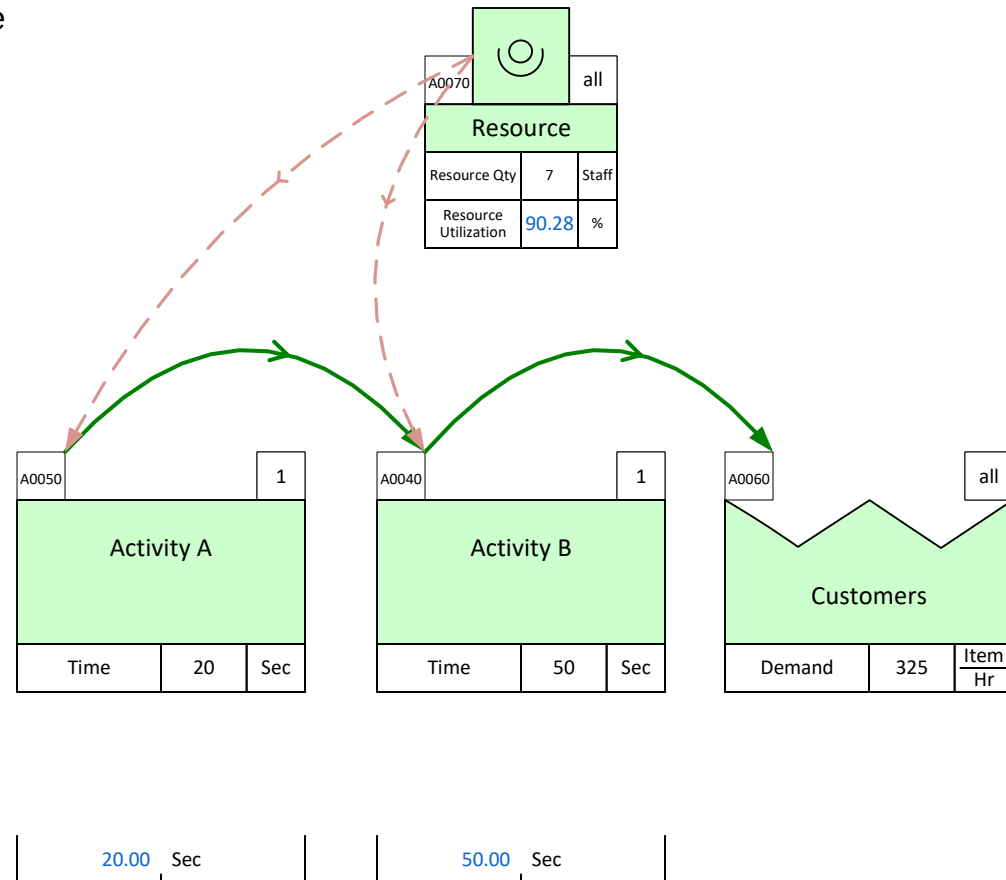
Resource Utilization

$$\text{Resource Utilization (\%)} = \frac{\text{Resource Used}}{\text{Resource Available}}$$

$$\begin{aligned} \text{Resource Used} &= \text{Demand} * \text{Total time per Item} \\ &= 325 * 70 = 22750 \text{ Seconds} \end{aligned}$$

$$\begin{aligned} \text{Resource Available} &= \text{Total Resource Time per hour} \\ &= 7 * 60 * 60 = 25200 \text{ Seconds} \end{aligned}$$

$$\begin{aligned} \text{Resource Utilization (\%)} &= \frac{22750}{25200} * 100\% \\ &= 90.28\% \end{aligned}$$



Z0010			#0
Summary			
Total Time	1.17	Min	
Value Added %	100.00	%	
Takt Time	0.18	Min Item	

Q. If management requires staff utilization to be >80%, but <90%, which one of these staff member quantities should the facility have?

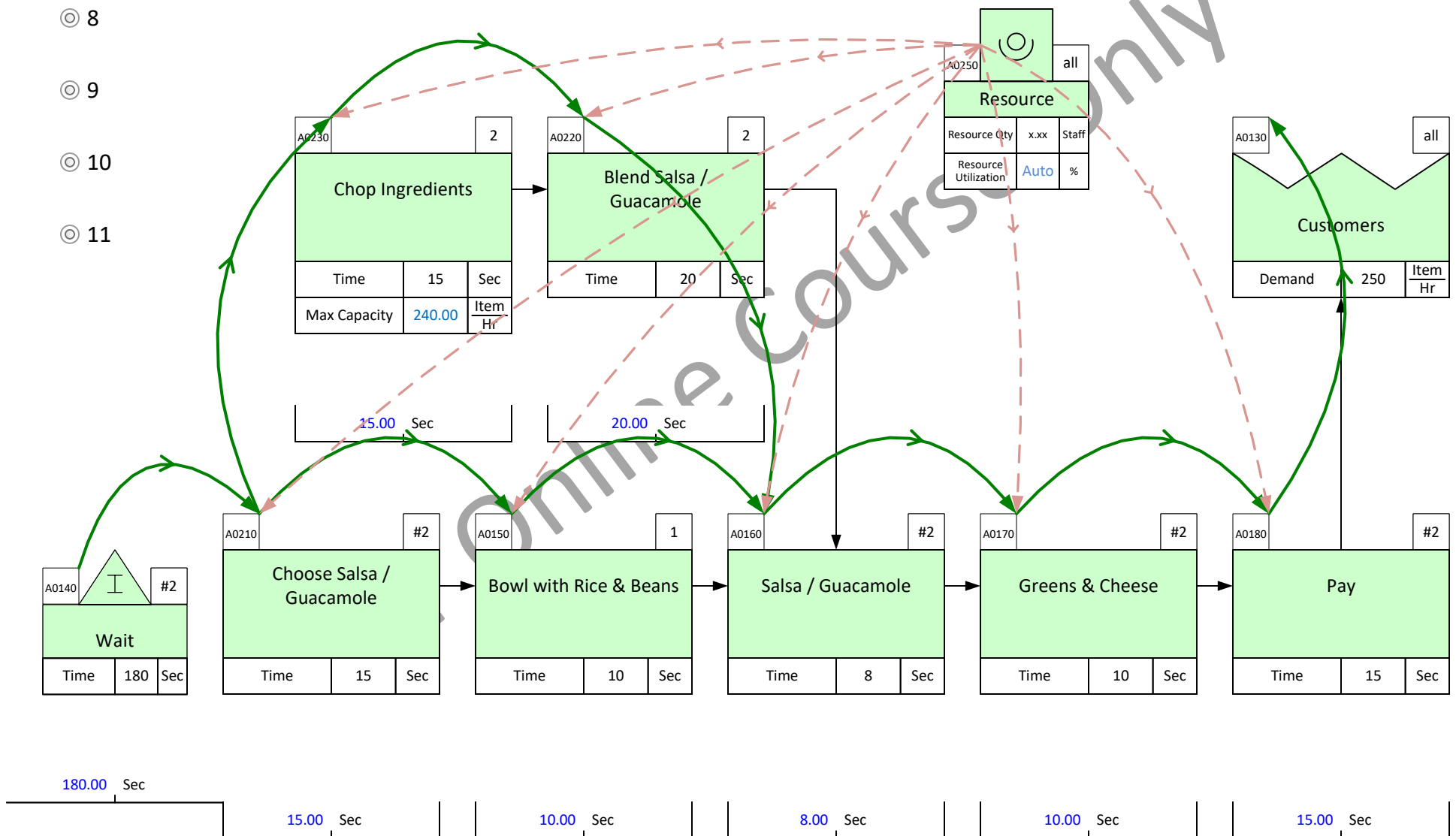
Ⓐ 7

Ⓑ 8

Ⓒ 9

Ⓓ 10

Ⓔ 11



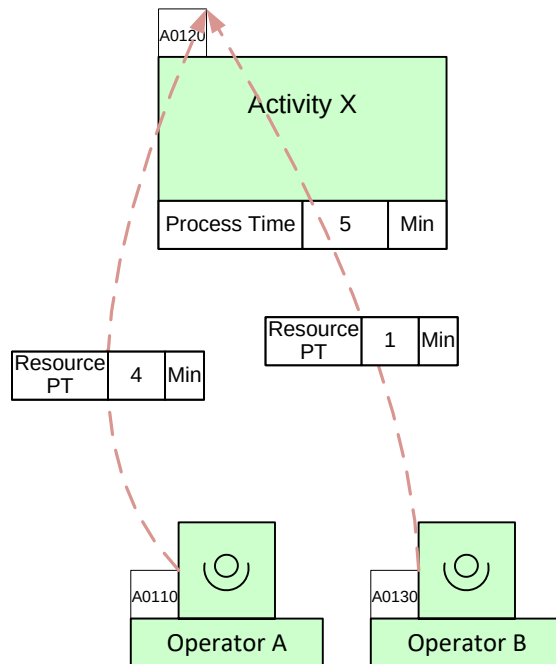
Resource Time vs Activity Time

Sometimes, the resource process time required is not the same as the activity process time. Below are some examples.

eVSM Mix Manufacturing Mix Processing, and the other main stencils (this is not available in Quick Time) allows you to input the Resource PT (process time) on the resource pipe arrow.

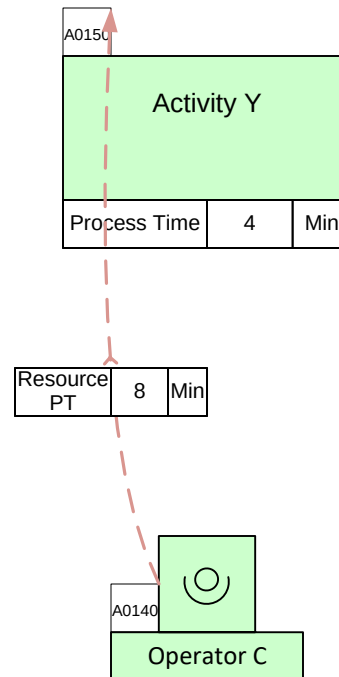
Example 1

Two different resources contributing to complete Activity X



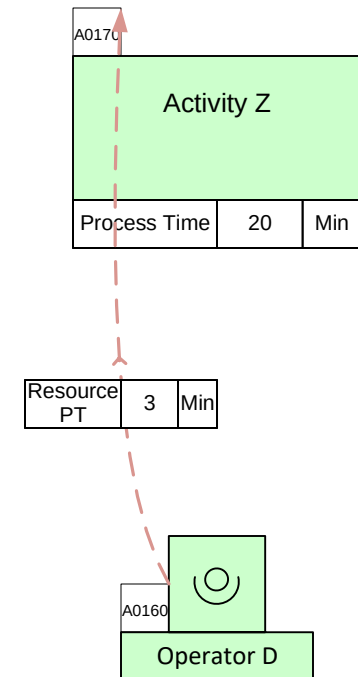
Example 2

Operator C completes this activity in 4 Min, but then has to spend an additional 4 Min for paperwork after the work item has gone



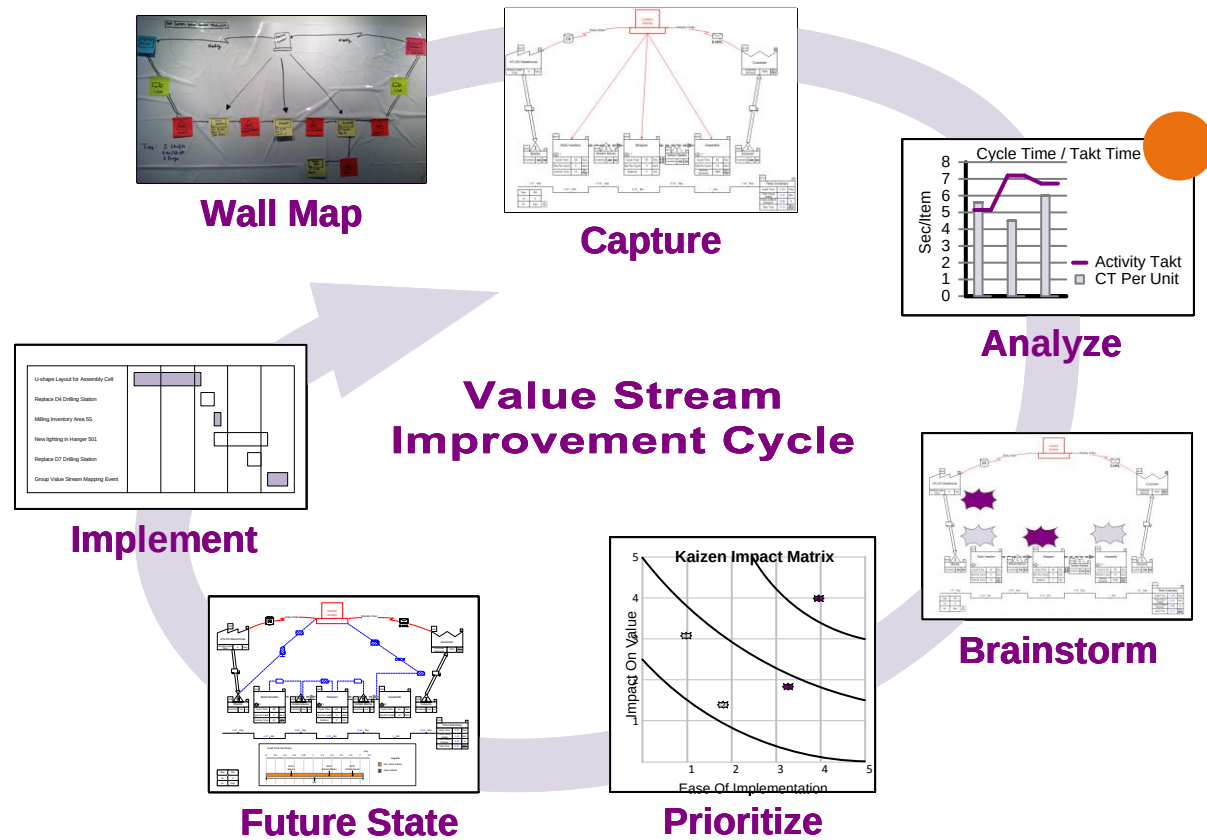
Example 3

Operator C starts this activity and then leaves it to complete on it's own, e.g. baking bread.



You learned that:

You can explore resource utilization directly on the map. This is important because resources represent a significant part of the operational cost of the store, which is impacted adversely by the new marketing “Fresh Salsa” initiative.



What's next:

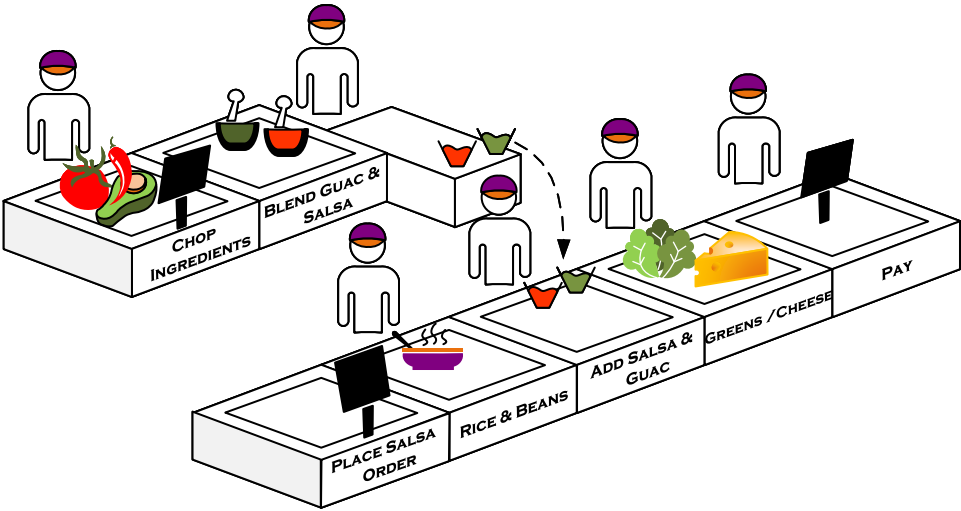
You have created a map to include marketing's concept of fresh salsa and it shows the staffing increase and hence cost has gone up. Next you review the map with the team and see what operational improvements can be made.

Brainstorm and Future State

You have seen that about 8 staff are currently needed to do the work at the store once the new salsa station has been implemented. You gather the team to review the map and generate ideas towards improving operation efficiency. You plan to use Kaizen Starbursts to show opportunities directly on the map so they can be discussed in context and prioritized.

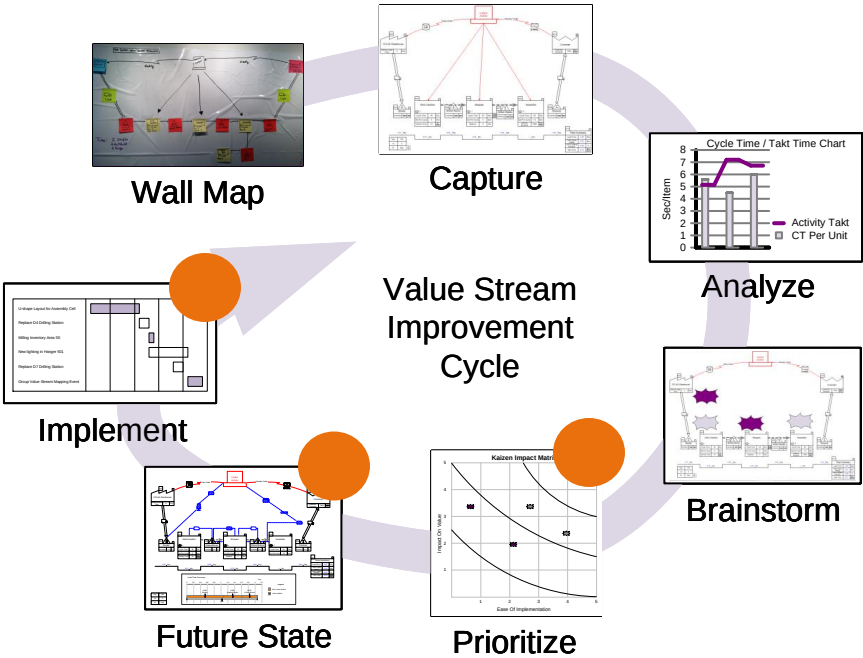
Brainstorm and Future State

Case Study



The planned headcount with the new salsa station is unacceptable. The team needs to come up with improvement ideas to reduce this whilst maintaining high levels of customer service.

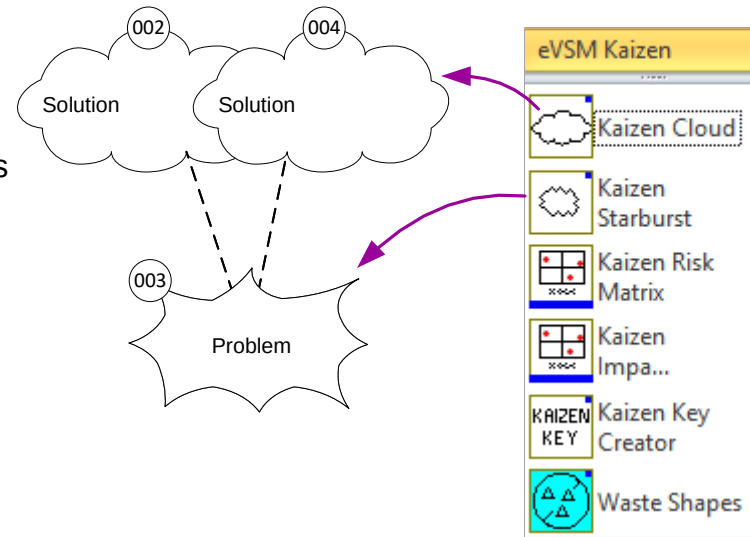
Improvement Cycle



Improvement ideas can be placed directly on the map and prioritized using what-if studies when they are useful. This would lead to a recommended future state and corresponding implementation plan.

Generating Improvement Ideas

Opportunities, risks or problems identified for the value stream can be put on the map using Kaizen bursts. Corresponding solutions or actions can be identified on the map with Kaizen clouds and leader lines from the bursts can connect to the corresponding clouds (one or more). For bursts and clouds, you can generate additional properties and then export them to a spreadsheet. This area was enhanced in v10.32



Selecting Kaizen Properties

You can select which data is important for Kaizen Starbursts and/or for Kaizen Clouds via the Kaizen Properties toolbar button



Kaizen Properties

Check or uncheck the boxes to add/remove properties of interest to you.

Double-click on 'Visible' to toggle all visibility on/off

Visible	Name	Description
☒	Description	
☒	Problem Description	Description of the problem being addressed
☐	Deliverable	What is the deliverable result of the project
☐	Resources	What resources are required to complete the project?
☐	Sustainment Plan	How will the project changes be sustained long-term?
☒	Finances	
☒	Comments	Misc comments about issues, etc...
☐	Estimated Cost	How much will the project cost?
☐	Cost Incurred	Cost incurred up to this point in project execution
☒	Estimated One Time Savings	One-time savings resulting from the project

Scroll Down to see all options

Edit Kaizen Properties

001 - Problem

Description
Problem Description

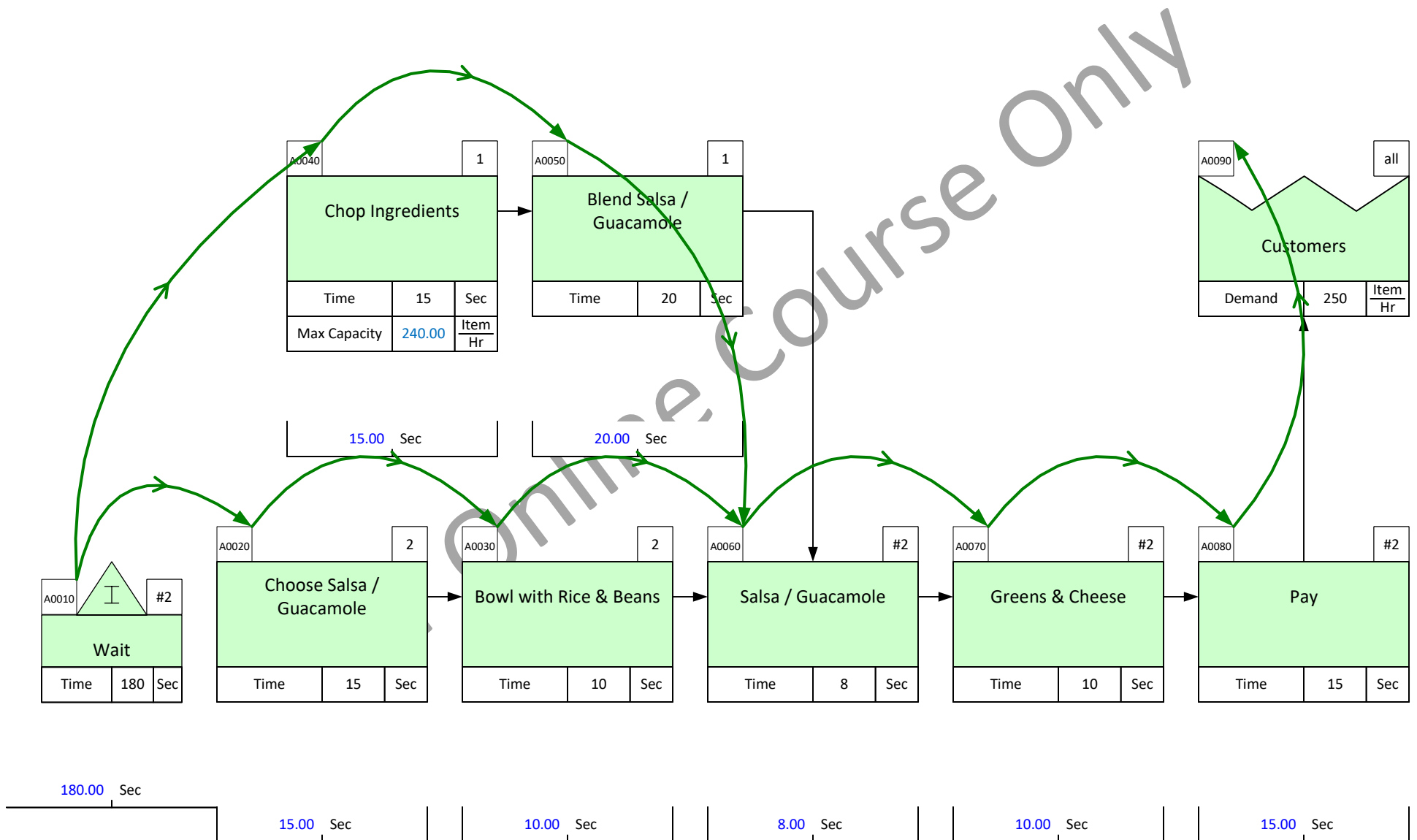
Comments

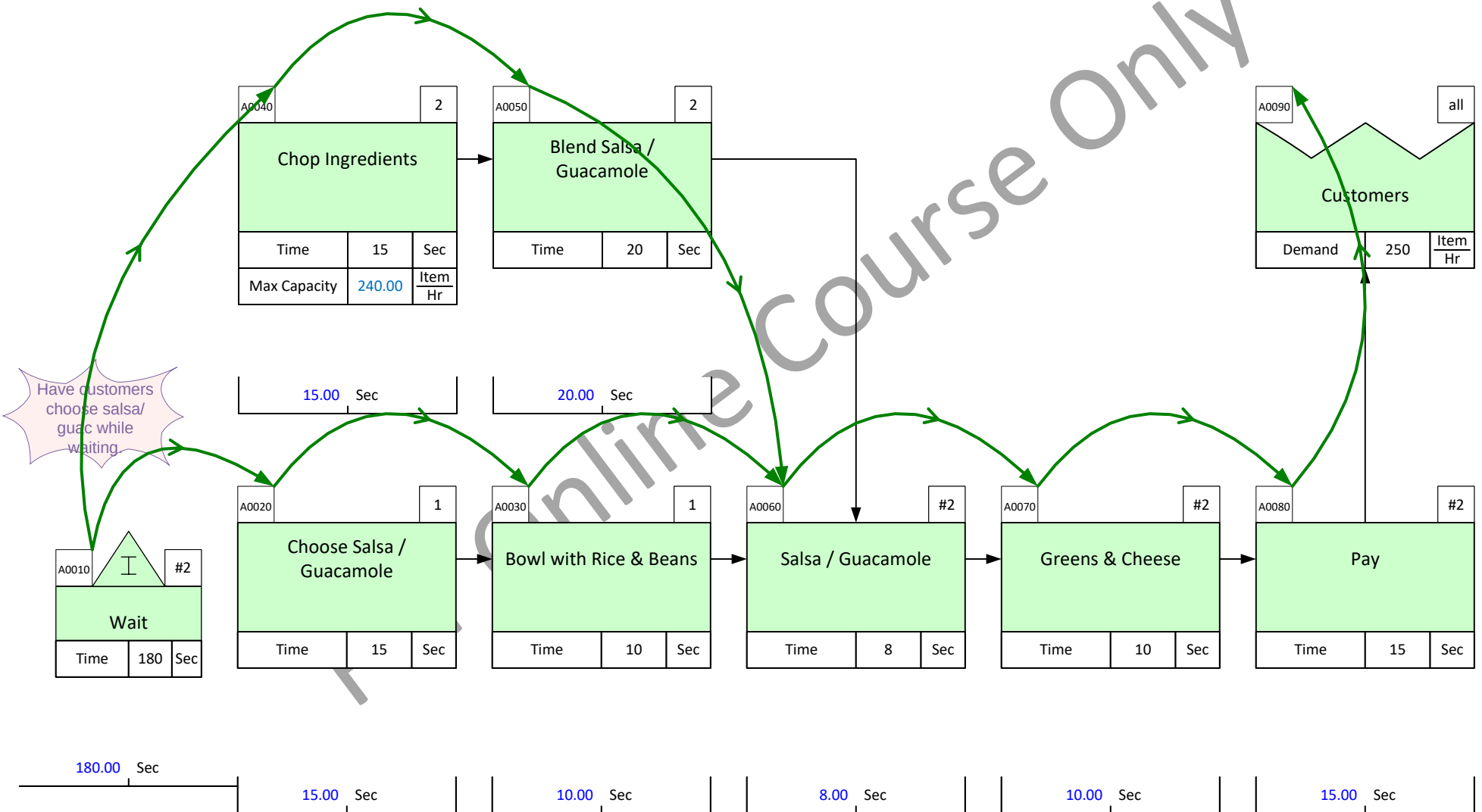
Finances
Estimated One Time Savings:
Estimated Annual Savings:

Priority
Impact: + -
Ease of Implementation: + -
Priority:

Cancel OK

For any Kaizen shape on your map, if you right-click and choose "Edit Kaizen", it will display an input form with just the properties you have selected.





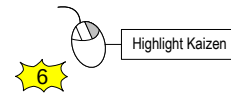
Prioritizing Improvement Ideas

Improvement ideas can be prioritized by using the Impact and Ease of Implementation properties of the Kaizen Starburst and plotting an Impact Matrix.

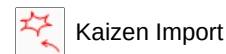
Kaizen Impact Matrix

The Kaizen Impact Matrix uses the “Impact” and “Ease of Implementation” custom properties of Kaizen Starbursts to plot an Impact Matrix. The objective of the matrix is to help select projects for implementation to improve the value stream.

On the matrix plot, each Kaizen Starburst is plotted with a mini-starburst. The mini-starbursts contain a copy of the custom properties from the Kaizen Starbursts. They are numbered based on the ID’s of the Kaizens on the map (ID’s are circles glued to the top left of the Kaizen shapes). A right-mouse click on any of the mini-Kaizens allows for selection and highlighting of the parent map Kaizen.



To edit the Impact and Ease of Implementation properties of each starburst, right-click the starburst and select “Edit Kaizen”. A form will appear allowing you to edit the various properties. Similarly, you could also edit the values via Excel spreadsheet and use the Kaizen Import button to bring the values back into eVSM.



Kaizen Import

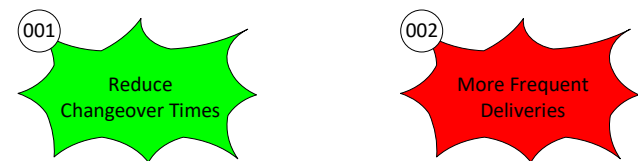
Impact (1-5): What impact this project will have on the value stream.

1 = Low, 5 = High

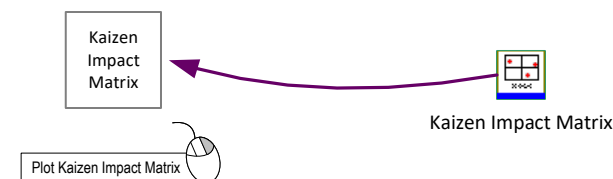
Ease of Implementation (1-5): How easy/difficult will it be to implement this measurable improvement to the Value Stream?

1 = Difficult, 5 = Easy

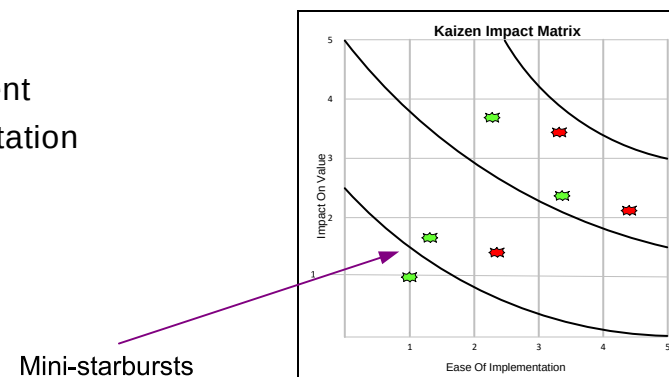
You can change the fill color of each starburst if you wish and the colors will be shown on the Impact matrix.



Drag out the Kaizen Impact Matrix icon from the eVSM Kaizen stencil and right-click to plot the matrix. Note that also on right click you can choose to plot just the bursts and not the clouds



Once the matrix has been created, you can drag the Starbursts to different locations on the matrix and the underlying impact and ease of implementation numbers will be automatically changed.



Using the Renumber Kaizen button in the toolbar, Kaizens can be renumbered in order starting at 001. Select the Kaizen Starbursts in the sequence you want using the Shift key, and then click the Renumber Kaizens button. Any un-selected Kaizen shapes will get numbered after the selected Kaizen shapes.  Renumber Kaizens

If bursts have leader lines that connect to the corresponding clouds then clouds will be renumbered to follow the burst shape. On Kaizen export, the report will then show each cloud under its corresponding burst shape. Very useful to create an action plan.

You can drag out the Kaizen Key Creator from the eVSM Kaizen stencil to differentiate each Kaizen Starburst by description. Such descriptions are typically placed below or to the side of the Impact Matrix.



Kaizen Key Creator

Generating Kaizen Report



Kaizen Export

The Kaizen Export toolbar button scans the map for all Kaizen Starbursts and exports their properties to an Excel spreadsheet. (Pagename_Kaizen). If you want to edit the properties data via Excel, you may do so and import the data back into eVSM using the Kaizen Import button.

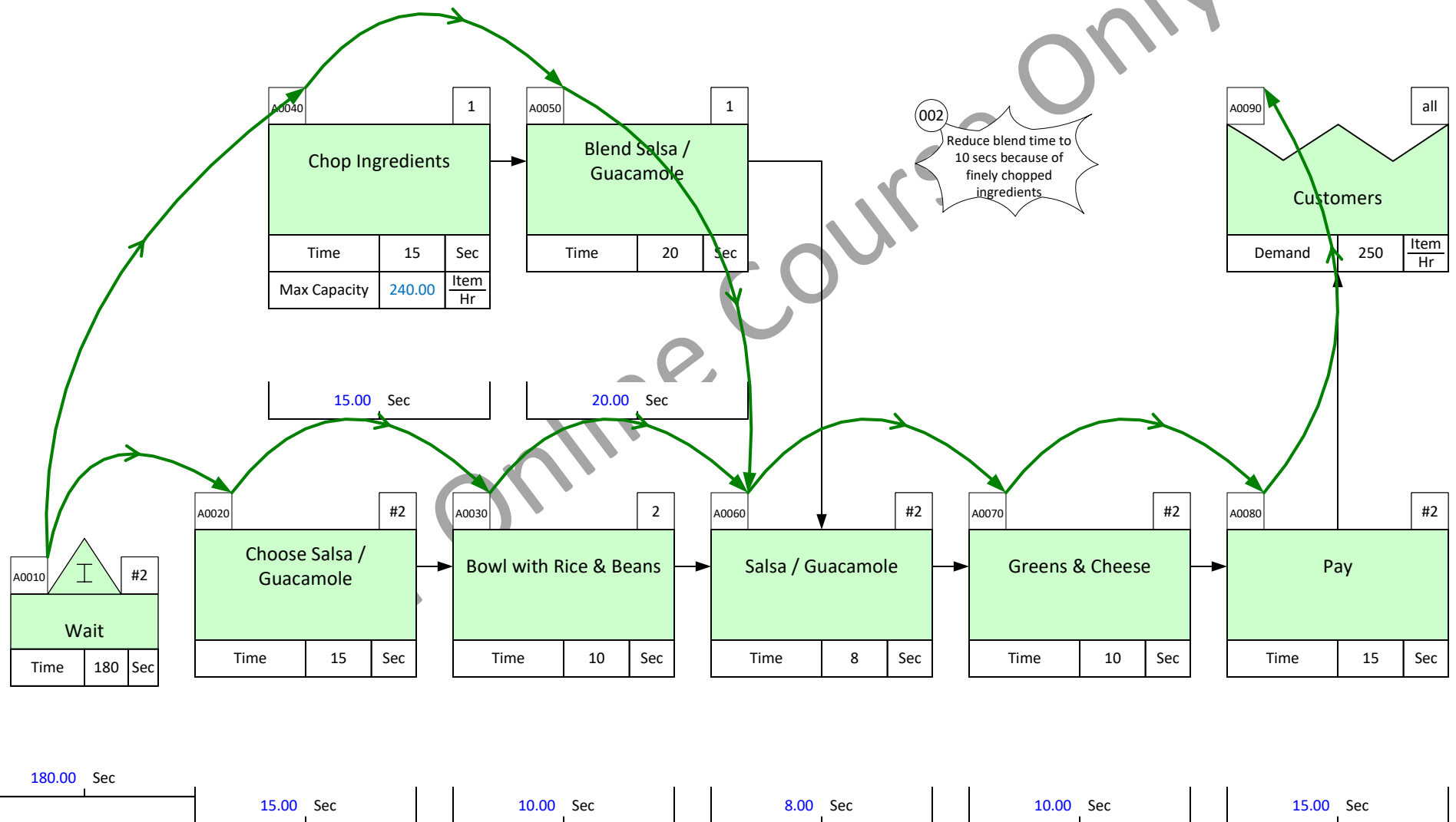


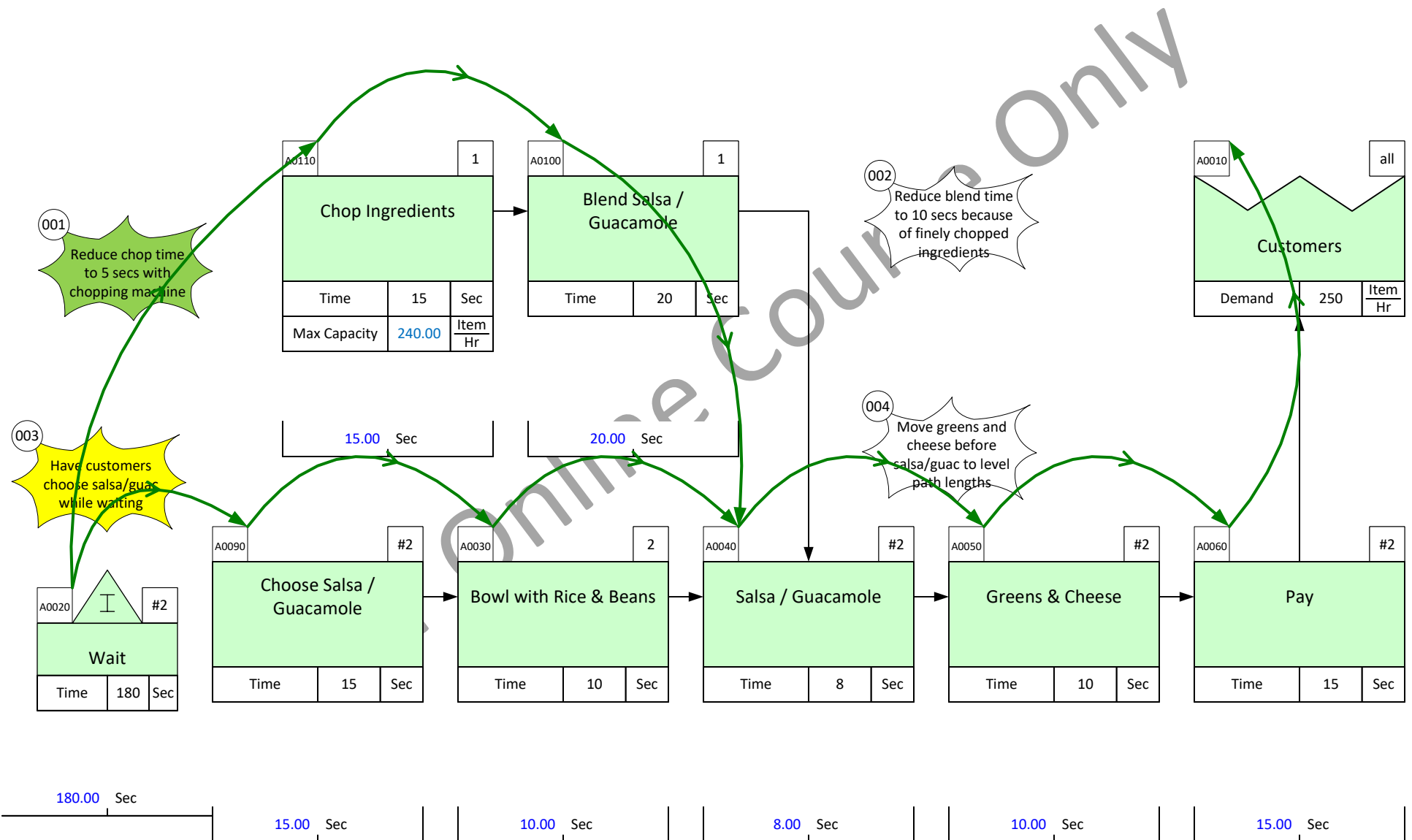
Kaizen Import

Note that new Kaizens can be added into the excel report below the blue line and then imported into the map. For each new Kaizen start by selecting the Kaizen Type in the spreadsheet cell pull-down

	B	C	D	E	F	G
	Kaizen No.	Kaizen Type	Kaizen	Problem Description	Comments	Estimated One Time Savings
2						
5	1	Starburst	Problem			
6						
7						
8						
9						
10						
11						
12						

Kaizen Type
Select a Kaizen type from the in-cell drop-down list





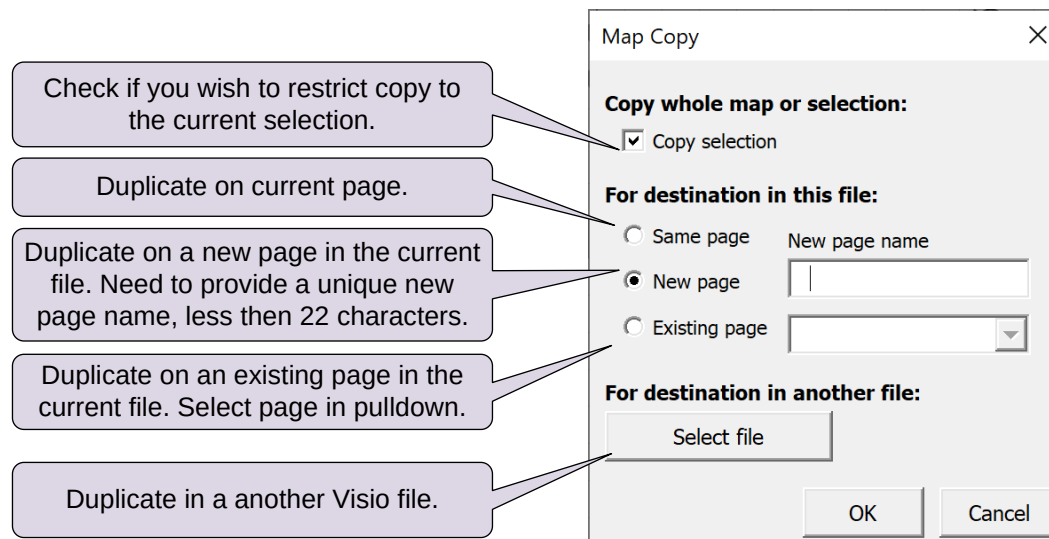
Doing What-If Studies

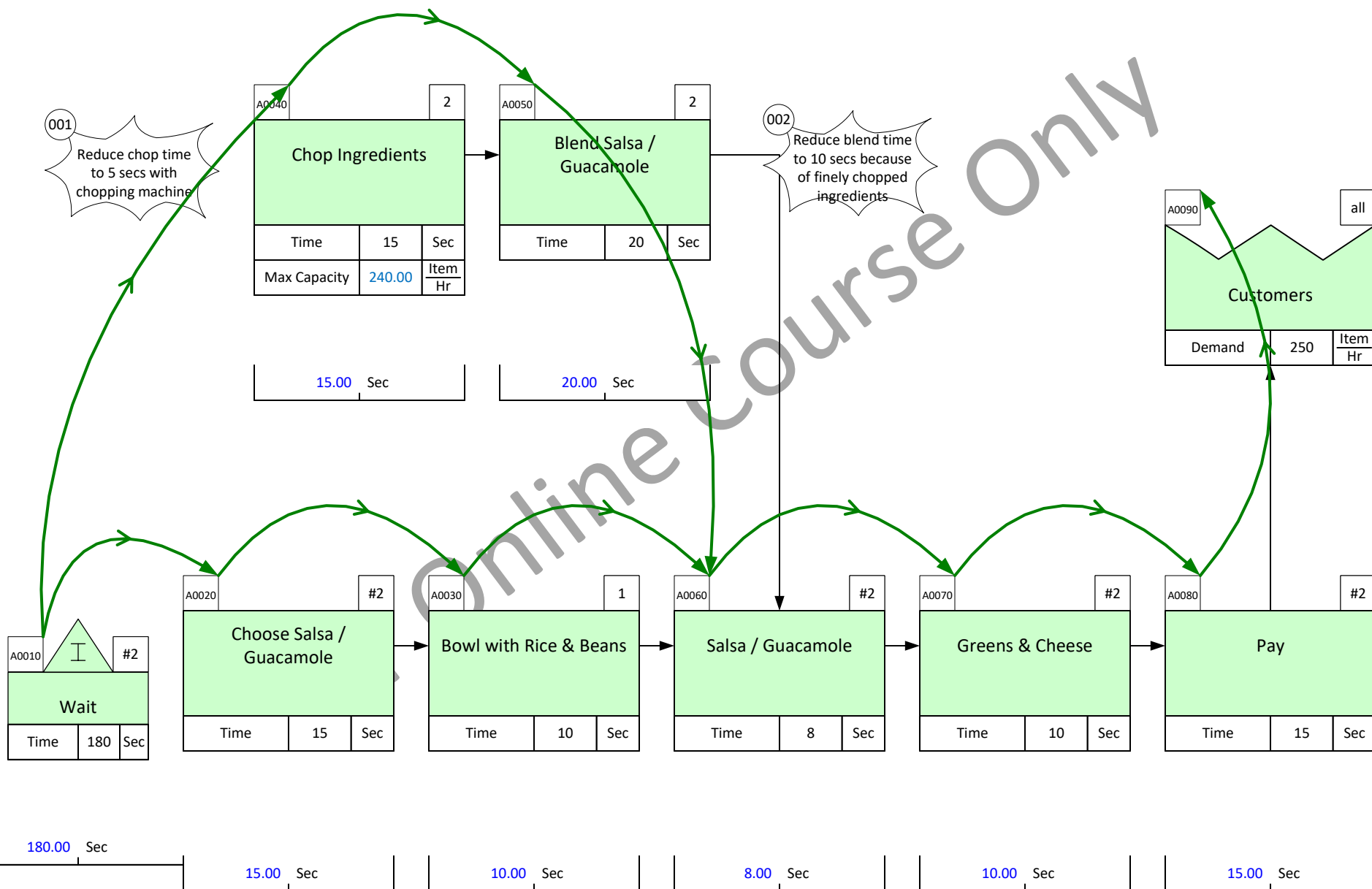
You can make a copy of the current state map to do “What-If” studies leading to the future state.

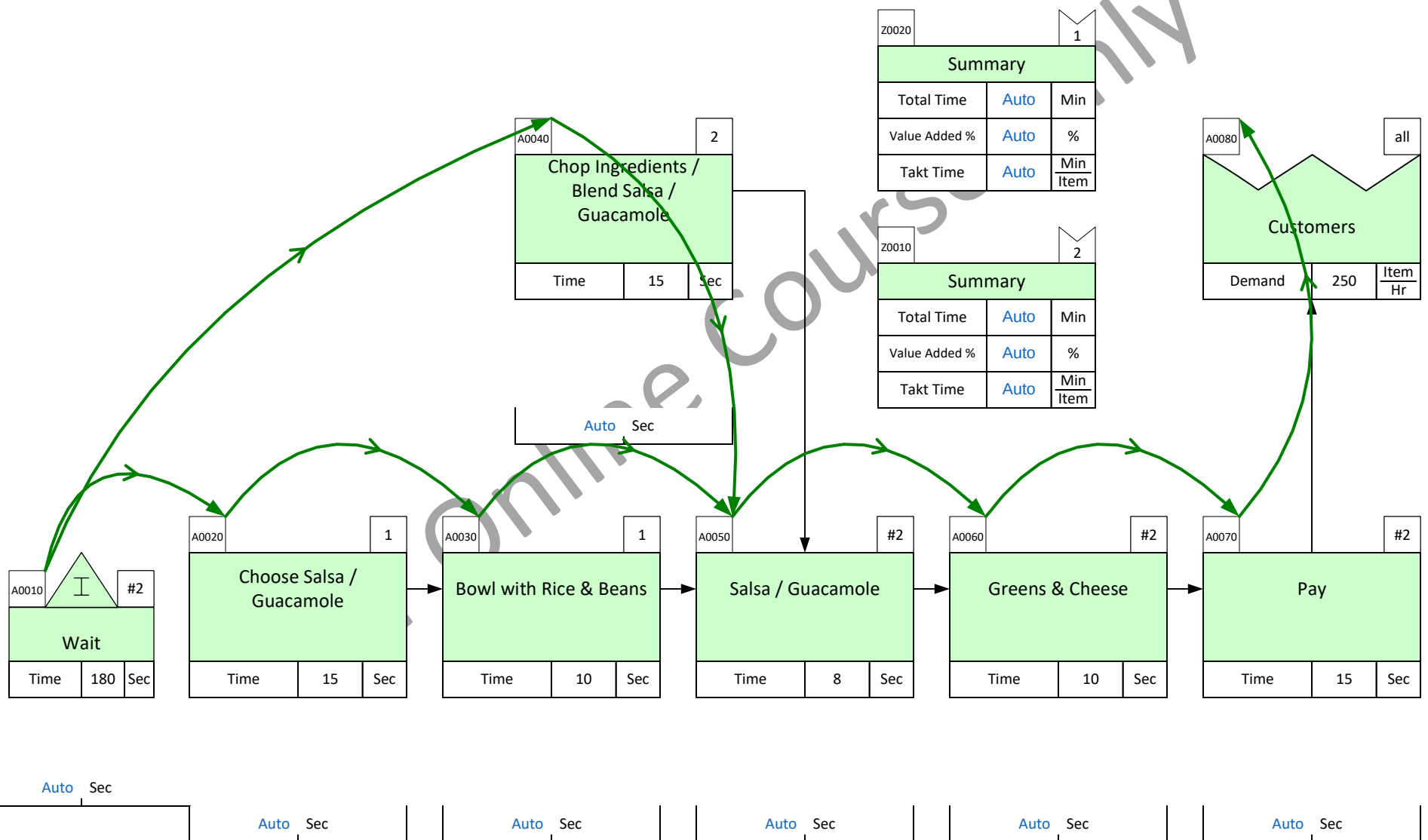
Using the Map Copy button in the toolbar, eVSM can copy your map to another page in the file. Important: Do not use Visio’s copy and paste function to duplicate a map. This will miss some of the hidden data associated with the map.

Map Copy Steps:

1. On the source map, select what you wish to duplicate. If you wish to duplicate the whole map, then select nothing.
2. Click the  Map Copy button to open the following dialog.



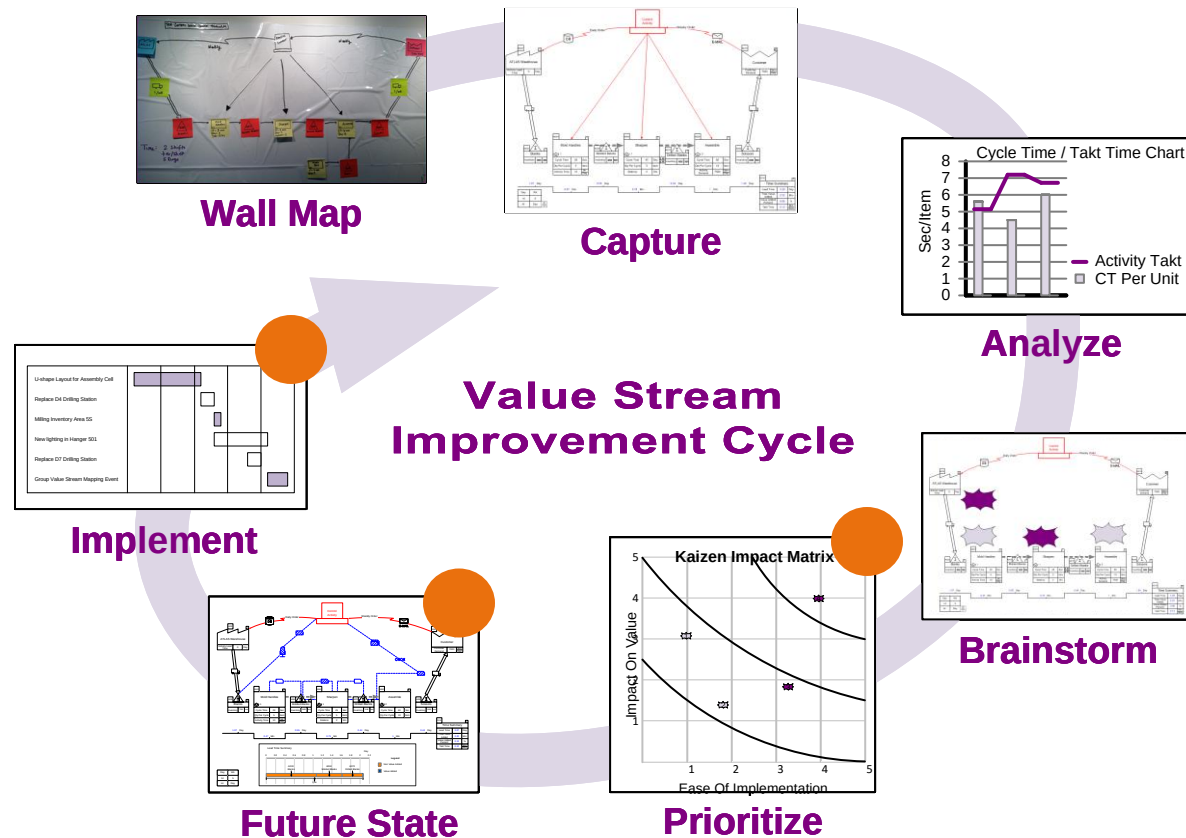




You have learned:

You have completed and passed through the improvement cycle and selected projects towards an improved future state, and in support of marketing's initiatives to have a fresh salsa station. You have learned how to use eVSM to help you through the steps of the improvement cycles as applied to the Customer Fulfillment value stream.

The map, analysis, and visuals have allowed team members to “see” the waste in the value stream and come up with improvement ideas. Finally, you have created a future state map that you can communicate any decisions with management and get support to related improvement resources.



—Useful Links—

eVSM Toolbar Guide

evsm.com/toolbarguide

eVSM Productivity Guide

evsm.com/productivity

eVSM Blogs

evsm.com/blog

eVSM Support FAQ

evsm.com/support

Download the Latest Version

evsm.com/install